Speech Tokenizers and Speech Language Models: A Survey of Representations, Modeling, and Systems

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Abstract

This survey maps the transition from cascaded ASR→LLM→TTS pipelines to unified Speech Language Models (SLMs) that reason and speak natively, lowering latency, preserving paralinguistics, and enabling duplex behaviors. We organize the modern speech-AI stack around token interfaces and a representation–latency–bitrate trilemma, synthesizing a taxonomy spanning semantic, acoustic, paralinguistic, mixed, and continuous pathways; quantization mechanisms (k-means, VQ/RVQ, BSQ); and rate/sequence controls (variable frame-rates, dedup/acoustic BPE). Architecturally, we review adapter bridges that fuse frozen SSL encoders to text LLMs, decoder-only SLMs with AR/NAR schedules (e.g., AR-on-first RVQ stream with NAR refinement), and interleaved speech–text modeling scaled by token-space pretraining, highlighting a robust 12.5 Hz operating point and ultra-low-rate viability at 6.25 Hz with text-aware diffusion. We systematize streaming and duplex design—chunked policies, wait-k/CIF gating, barge-in/IRQ control—and connect them to practical corpora (e.g., J-CHAT) and bandwidth ladders. Rate-aware evaluation is consolidated across datasets (LibriSpeech, Common Voice, CoVoST2, LibriTTS, VoxCeleb) and suites (DASB, STAB, Dynamic-SUPERB, VoxEval), with metrics for semantics (WER/CER, BLEU), perception (MOS/UTMOS, FAD), unit discriminability (ABX), identity/style, and streaming latency (RTF, AL/LAAL). Contributions include: (i) a unified token taxonomy with explicit rate/bitrate attributes; (ii) a comparative design analysis linking quantization, codebook usage, and rate control to intelligibility, naturalness, and style; and (iii) standardized, bitrate-controlled training/evaluation practices for reproducible cross-paper comparison. We close with open challenges—unified vs multi-stream tokens, long-context conversational memory, multilingual scaling, alignment without forgetting, and a rate–capacity theory—and provide recommendations for duplex-ready, safe, and efficient deployment.

1 Introduction

1.1 Why now: From cascaded pipelines to unified end-to-end Speech Language Models

Cascaded ASR→LLM→TTS pipelines accumulate recognition and translation errors, inflate interaction latency, and discard prosody, emotion, and non-verbal cues that matter for human–AI conversation. Duplex dialogue and socially aware assistance require agents that can listen, reason, and speak without collapsing paralinguistic information into text-only intermediates, a gap explicitly articulated in beyond-semantic taxonomies for speech agents and addressed by native speech-in/speech-out modeling . End-to-end Speech Language Models (SLMs) directly optimize for these user needs by integrating understanding and generation within a single generator capable of barge-in, interruption handling, and continuous backchanneling [1].

Empirical evidence shows that unified SLMs can surpass cascaded systems while training far fewer parameters. Bridging a frozen USM encoder and a frozen mT0-MT text LLM with a lightweight

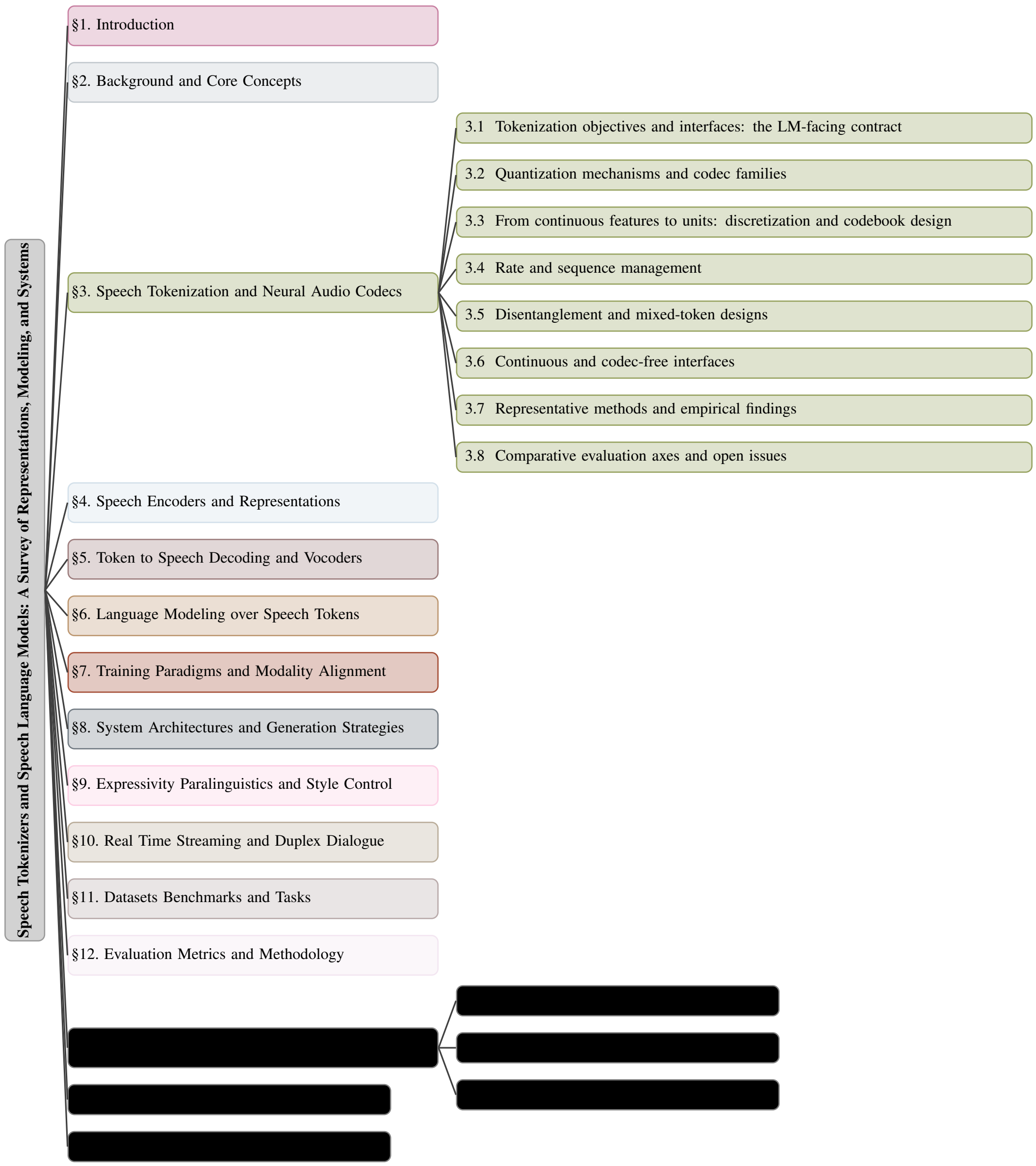

Figure 1: chapter structure

adapter improves CoVoST2 Fr→En BLEU from roughly 32 in a cascade to about 38 in a single end-to-end model, while training only 1

User-facing requirements amplify this shift. Emotion and empathy are central to coaching and care scenarios, yet prior training pipelines relied on TTS-synthesized emotional speech that fails to generalize to natural human audio and incurs significant data and compute costs; end-to-end SLMs that natively ingest and emit speech are positioned to learn robust paralinguistic cues and deliver empathetic dialogue capabilities envisioned at the higher levels of the BoSS capability taxonomy . Discrete token-based SLM TTS has also matured: such models enable probabilistic, context-aware generation, zero-shot voice prompting, and multi-task behavior, though current evaluations underweight intelligibility robustness, speaker consistency, prosodic control, and spontaneity, motivating comprehensive end-to-end assessment aligned with SLM use cases [2].

Technical catalysts make unified SLMs tractable at scale. Modality projectors and adapters allow frozen, self-supervised speech encoders to be fused with large text LLMs while preserving text competence and minimizing trainable parameters, unlocking multilingual speech tasks and spoken instruction following without extensive paired instruction data [3]. Synthetic interleaving of speech tokens within text corpora enables trillion-token pre-training without the need to synthesize full waveforms, removing a major computational bottleneck and supporting interleaved speech–text learning at language-model scale [4]. On the generation side, decoder-only SLMs that produce acoustic tokens or codec units consolidate text-to-speech functionality into the same model that performs reasoning and translation, simplifying deployment and reducing latency [5].

Streaming and duplex operation further favor unified models. A language model that listens while speaking natively handles barge-in and immediate interruption during speech generation, cutting time-to-first-token and aligning emission with incoming audio—behaviors that are difficult to realize in cascades with rigid stage boundaries [1]. These capabilities are increasingly indispensable for assistants expected to manage overlaps, backchannels, and rapid turn-taking.

Data availability is catching up to these modeling advances. Large, open, spontaneous, and acoustically curated spoken dialogue corpora beyond English have been scarce; an automated, language-independent construction pipeline has produced a 68,892-hour Japanese multi-domain corpus whose scale and cleanliness support end-to-end SLM training and evaluation, with multi-domain data improving naturalness and meaningfulness in downstream speech LMs [6]. Such corpora enable cross-lingual generalization and reduce the reliance on synthetic data that previously constrained paralinguistic learning [7].

Recent results collectively supply the missing pieces for replacing cascaded ASR→LLM→TTS with a single end-to-end SpeechLM: scalable pretraining on 1T tokens built from synthetic interleaved text–speech-token data (including 600B synthetic) that bypasses waveform generation and uses ASR-derived VQ tokenizers to preserve semantics at low frame rates; dual-token SpeechLMs that align with text via an understanding-driven tokenizer while generating acoustic tokens with stabilized training; simple, robust, training-free dMel discretization that supports a unified LM-style architecture for both ASR and TTS; and joint speech–text LMs that enable zero-shot cross-modal transfer and competitive spoken chatbots—evidenced by large gains in spoken QA (e.g., 13

1.2 Scope, contributions, and taxonomy

This survey focuses on the end-to-end speech–AI stack that converts waveforms into model-consumable representations, performs language modeling over speech and mixed modalities, and renders tokens back to audio under practical latency, bitrate, and deployment constraints. The covered components include: speech tokenizers and neural audio codecs; self-supervised and supervised encoders; token-to-speech decoders and vocoders; language models over discrete and continuous interfaces; training and alignment pipelines spanning pre-training, modality adaptation, supervised fine-tuning, instruction tuning, and post-alignment; system architectures from cascaded to interleaved and duplex; datasets and benchmarks for recognition, generation, and paralinguistic evaluation; and methodology for rate-aware, streaming, and duplex assessment as well as efficiency and safety. Following established delineations, classic signal-processing codecs are not treated in depth, continuous-only SSL features are considered primarily as sources for quantization, and broader speech tasks are used as evaluation targets rather than as objects of methodological analysis [8].

Token interfaces and rates constitute the central organizing axis. We adopt a definition of discrete speech tokens that distinguishes acoustic tokens—neural audio codec indices optimized for waveform reconstruction—from semantic tokens—quantized SSL or supervised intermediates optimized for phonetic and linguistic content [8]. Building on community taxonomies, we categorize tokenizers into semantic, compression, and hybrid families (e.g., Discrete HuBERT/WavLM/Wav2Vec2; EnCodec/DAC; SpeechTokenizer), enabling comparisons under unified bitrate controls and standardized downstream use [9]. Quantization mechanisms span k-means and learnable vector quantization variants, including Gumbel VQ, finite/state/product quantization, and residual/grouped/clustered schemes such as RVQ, GVQ, GRVQ, and CSVQ, with attention to codebook utilization, collapse prevention, and stability under training-time objectives like semantic distillation and disentanglement [8]. Rate- and sequence-management techniques (variable frame-rate tokenization, multi-resolution or variable-bitrate coding, deduplication, and acoustic BPE) are included to surface the representation–latency–bitrate trade-offs that drive downstream modeling and system latency [8]. Token vocoders for semantic sequences and codec decoders for acoustic tokens provide the rendering endpoints needed for controlled generative evaluation .

Modeling and generation are surveyed across decoder-only autoregressive and non-autoregressive regimes, dual-branch and interleaved text–speech modeling, and parallel or hierarchical token generation. Architectures covering hierarchical residual quantization, multi-token prediction, and efficient long-horizon planning (e.g., Groupformer- or RQ-Transformer-style schedules) are discussed with respect to token-rate sensitivity, context budgeting, and duplex interaction. Concretely, we analyze alignment and modality transfer by (i) using modality projectors such as discrete speech tokenizers and speech captioners to map audio into text-aligned units, (ii) jointly training on mixed speech-unit and text sequences with automatic metrics to assess mixing quality, and (iii) coupling the tokenizer, LM, and decoder via shared or aligned token spaces—an approach that preserves text competence while adding speech understanding, enables zero-shot cross-modal transfer and instruction following without explicit speech instruction tuning, and improves SLU and Dynamic-SUPERB generalization. [10, 11]

Evaluation and methodology emphasize rate-aware, apples-to-apples comparisons. We align with bitrate-controlled, multi-task benchmarking practices that standardize downstream models and report the best among reference architectures per task, enabling isolation of tokenization effects from backbone choices [9]. Consistent rendering via a scalable HiFi-GAN-based semantic token vocoder supports generative assessments across naturalness, intelligibility, and speaker/style preservation [9]. Metric families include reconstruction- and recognition-oriented measures (e.g., ABX, codebook usage, WER), semantic fidelity (BLEU, BERTScore), perceptual and distributional audio quality (MOS, Fréchet Audio Distance), and streaming/duplex latency diagnostics; we additionally advocate bitrate- and frame-rate–controlled ablations, sequence-length reduction analyses, and cross-domain robustness checks .

The survey's contributions are threefold. First, a token taxonomy is articulated that unifies acoustic, semantic, paralinguistic, and hybrid/multi-stream interfaces with explicit token-rate, bitrate, and sequence-length attributes, consolidating LLM-era perspectives on discrete speech tokens [8] and harmonizing them with benchmark-driven families used in practice [9]. Second, a comparative design analysis integrates quantization mechanisms, codebook designs, disentanglement strategies, and rate control with downstream SLM performance, including reconstruction, voice conversion, and expressive generation [8]. Third, unified training and evaluation practices are proposed: bitrate-controlled, reference-architecture–normalized experiments; standardized token-to-speech rendering pipelines; and diagnostics for modality alignment and forgetting, with recommendations that reflect scalable benchmarking tenets and enable reproducible, cross-paper comparison . The agenda highlights actionable directions such as decoupled and variable frame-rate token designs, variable-bitrate coding, and sequence compression that preserve semantic fidelity while meeting real-time and duplex constraints [8].

1.3 The representation–latency–bitrate trilemma

Speech language models must balance three competing objectives: semantic fidelity for reasoning and instruction following, acoustic detail for naturalness and speaker/style preservation, and bitrate/latency for real-time and duplex operation. The operating point is jointly determined by the token interface (acoustic vs semantic vs mixed), token rate and codebook design, the decoding/vocoding family, and the generation schedule. Acoustic codec tokens typically deliver higher reconstruction quality at higher bitrates, whereas semantic tokens compress aggressively and align with linguistic objectives but shed prosody and timbre; diffusion-based decoders often improve perceptual fidelity at the cost of higher inference latency relative to GAN or direct codec decoders [8]. Variable frame-rate tokenization, acoustic BPE, and deduplication redistribute bits and shorten sequences, yet can erode fine temporal detail if applied aggressively [8].

Codec-centric designs make the fidelity–bitrate–latency surface concrete. Residual VQ operating near tens of frames per second inflates sequence length and latency for long-horizon TTS and S2S generation; using only early EnCodec RVQ codebooks lowers token rate and speeds decoding but reduces fidelity, which can be partially recovered by a non-autoregressive refinement transformer that improves Fréchet Audio Distance without reintroducing full-rate latency . Language-Codec attains high-fidelity reconstruction around 3 kbps with four quantizer channels and deliberately redistributes information away from the first channel to reduce the predictability burden on text-conditioned generators, improving end-to-end usability without increasing bitrate [12]. Across codec families, acoustic tokens lead on PESQ/STOI/GPE at higher bitrates, while diffusion backbones deliver additional quality at a latency premium [8].

Hybrid tokenization and improved decoders can shift the frontier toward lower bitrates without sacrificing intelligibility. DiffSoundStream combines semantic-conditioned hybrid tokens with latent diffusion decoding to match or exceed perceptual quality at 50 tokens/s compared to a standard SoundStream at 100 tokens/s, reconstructing perceptually important microstructure from coarse tokens; moment-matching step-size distillation reduces sampling from 100 steps to 4 with only slight DNSMOS degradation, shrinking decoding latency while holding WER [13]. Ultra-low-rate designs push further: TaDiCodec reaches 6.25 Hz and 0.0875 kbps (24 kHz) while maintaining intelligibility (SeedTTS test-en reconstruction WER 2.73), speaker similarity (SIM 0.69), and perceived quality (UTMOS 3.73), with a small reconstruction–generation gap that suggests generation-friendliness even at extreme compression [14].

Low-bitrate semantic or coarse linguistic units target the latency corner while stressing semantic–acoustic trade-offs. SyllableLM operates at 5.0–8.33 Hz (roughly 60–91 bps), maintaining intelligibility at 5.0 Hz but reducing lexical sensitivity on sWUGGY; a 6.25 Hz, 8192-unit configuration reaches LibriSpeech test-clean WER 7.0 and CER 3.2, approaching TWIST tokenizer upper bounds at similar bitrate [15]. A single-codebook tokenizer at 12.5 Hz (175 bps) enables low-latency generation, yet residual 0.8 s block-wise streaming in the speech decoder reveals an acoustic-quality–latency compromise; moreover, direct speech-to-speech reasoning underperforms speech-to-text prompting on semantic tests (StoryCloze S$\rightarrow$S 62.4 vs S$\rightarrow$T 76.3), underscoring autonomy–quality tension when divorcing from text [16]. Codebook and frame-rate choices sharpen these tensions: enlarging the primary codebook (e.g., 200$\rightarrow$500 units) injects phonetic detail at higher bitrate and compute, while reducing token frame rate from 50 Hz to 25 Hz lowers sequence length but can degrade resynthesis quality [17]. Moderately coarse fixed-width segmentation with larger K-means vocabularies improves zero-shot spoken language understanding and halves token counts, cutting training runtime by about 70

Sequence compression at the encoder–LM interface is an additional lever for latency and compute. Uniform 4× reduction by discarding frames reconciles long speech sequences with LLM context limits, trading some acoustic detail for manageable bitrate/latency and enabling effective end-to-end modeling; light adapter finetuning mitigates the loss and improves out-of-domain ASR [3]. Temporal pooling projectors compress Whisper large-v3 frames by 4:1–5:1 (e.g., 1,500$\rightarrow$375/300 per 30 s), reducing conditioning length and compute while maintaining competitive accuracy, as evidenced by a 16.63

Compression, selection, and adaptive-rate mechanisms further modulate the frontier. Acoustic BPE and deduplication shorten sequences and improve LM efficiency at the potential expense of temporal detail and prosody [8]. Selection-based, token-granular compression that maps n frame-level features to m transcription tokens sharply lowers bitrate and latency while preserving semantic fidelity, but deprioritizes explicit modeling of prosody, speaker, and emotion [18]. Variable frame-rate tokenization is underexplored but is proposed to allocate bits to information-dense regions and relax elsewhere [8]. Learnable query pooling in Q-former-style modules compresses audio/text into compact embeddings, lowering effective bitrate and latency while retaining salient cues, and chunk-wise streaming tokenization with expanding context windows bounds emission delay while preserving local fidelity .

Decoding and scheduling choices close the loop by trading wall-clock latency for acoustic detail. Diffusion-based codec/vocoder pipelines introduce higher inference time than GAN or direct decoders [8], but non-autoregressive refinement stages can claw back quality after coarse token generation at little incremental latency [19]. Residual block-wise streaming in end-to-end speech decoders imposes latency floors even at very low token rates [16]. Information redistribution across quantizer channels reduces LM predictability pressure and aligns codec design with text-conditioned generation [12]. Practical design axes follow: choose token interfaces and codebooks to balance detail and bitrate , set token rates and adaptive schedules that respect LM context and streaming constraints , and pair compression/selection with decoding backbones that recover perceptual quality without forfeiting real-time operation .

1.4 Survey roadmap

Section 2 formalizes the end-to-end speech LM pipeline—from waveform to learned representation to tokenization to LM and back to audio—detailing component roles and interfaces while anchoring definitions in current tokenizer and benchmark taxonomies: it contrasts discrete, LM-aware tokenizers such as LAST (which align speech tokens to a pre-trained text LM and enable a single LM to handle both speech and text) with supervised phonetic-acoustic tokenizers like PAST (including a streamable causal variant), and with continuous tokenizers such as Cont-SPT (which reduce information loss and improve TTS continuity and MOS), and it previews the evaluation families used throughout, including spoken language modeling and speech-to-text task metrics, phonetic fidelity and reconstruction quality for tokenizers, and TTS continuity/MOS for synthesis. [20, 21, 22]

Section 3 reviews speech tokenization and neural audio codecs from both compression and semantic angles: it covers quantization schemes (VQ/RVQ and masked-channel RVQ), channel allocation and codebook budgeting, sequence/rate control and the Discrete Representation Inconsistency problem, disentanglement and continuous/semantic interfaces (e.g., tokenizing SSL features with a universal vocoder and attention-based layer selection), and representative designs such as Language-Codec, RepCodec, and X-Codec. It highlights codec–model co-designs that rebalance information across RVQ channels to ease the first-channel overload, inject semantic encoders and losses around RVQ to preserve content, speaker, and paralinguistic cues, and mitigate DRI to curb omissions/repetitions—collectively making tokens more predictable for speech LMs, lowering WER, and improving fidelity at competitive bitrates. [23, 24, 12, 25, 26]

Section 4 reviews encoder families—ranging from self-supervised encoders like HuBERT and data2vec to Whisper-style ASR-supervised models—alongside layer-wise specialization and fusion/selection strategies, with considerations of robustness and streaming; it also covers diagnostics of representation quality and deployment-oriented compression at the encoder–LM interface, highlighting semantic tokenizers such as RepCodec that learn a vector-quantized codebook to reconstruct encoder representations (retaining more information than k-means across encoders and languages), insights from neural audio codecs showing that better waveform reconstruction does not necessarily improve SLM-based speech generation (with naturalness driven by decoder quality and intelligibility by the quantizer), and hybrid continuous–discrete approaches like Whisper-GPT that combine spectrogram features with discrete tokens to mitigate context-length blow-up and improve perplexity and negative log-likelihood. [27, 28, 25]

Section 5 deep-dives into token-to-speech decoding and vocoders across GAN, diffusion, and autoregressive/non-autoregressive backbones, showing for example that latent-diffusion decoders like DiffSoundStream can synthesize SoundStream-level audio from semantic and coarse acoustic tokens at about 50 tokens/s and, after step-size distillation, with as few as four sampling steps. It also surveys hybrid continuous–discrete pathways—spotlighting continuous tokenizers such as Cont-SPT that preserve low- and high-frequency detail and improve MOS, LM-aware (LAST) and noise-aware (NAST) discrete tokenization that enable a single LM for both speech and text while adding robustness to noise, reverberation, pitch-shift, and time-stretch—and contrasts acoustic versus semantic tokens for controllability alongside the latency–quality trade-offs of non-autoregressive refinement. [20, 22, 13, 29, 8]

Section 6 deepens the survey of speech-token language modeling by contrasting autoregressive and non-autoregressive decoding with interleaved text–speech training built from large-scale synthetic text-to-token data that bypass audio generation; it details modality transfer via LM-aware tokenization (e.g., LAST, which uses a text-LM objective to learn discrete speech units so one pre-trained LM can process both speech and text, with trade-offs in speech vocabulary and LM size) and via continued pretraining of a text LM to 1T tokens (including 600B interleaved speech–text), enabling duplex speech I/O and revealing scaling under token-rate constraints, where ASR-derived VQ tokenizers preserve semantics even at low frame rates (12.5 Hz) while maintaining reconstruction quality and delivering state-of-the-art spoken LM and QA performance (e.g., improving spoken QA from 13

Section 7 organizes end-to-end SLM training as a multi-stage continual pipeline—LLM pretraining, speech-modality adaptation across tasks like ASR, TTS, and spoken QA, followed by supervised/instruction tuning and post-alignment—and details: (i) speech–text alignment mechanisms such as descriptive speech captioning that inject paralinguistic understanding, preserve the LLM's language skills, and enable zero-shot instruction following; (ii) synthetic/programmatic supervision that uses LLMs and TTS to generate paired speech–text data, including from unlabeled audio to expand language coverage; and (iii) diagnostics and mitigations for catastrophic forgetting across stages, highlighting model merging, discounting the LoRA scaling factor, and especially experience replay as the most effective approach, with further gains from combining methods and corroborated by results on Dynamic-SUPERB and AIR-Bench-Chat—alongside efficiency techniques such as sequence compression. [30, 31, 10, 32]

Section 8 contrasts cascaded ASR/TTS+LLM pipelines, fully end-to-end speech–text LMs, and interleaved prompting setups; situates them within the discrete speech-token landscape (acoustic vs semantic tokens) and LM-aware tokenization (e.g., LAST, which aligns token learning with a pre-trained text LM so one LM can process both speech and text); catalogs generative regimes such as hierarchical RVQ codebooks, multi-token prediction, and prompt-driven generation (SpeechGen); and formalizes sequencing/scheduling taxonomies for parallel or chunked decoding to improve latency and throughput in practical agents. [22, 33, 8]

Section 9 surveys how LM-aware tokenization and conditioning choices shape expressivity, paralinguistics, and style control: LM-aware speech tokenizers (e.g., LAST) align speech units with textual LMs—enabling a single LM to process both speech and text and improving spoken language modeling and ASR—while prosody-aware tokenization (e.g., ProsodyLM's word-level prosody tokens) preserves content–prosody interdependence and yields emergent abilities such as contrastive focus, emotion/stress understanding, and long-context prosody consistency; it further details control interfaces that inject explicit paralinguistic metadata or implicitly generated QA pairs (boosting LLM-judged empathetic reasoning by 38.41

Section 10 treats real-time streaming and duplex dialogue, detailing latency-aware design and streaming policies for backchannels, barge-in, and overlapping speech; planning-based methods such as TurnGuide that generate turn-level text before speaking to keep double-channel audio precisely time-aligned; and duplex S2S architectures (e.g., SALM-Duplex) that directly model simultaneous user and agent streams using channel fusion, a pretrained streaming encoder, separate user/agent paths, and low-bitrate codec outputs (about 0.6 kbps), enabling listening-while-speaking, stronger turn-taking and interruption handling, and deployment with less speech pretraining. [34, 35]

Section 11 surveys concrete resources and benchmarks—ranging from IndicSUPERB built on the 1,684-hour, 12-language Kathbath corpus covering ASR, speaker verification/identification, language ID, query-by-example, and keyword spotting; to spoken versions of StoryCloze for generative evaluation; paralinguistic QA datasets that inject emotion/context for empathetic dialogue; and large-scale synthetic interleaved speech–text pretraining corpora scaled to about 1 trillion tokens (including 600B synthetic) enabled by supervised and LM-aware tokenizers such as LAST—and maps them to ASR, S2ST, TTS, and dialogue/expressivity evaluation, highlighting multilingual and spontaneous dialogue resources that support large-scale training and end-to-end spoken chatbots. [22, 36, 37, 4, 38]

Section 12 specifies concrete metrics and end-to-end protocols: it adopts bitrate- and token-rate–aware accounting; measures streaming/duplex latency; evaluates streaming speech-to-speech on CVSS-C with BLEU, BLASER 2.0, and average latency; assesses acoustic-language-model awareness via SALMon (noise, emotion, speaker identity, room impulse response); benchmarks discrete tokenizers with STAB and prosody-perturbation sensitivity analyses; and includes multilingual SLU tasks from IndicSUPERB (ASR, speaker verification/identification, language ID, query-by-example, keyword spotting), alongside human and LLM-based judging with statistically rigorous reporting. [39, 40, 36, 41, 42]

Section 13 analyzes efficiency and deployment in concrete terms, showing how parameter-efficient finetuning and sequence compression can be paired with semantically oriented tokenizers—e.g., RepCodec, which learns a vector-quantized codebook over HuBERT/data2vec representations to preserve information and outperform k-means across encoders and languages, and X-Codec, which injects semantic features before RVQ and adds a post-RVQ semantic reconstruction loss to reduce WER and improve TTS, music, and sound generation—while also detailing fast-decoding strategies and practical edge–server bandwidth trade-offs, with step-by-step recipes for production-style pipelines that convert waveforms to semantic tokens via a speech encoder, codec encoder, and VQ codebook. [24, 25]

Section 14 surveys safety, robustness, and ethics for speech agents, covering threat models and defense-in-depth—including white- and black-box audio jailbreaks with average success rates of 90

Section 15 reviews open toolkits and frameworks through a practical 5C reproducibility lens (Code, Config, Corpora, Comparison, Compute), using recent systems as case studies—for example, RepCodec's HuBERT/data2vec-based vector-quantized semantic tokenization that retains more information and outperforms k-means for speech understanding and generation across encoders and languages, and C3LLM's LLM-bridged, discrete "acoustic vocabulary" pipeline that hierarchically generates semantic then acoustic tokens for video-to-audio, audio-to-text, and text-to-audio—thereby motivating bitrate-controlled, standardized evaluation, explicit baselines, and fully documented training/inference pipelines with reported compute. [19, 25]

Section 16 sharpens the open problems at the speech–text boundary: whether to adopt a single LM-aware token stream or a dual stream that separates high-level semantic tokens from fine-grained acoustic tokens (as in LAST and DualSpeechLM); how to extend conversational access over long horizons via direct audio retrieval instead of error-prone ASR transcripts (SpeechRAG); how to ensure multilingual robustness with standardized SLU benchmarks and diverse data (IndicSUPERB); how to align speech tokens with text knowledge so one LM can handle both modalities without degrading text skills (LAST, DualSpeechLM); and how to set principled trade-offs among token rate, vocabulary size, and model/decoder capacity, leveraging tokenizer and scaling analyses (e.g., LAST's vocabulary-size vs LM-size findings). [43, 22, 44, 36]

Section 17 closes with concrete guidance: define token interfaces that preserve phonetic/semantic content, paralinguistics, and speaker identity; favor semantic tokens over compression tokens for most tasks and consider RepCodec-style VQ over k-means to improve information retention, while noting continuous tokenizers (e.g., Cont-SPT) can yield superior TTS continuity and MoS; align speech tokens to LLM text units via encoder-informed pipelines (HuBERT/data2vec) and streaming protocols that support low-latency, full-duplex ASRTTS; evaluate using standardized, bitrate-normalized benchmarks (STAB, DASB) that correlate intrinsic metrics with downstream performance across ASR, speaker ID/verification, emotion, keyword spotting, intent, enhancement, separation, and TTS; and adopt deployment practices that track bitrate/latency trade-offs, cross-lingual robustness, and leaderboard-based comparisons. [20, 9, 40, 25]

1.5 Out-of-scope

This survey centers on speech-in/speech-out modeling with tokenizers/codecs, encoders, speech LMs, and token-to-speech decoders/vocoders. Out of scope are latency-optimized "think-while-speak" generation for SLMs (e.g., STITCH), unsupervised ABX probing of cross-lingual speech embeddings, beyond-semantic affect and social-interaction capability frameworks (BoSS), large multilingual SLU benchmarks for Indic languages (IndicSUPERB), and end-to-end speech QA evaluations of knowledge and robustness (VoxEval)—except where used narrowly to motivate design choices or to borrow transferable methods such as chunked reasoning, ABX-style diagnostics, capability taxonomies, or benchmark protocols. [45, 46, 47, 36, 48]

We present two complementary advances: IndicSUPERB, a 1,684-hour, 12-language SLU benchmark built from the Kathbath corpus (1,218 speakers from 203 Indian districts) covering six tasks—ASR, speaker verification/identification (mono/multi), language identification, query-by-example, and keyword spotting—where language-specific self-supervised models outperform FBANK baselines (e.g., a 76

We cover purely text-only LLMs, their instruction-tuning corpora, and alignment methods without speech I/O only as they inform modality transfer or post-alignment for SLMs—for example: leveraging frozen LLMs with lightweight adapters to bridge speech and text while retaining foundation capabilities and enabling zero-shot instruction following (SLM); automatically generating speech–text pairs to inject paralinguistic understanding without costly speech instruction-tuning or catastrophic forgetting (DeSTA2); and adopting LM-aligned speech tokenizers that produce semantically compact, low–frame-rate tokens for more efficient cross-modal modeling (LM-SPT). [3, 49, 32]

Legacy speech pipelines—HMM/GMM-based ASR, unit-selection TTS, and waveform codecs like MP3/Opus—do not expose LM-aligned neural token interfaces. In contrast, modern tokenizers learn phonetic-aware, streamable representations (PAST), use continuous tokens to reduce information loss and improve naturalness (Cont-SPT), and align tokenization with text-LM objectives to enable a single LM to process both speech and text (LAST). [20, 21, 22]

Non-speech audio tasks such as environmental audio tagging, sound event detection/localization, music analysis/synthesis, and automated audio captioning (AAC). Evidence from AAC suggests concrete avenues for token design and LM conditioning: using a CED-distilled audio encoder whose acoustic tokens are compressed and bridged to an LLM via a Q-Former, together with LoRA-tuned LLaMA-2-7B decoding and an auxiliary LLM for text correction, delivers 33.0 SPIDEr-FL—surpassing the DCASE 2023 Task 6A winner—while complementary advances in discrete speech tokens and LM-aware tokenization (e.g., LAST) show that compact acoustic/semantic tokens and text-LM–supervised tokenizers align speech with the language-modeling interface, enabling a single LM to handle both speech and text and improving spoken language modeling and speech-to-text. [8, 50, 22]

Audio–visual or cross-modal works are cited only when they clarify how speech is aligned with, or compressed for, non-speech modalities—for example, early-fusion systems that quantize speech into discrete tokens for unified speech–text reasoning (Ichigo), LLM-bridged models that introduce a discrete "acoustic vocabulary" and hierarchical tokenization to align videoaudio and audiotext generation (C3LLM), and speech–language models showing that frozen speech/text backbones can be aligned with lightweight adapters (SLM) or trained end-to-end for instruction following (LLaSM); we also use analyses of representation granularity that reveal which linguistic vs. extra-linguistic information is retained at different snippet lengths (ABX closeness). Vision-first tasks (lip reading, AV event grounding, video captioning) or other modalities (e.g., haptics, sensors) appear only when they illuminate alignment or compression choices relevant to speech tokens. [51, 46, 52, 3, 19]

Speaker biometrics—verification and identification—and diarization are largely treated as standalone evaluation tasks that probe whether speech representations preserve paralinguistic information (speaker identity, prosody), as evidenced by DASB and fastabx-style analyses; in most modeling setups (e.g., dMel and speech LMs), speaker labels are used for conditioning or as auxiliary signals rather than primary optimization targets, with benchmarks showing semantic tokens outperform compression tokens on these tasks yet still trailing standard continuous features. [9, 53, 39, 54]

Speech enhancement, source separation, dereverberation, acoustic echo cancellation, beamforming, and microphone-array processing—treated here as optional front-ends rather than core methodology; nonetheless, recent SLM-based systems increasingly model these acoustic factors directly (e.g., direct speech RAG and unsupervised acoustic-semantic segmentation exploiting noise, speaker, and room cues), while new benchmarks provide systematic evaluation (DASB for enhancement/separation; SALMon for background noise, speaker identity, emotion, and room impulse response), alongside improved tokenization (e.g., PAST) that strengthens SLMs' acoustic representations. [21, 9, 43, 41, 55]

We do not provide a formal low-level signal-processing or information-theoretic treatment—such as rate–distortion analyses of waveform coders that lack learned token interfaces—beyond brief intuition about bitrate–latency–fidelity trade-offs; instead, our focus is on learned discrete audio tokens (semantic tokens from SSL quantization vs. neural-codec tokens) that aim to preserve content, paralinguistics, and speaker identity, including recent context-aware compression strategies (e.g., query-based compression with MAE objectives, semantic-prior vector quantization, and autoregressive losses) that achieve lower bitrates while retaining semantic richness. [26, 56]

Systems and hardware engineering details (telephony stack integration, networking protocols, device firmware, accelerator-specific kernels) are out of scope; instead we report runtime/latency and model-scaling results for interleaved speech–text LMs—reflecting prior findings that they scale more efficiently with compute, merit allocating more budget to model size than to training tokens, and can benefit from synthetic data and TextLM initialization—and we use contamination-aware evaluation on held-out speech sets (avoiding benchmarks known to appear in LLM pretraining such as LibriSpeech/Common Voice); we do not cover hardware design or networking. [57, 58, 32]

We do not survey general HCI/UX or conversational policy learning; our scope is limited to mechanisms essential for duplex turn-taking and streaming evaluation in end-to-end spoken language models—for example, chunked "think-while-you-speak" generation for simultaneous reasoning and response (as in STITCH) and speech-in/speech-out benchmarking that probes latency, robustness to varied input audio conditions, and spoken mathematical reasoning (as in VoxEval); broader social-agent design frameworks are also out of scope. [45, 48]

Security and broader AI safety literatures not specific to speech; only speech-relevant safeguards and evaluation are discussed in the designated safety section.

Together, IndicSUPERB provides a 1,684-hour, 12-language benchmark spanning six speech tasks (ASR, speaker verification/identification, language identification, query-by-example, and keyword spotting) that highlights the advantages of language-specific self-supervised models, while fastabx offers a high-performance Python library to rapidly build and compute ABX discriminability tasks for analyzing learned speech representations. [53, 36]

Where adjacent work offers transferable lessons, we cite it selectively and map it to speech practice: ABX analyses of multilingual speech embeddings show that short snippets better capture segmental/phonetic information while longer windows capture extra-linguistic factors (room acoustics, genre), informing token-rate and windowing; studies of discrete representation inconsistency in neural audio codecs motivate conditioning and regularization to stabilize token sequences and reduce omissions/repetitions; LM-aware (LAST) and phonetic-acoustic (PAST) tokenizers improve audio–text alignment and SLM conditioning (with streamable variants for real time); and modality-evolution results indicate that long sequence length and paralinguistic variability—more than limited semantic content in speech tokens—are the main obstacles to coherent speech generation. [21, 22, 46, 59, 23]The following sections are organized as shown in Figure 1.

2 Background and Core Concepts

2.1 From waveform to language model and back: pipeline and terminology

A modern speech–AI stack maps waveforms to model-facing representations—continuous features or discrete tokens—passes them through a sequence model, and renders predictions back to audio with token-to-speech decoders or vocoders [21, 22, 54, 8]. The canonical stages are: (i) an encoder producing time-aligned embeddings; (ii) a tokenizer/quantizer that yields discrete indices or compressed continuous vectors at a chosen frame/segment rate; (iii) a language model (LM) operating over discrete and/or continuous sequences; and (iv) a decoder/vocoder that meets bitrate, latency, and streaming constraints. Discretization choices span LM-aware tokenizers aligned to shared text/speech LMs (LAST), supervised phonetic tokenizers with streamable causal variants (PAST), and training-free mel-channel discretization with efficient parallel encode/decode and strong OOD robustness (dMel) [22, 21, 54].

Encoders and continuous interfaces inject speech into pretrained text LLMs via lightweight adaptors, enabling speech-conditioned instruction following without degrading text competence; Whisper-class encoders plus projector/MLP and temporal pooling are common, with windowed segmentation and downsampling to fit context budgets [3, 60, 61]. CTC-compressed continuous prompts remove blanks and linearly map remaining features into decoder embeddings for decoder-only conditioning [62], while mixed text–audio inputs can be interleaved in a single LLM context using boundary tokens and fixed audio-patch lengths [52].

Tokenizers, quantizers, and codecs provide the discrete interface. Acoustic tokenizers are neural audio codecs (encoder–quantizer–decoder) optimized for waveform reconstruction under explicit bitrate control; VQ selects nearest codebook entries and RVQ stacks codebooks to encode residuals, enabling multi-stream generation and clear rate–fidelity control [8]. End-to-end S2S stacks often autoregressively generate the most informative codec stream, then non-autoregressively predict the others to reduce latency [5]. Semantic tokenizers discretize SSL features (e.g., HuBERT/WavLM) via k-means or related schemes with segment pooling/deduplication; Segment-Pool K-means Tokenization (SPKT) formalizes this pipeline [63]. Beyond k-means and VQ/RVQ, binary spherical quantization yields mel-level codes decoded by a text/prompt-conditioned flow-matching Transformer to mels, then a neural vocoder [14].

Information taxonomy separates semantic (phonetic/lexical), acoustic (prosody/timbre), paralinguistic (speaker/emotion/style), and mixed tokens, motivating control vectors for lexical, affective, contextual-dynamics, and implicit-semantics channels [8, 47]. Paralinguistic control can be supplied by speaker/style enrollment or short codec prompts in S2S [64, 5], and mixed interfaces interleave text subwords with discrete speech tokens [64].

A mixed semantic–acoustic pathway illustrates the composition: DiffSoundStream extracts WavLM semantic tokens (downsampled to 12.5 Hz, 2048-entry codebook) and SoundStream acoustic tokens (8×RVQ at 12.5 Hz), FiLM-conditions encoder/decoder on semantics to decorrelate content from acoustics, and synthesizes 24 kHz audio with a WaveNet-based latent diffusion model and a continuous-latent SoundStream decoder [13].

Language modeling over discrete and continuous streams typically uses decoder-only LMs with modality tags for unified speech understanding, TTS, and S2ST; AR prediction covers information-bearing streams, with non-AR refinement for the remainder to curb latency and exposure bias [64, 5]. Large-scale pretraining couples a tokenizer, token-to-waveform decoder, text-to-token transformer, and text-initialized LM trained on mixed text/speech/interleaved corpora with block-wise streaming generation [4]. Continuous-interface systems retain adapted encoder embeddings inside the LLM context with boundary tokens and chunked audio patches [60, 52].

Token-to-speech decoders and vocoders invert RVQ streams at codec rates (EnCodec/SoundStream) or synthesize from semantic units/quantized mels using GAN, flow, or diffusion backbones; discrete-token HiFi-GAN variants are prevalent for HuBERT-k-means unit inversion and interleaved pretraining [14, 4].

Rates, segmentation, and streaming control determine sequence length, bitrate, and latency. Codec pipelines use fixed frame rates; semantic-unit systems lower effective rates via segment pooling/deduplication; continuous interfaces downsample encoder outputs and segment audio to meet LLM contexts. Typical practice uses 30 s windows with 15 s stride (or 8 s in production), projector compression to 10 Hz, and block-wise token decoding for low-latency generation [60, 61, 4]. Together, tokenizers/quantizers (k-means, VQ/RVQ, BSQ), neural codecs, continuous adapters, mixed-stream LMs, and token-to-speech decoders/vocoders define interoperable, rate-aware interfaces [8].

2.2 Canonical building blocks and exemplars

Canonical components include encoders, speech tokenizers and neural codecs, LM backbones with modality adapters, and token-to-speech decoders/vocoders, assembled into interoperable stacks [8]. Self-supervised encoders (HuBERT, wav2vec 2.0, data2vec) are standard upstream feature spaces for semantic tokenization and continuous conditioning; compute-efficient HuBERT reproductions retain strong ASR/SUPERB performance, and probing guides layer selection/fusion for syntactic/semantic content [65, 66]. Whisper-class encoders serve as robust front ends in Encoder–Adaptor–LLM stacks, while modality-projector SLMs align frozen encoders (e.g., USM) to frozen text LLMs via lightweight adapters [3, 60, 8].

SoundStream/EnCodec expose RVQ acoustic tokens with explicit rate–fidelity control and multi-stream generation, whereas semantic tokenizers discretize SSL features via k-means/VQ with segment pooling, deduplication, and acoustic BPE compression [8]. GSLM exemplifies textless modeling over HuBERT units [67]. Unified codec–tokenizer designs redistribute information for LM predictability, hybrid stacks pair semantic tokens with RVQ acoustics to balance fidelity and rate, and continuous-token/embedding tokenizers keep SLMs in continuous spaces for VALL-E–style conditioning [20].

Decoder-only Transformers dominate speech-token LMs, from compact acoustic-BPE LMs to SLMs that integrate AR semantics with non-AR RVQ refinement for low latency; parameter-efficient adaptation (e.g., LoRA) composes RVQ codecs and contrastive audio encoders with lightly adapted LLMs [68, 5, 19]. Interleaved pretraining injects speech tokens into LLM corpora, while encoder–adaptor bridges align continuous audio embeddings to frozen LLMs [4, 3].

Codec decoders invert RVQ streams and support hierarchical/masked decoding for fast high-fidelity synthesis; unit/latent vocoders (e.g., CTX-vec2wav) synthesize from semantic tokens or quantized mels, with HiFi-GAN a common final stage [68, 4, 69]. Representative compositions include acoustic-BPE LMs for text-free generation [68], multimodal generators combining RVQ codecs, contrastive encoders, and LoRA-adapted LLMs [19], and end-to-end SLMs that jointly predict semantic and codec tokens; strong cascaded S2ST baselines remain useful references [5, 70].

2.3 Token interfaces and their rates

Token interfaces determine what information the LM sees and at what cadence, shaping sequence length, bitrate, latency, and compatibility with text-token context windows. Five families recur: (i) semantic units that prioritize linguistic content, including representation–codec tokenizers that reconstruct SSL encodings; (ii) acoustic tokens from neural codecs for prosody/timbre; (iii) explicit paralinguistic controls (e.g., word-aligned prosody tokens) for speaker/emotion/style; (iv) mixed speech–text designs for cross-modal transfer; and (v) continuous embedding interfaces from multilingual encoders with temporal compression [71, 46, 11, 25].

Semantic interfaces typically operate at 25–50 Hz; supervised tokenizers at 12.5 Hz cut sequences by 2–4× with minimal semantic loss, whereas sub-12.5 Hz rates degrade content [4]. Textless LMs often consume HuBERT+k-means units (e.g., 100 clusters) with utterance-level normalization and fixed token counts for stable context [67]. Coarser syllable-like units (SyllableLM, 5.0–8.33 Hz, 2k–16k codebooks, 60–91 bps) improve throughput [15]. In multimodal LMs, finite/state quantization with a 4096-entry codebook at 25 tps constrains sequence length while style is carried separately [72].

Acoustic interfaces expose higher-rate RVQ streams for micro-prosody and timbre with explicit bitrate control. A representative EnCodec setting uses eight 1024-way codebooks at 50 Hz, yielding 4 kbps and 500 steps per 10 s per stream [5]. Alignment with visual rates (25 fps) facilitates audio-visual pretraining and motivates hierarchical scheduling or non-AR refinement [73, 5].

Paralinguistic controls decouple content and style: a 25 tps discrete semantic path (FSQ, 4096 entries) paired with a continuous prototype embedding improves controllability and stability [72].

Mixed and scheduling-aware interfaces coordinate content and acoustic streams at harmonized cadences. DiffSoundStream emits semantic and acoustic tokens at 12.5 Hz with N$_s 0, 1, N_a [1, 8]; a common point uses 1 semantic + 3 acoustic tokens per frame (50 tps) with 2048-way codebooks for 0.55 kbps, with a maximum 1.2375 kbps [13]. In S2ST SLM$
$autoregressively predicting others at 50 Hz; semantic units can be modeled in parallel with 1000$-$way vocabularies [5]. TaD$
$entry codebook (87.5 bps) and short LM sequences [14]$.

Continuous interfaces project encoder features into the LM embedding space and control sequence length via temporal downsampling and boundary-aware chunking: Whisper large-v3 yields 1500 frames per 30 s, compressed to 300–375 vectors (4–5× reduction) [60]; production stacks downsample to 10 Hz and constrain VAD segments to 8 s [61]. Audio-visual models synchronize rates (e.g., 50 Hz audio, 25 fps video) for efficient fusion [73].

Implications for bitrate, sequence length, and LM context follow directly. For 10 s of audio, sequence length is 500 steps at 50 Hz, 125 at 12.5 Hz, and 62 at 6.25 Hz; bitrate scales with vocabulary entropy and streams: 8 EnCodec streams at 50 Hz with 1024-way codebooks yield 4 kbps [5]; mixed 50 tps with 2048-way codebooks reach 0.55 kbps [13]; syllabic units at 5.0–8.33 Hz deliver 60–91 bps [15]; TaDiCodec runs at 87.5 bps [14]. Reducing semantic rates from 50 to 12.5 Hz provides large efficiency gains with little semantic loss, whereas sub-12.5 Hz rates or aggressive stream pruning erode prosody and intelligibility; projector-based downsampling and short, streaming-aligned windows offer comparable reductions for continuous interfaces [4, 60, 61].

2.4 Datasets and evaluation families: a primer

Core corpora span read, conversational, and multilingual speech (LibriSpeech, Common Voice, VoxPopuli, LibriTTS, CoVoST2, VoxCeleb, GigaSpeech, AISHELL, Fisher) across ASR, TTS, S2T/S2ST, speaker ID/verification, and keyword spotting, augmented by task-specific sets such as GigaST (S2ST), SLURP (intent/slot), and Speech Commands. Large-scale spontaneous dialogue is increasingly central for end-to-end SLMs; J-CHAT provides 68,892 hours of Japanese dialogue with diarization and no transcripts, built by an automated pipeline and hosted on Hugging Face [6].

Benchmark suites standardize coverage and bitrate awareness. DASB spans recognition and generation—ASR (LibriSpeech, Common Voice), speaker ID/verification (VoxCeleb1), emotion (IEMOCAP), keyword spotting (Speech Commands), intent classification (SLURP), enhancement (VoiceBank), separation (Libri2Mix), and TTS (LJSpeech) [9]. Dynamic-SUPERB Phase-2 unifies classification, regression, and sequence generation under natural-language interfaces and couples accuracy/WER with LLM-judged mappings for free-form outputs [74].

Unsupervised and unit-level diagnostics include ABX discriminability, with fastabx offering dense/discrete support, ON/BY/ACROSS conditions, DTW alignment, and a zerospeech_abx CLI that reproduces canonical ABXpy/Libri-Light results [53]. Unit-oriented tokenizers are additionally probed via PER/UER, while lexical/syntax probes draw on sWUGGY/sBLIMP and sStoryCloze.

Metric families and recommended reporting cover recognition/alignment (WER/CER/WCER, PER/UER), translation/semantic fidelity (BLEU/BERTScore), perception/realism (MOS/UTMOS, DNSMOS, FAD), and system/identity measures (task accuracy, EER, dWER, speaker similarity) [9]. Deployment reporting should include real-time factor, end-to-end latency, and token/bandwidth budgets; natural-language interfaces benefit from Dynamic-SUPERB's combined criteria [74].

Mapping datasets to evaluation families is therefore straightforward: ASR/AST rely on LibriSpeech, Common Voice, VoxPopuli, AISHELL, Fisher with WER/CER and BLEU/BERTScore; speaker-centric tasks use VoxCeleb with accuracy/EER; TTS/generation use LibriTTS and LJSpeech with MOS/UTMOS, DNSMOS, FAD, dWER, and speaker similarity. DASB enables bitrate-aware comparisons across tokenizers/decoders [9], Dynamic-SUPERB and fastabx provide standardized transfer and zero-resource discriminability [74, 53], and large, diarized, multi-speaker resources such as J-CHAT support end-to-end SLM evaluation beyond text-bound intermediates [6].

3 Speech Tokenization and Neural Audio Codecs

Tokenizers and neural codecs define an information contract between waveforms and language models: they decide what attributes are exposed (lexical content, prosody, speaker/style), at what cadence and bitrate, and with what predictability for AR/NAR decoders, thereby governing semantic retention, prosodic expressivity, reconstruction fidelity, and LM difficulty [8]. STAB-like diagnostics and LM-aware tokenizers make these trade-offs measurable and comparable across designs [75, 71, 22, 40]. The analysis below targets four objectives: preserve intelligible semantics at low rate, expose controllable paralinguistics, maintain codec-grounded fidelity, and enable streaming-friendly predictability.

3.1 Tokenization objectives and interfaces: the LM-facing contract

A usable contract offers low-rate, predictable content tokens; separable, controllable prosody/identity; codec-compatible fidelity; and AR/NAR schedules that support streaming [8]. Robust, low-rate discretizers—e.g., Whisper features with average pooling and VQ at 12.5 Hz under block-causal attention, or segment-pool k-means with moderate vocabularies—bound LM entropy while retaining semantics on sWUGGY/sBLIMP/ProsAudit/StoryCloze and stabilize streaming inputs [4, 63]. Coarser syllable-synchronous units at 5–8.33 Hz (60–91 bps) sustain intelligibility via loss-guided boundaries and student–teacher refinement, and continuous lexical embeddings (200 ms, PCA + per-dimension k-means) offer a low-entropy alternative that eases multi-step prediction [15, 76].

Prosody and paralinguistics benefit from decoupling. Semantic streams derived from HuBERT/ContentVec/data2vec emphasize lexical cues, while prosody-sensitive objectives (e.g., emotion2vec-style) plus light smoothing increase prominence sensitivity and reduce speaker leakage [39]. With codec tokens as the interface, quantizer/decoder co-design matters: decoder quality drives perceived naturalness, whereas quantizer geometry and code utilization shape intelligibility (WER) and LM predictability [28]. Distilling text and SSL features into residual-quantized tokens (DM-Codec) improves WER/WIL, ViSQOL, and MOS at similar complexity, loading semantics into early residual levels [77].

Hybrid contracts that condition acoustics on semantics reduce redundancy and align well with AR-on-content plus NAR refinement: FiLM-conditioned SoundStream lowers overlap and improves rate–predictability trade-offs, while text-aware diffusion/flow decoding narrows the reconstruction–generation gap and preserves intelligibility and similarity at ultra-low rates without adversarial losses [13, 5, 14]. Practical knobs include token/frame rate, vocabulary size and geometry, training data, decoder conditioning, and channel allocation; strong, repeatable choices comprise 12.5 Hz VQ or segmental k-means for semantics, syllabic units where throughput dominates, prosody-aware preprocessing, AR-anchored codec scheduling, FiLM-conditioned hybrids, and text-aware diffusion/flow decoders, with reporting that couples WER/WIL and prosody sensitivity (e.g., PNMI/CNMI, perturbation tests) to perceptual/distributional audio quality [4, 63, 15, 39, 28, 13, 14, 8].

3.2 Quantization mechanisms and codec families

Quantizer geometry (VQ/RVQ, grouped/product, scalar/binary), cadence (frame vs segment), and channel layout jointly determine fidelity, semantic retention, predictability, and streaming stability;

pitfalls include codebook collapse, depth imbalance, and overloaded early levels that inflate AR entropy [4]. For content-centric interfaces, single-codebook VQ and segmental k-means are strong baselines—pre-VQ pooling stabilizes rates and enables streaming in Whisper-based stacks [15, 4]. RVQ dominates acoustic codecs for coarse-to-fine residualization and explicit rate control; in S2ST, EnCodec (8×1024 at 50 Hz) commonly runs AR on the first stream for intelligibility with NAR refinement to reduce latency [5]. Hybrids such as DiffSoundStream shift RVQ to 12.5 Hz and rely on balanced per-level utilization to avoid AR uncertainty from overloaded early levels and bitrate waste in deeper levels [13].

Early-level semantic distillation (DM-Codec) improves content preservation and predictability; eight RVQ levels at 50 Hz with RVQGAN-style discriminators and cosine alignment substantially raise early-layer utility [77]. Scalar/binary schemes offer deterministic, entropy-predictable assignments: BSQ underpins TaDiCodec's 6.25 Hz stream (implicit 2^{14} codes), avoiding adversarial training and pairing naturally with text-aware decoders at extreme compression [14]. Gr

Cadence should match geometry and scheduling: 50 Hz RVQ maximizes fidelity but pressures context and latency, motivating AR-first/NAR-rest schedules; 12.5 Hz RVQ or segmental tokens trim sequences with limited semantic loss; BSQ yields ultra-compact tokens with minimal scheduling overhead. Health checks include per-level perplexity/entropy and dead-code rates, aided by pre-VQ pooling, early-level distillation, AR-on-first/NAR-on-rest scheduling, and 12.5 Hz cadence; semantic-only interfaces benefit from jointly tuning rate and vocabulary size (e.g., 5–8.33 Hz with 2k–16k codes) [5, 4, 13, 14, 77, 15].

3.3 From continuous features to units: discretization and codebook design

Design choices span encoder/layer selection, temporal granularity, and codebook geometry/size with stringization for LM compatibility [8]. Later HuBERT/ContentVec layers with small k (100–500) yield strong PNMI, while data2vec/emotion2vec benefit from larger k (2000); discretizer data and smoothing balance prosody sensitivity against speaker invariance [39]. Common recipes reflect these trends: HuBERT layer-6/9 with k500,1000 (SpeechComposer/SHuBERT), HuBERT+k=100 (textless SLMs), and WavLM with k=2048 at 12.5 Hz as a single semantic stream in hybrids [64, 67, 13].

Cadence governs bitrate and abstraction: 50 Hz preserves phonetic detail but lengthens sequences; segmental pooling shortens sequences but often requires larger K; text-synchronous units (TASTE/BEST-STD) align with ASR/SLU; syllable-synchronous designs compress further (e.g., 6.25 Hz, 8192 units, WER 7.0/CER 3.2); AV discretization benefits from small token patches that expose a clean granularity–expressivity trade-off [63, 78, 15, 73]. Vocabulary size and geometry set capacity and entropy: 200–500 is effective for phonetic/speaker-invariant units; larger k mitigates loss at coarser cadences or with prosody-sensitive SSLs; operational ranges include k=100 (textless), k1000 (cross-modal AR), and k=2048 at 12.5 Hz in hybrids; BSQ can outperform VQ at ultra-low rates, and grouped/product VQ stabilizes multimodal discretization [17, 39, 67, 64, 13, 14, 73]. Place coarse, predictable codes early, reserving refinements for later channels to lower LM uncertainty [8].

Stringization—run-length de-dup plus BPE/SentencePiece—reduces duration variance and reuses text-LM infrastructure; mixed vocabularies interleave speech units with text subwords [8, 64, 67]. Task-aligned guidance: (i) phone-level probes use shallow/mid SSL layers with k200–500 or noise-aware local units; (ii) SLU/ASR favors text/segment-synchronous units and shared vocabularies; (iii) sentence-level modeling under tight budgets prefers moderately coarse segmentation with larger k, interleaved speech–text training with sufficient model capacity, and ultra-low-rate codebooks or multi-token combos for precision [79, 80, 63, 57, 81].

3.4 Rate and sequence management

A hierarchical overview is provided in Figure 2: the left panel maps the strategy space balancing frame rates and decoder strength (highlighting the 12.5 Hz sweet spot, depth-trimmed hybrids, and ultra-low 6.25 Hz regimes with text-aware diffusion); the middle panel consolidates sequence compression and alignment mechanisms (projector pooling, CTC blank removal, acoustic BPE/unigram, alignment-aware connectors, and block-causal streaming); and the right panel codifies a standardized reporting protocol (rates and bitrate; pre/post-compression statistics; codec specifics; latency; and paired accuracy–perceptual metrics with language/BPE notes).

Reflecting the left panel, cadence, bitrate, and emission schedules are co-designed to meet wall-clock constraints without sacrificing semantics or perceptual quality [42, 23, 81, 25]. Large-scale interleaved pretraining converges on 12.5 Hz as an accuracy–efficiency sweet spot; going lower risks content loss unless countered by stronger decoders [4]. Hybrids at a fixed 12.5 Hz vary acoustic depth per frame (12.5–112.5 tps) and, with latent diffusion, retain low WER and flat DNSMOS as depth decreases, outperforming a 100 tps SoundStream baseline at 50 tps on MUSHRA [13]. Ultra-low-rate TaDiCodec (6.25 Hz; few diffusion steps) remains intelligible across languages under text-aware decoders [14].

Echoing the middle panel, temporal compression without changing discretization uses projector pooling for continuous features (4–5× with Whisper) and CTC blank removal for label-synchronous streams, shrinking sequences to 14.55

Aligned with the right panel, for fair comparison, report frame/token rates; bitrate; tokens/s pre/post compression (BPE/unigram, CTC, projector pooling) with ratios; codec specifics (RVQ depth/code sizes, semantic conditioning); streaming/duplex latency; and pair WER/CER with perceptual metrics, noting language-specific frame-rate choices and acoustic-BPE settings and their impact on semantic fidelity [82, 24, 81].

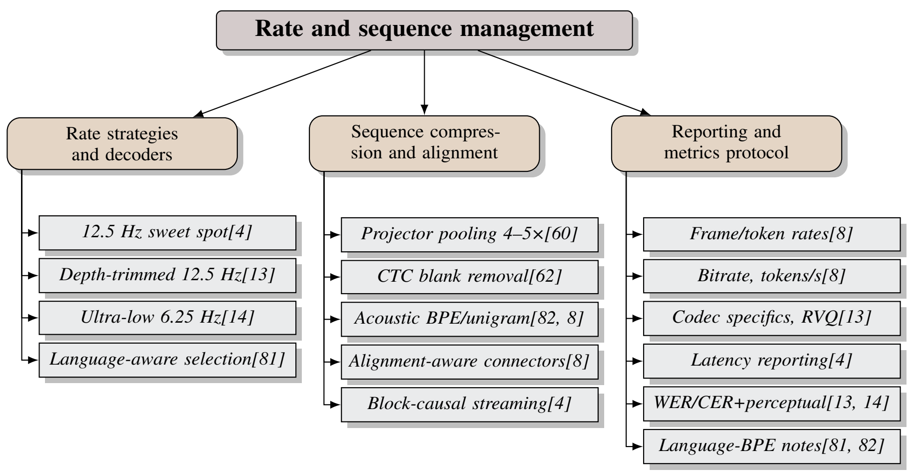

Figure 2: Figure: Hierarchical overview of rate and sequence management. Left: strategy space balancing frame rates and decoder strength (12.5 Hz sweet spot, depth-trimmed hybrids, and ultra-low 6.25 Hz with text-aware diffusion). Middle: sequence compression and alignment mechanisms (projector pooling, CTC blank removal, acoustic BPE, alignment-aware connectors, and block-causal streaming). Right: standardized reporting protocol covering rates, bitrate, compression stats, codec specifics, latency, and coupled accuracy–perceptual metrics with language/BPE notes.

3.5 Disentanglement and mixed-token designs

Disentanglement routes low-entropy content to the LM while prosody, timbre, affect, and non-verbal cues are handled by orthogonal controls, aligning with multi-stream capability frameworks [47, 8]. RVQ codecs naturally factorize: SpeechTokenizer distills RVQ-1 to linguistics (high MI with text; competitive ASR) and reserves RVQ-2:8 for prosody/timbre, enabling style editing by swapping residual levels [83]. Two-branch designs such as EMOVA combine discrete semantics with a compact prototype style codebook predicted by an LLM (gender × emotion × pitch), improving controllability and outperforming entangled HuBERT units in vision–language and ASR [72]. NAST's global–local split uses a single global residual vector (speaker) plus local discrete units (content), reducing speaker leakage [29].

Hybrid interfaces condition acoustics on semantics: DiffSoundStream applies FiLM to disentangle content and residual detail, with WER/DNSMOS dropping without semantic conditioning at low rates; end-to-end S2ST combines HuBERT-like units (content) with EnCodec streams (timbre/prosody) and short prompts for identity; ultra-low-rate pipelines shift identity and long-range prosody into prompts/text-aware decoders, keeping discrete tokens generation-friendly; AV stacks use coupled masking to bias codes toward shared emotional content [13, 5, 14, 73].

Training signals enforce invariance and stability: speaker-perturbed swapped-prediction with F0/formant scaling yields speaker-invariant tokens and lower unit edit distance under noise/reverb/time-stretch; lexical bottlenecks via continuous embeddings improve ABX/position discriminability and generation; early-codebook semantic distillation increases LM predictability; factorized codecs allocate content, prosody, identity to disentangled subspaces at low bitrate; fixed windows typically match phoneme/syllable/word segmenters at the same duration [17, 76, 8, 63]. Editing/control interfaces mix discrete semantics with orthogonal style controls (prototype selection, residual-level swaps, short speaker prompts), and schedules align information to latency budgets: AR on semantics or early RVQ levels plus NAR refinement, with prompts/later channels preserving style; synchronized AV rates and coupled masking support parallel prosody/facial recovery without inflating content [72, 83, 5, 47, 13, 73]. Evaluate streams separately: content WER/CER and MI with text; speaker similarity/verification and leakage via speaker classification and ABX; and perceptual/distributional metrics under bitrate constraints [83, 8].

3.6 Continuous and codec-free interfaces

Codec-free interfaces pass spectrograms or continuous encoder embeddings to LLMs/decoders, avoiding quantization error and helping prosody/timbre retention at the cost of longer sequences that require projector compression, cross-attention gating, or label-synchronous selection [8]. Thin adaptors between frozen encoders and LLMs (e.g., USM→mT0 via a 2-layer Transformer; Whisper-large via a convolutional subsampler) validate a small modality gap for multilingual ASR, speech-conditioned reasoning, and empathetic dialogue [3, 7, 60]. CTC blank removal yields label-synchronous continuous prompts that outperform convolutional downsampling, integrate with label-synchronous beam search, and markedly reduce attention cost for decoder-only ASR [62].

For generation, continuous latents replace RVQ stacks to retain high-frequency detail and support diffusion/flow training; SoundStream with a 24-d Gaussian latent at 50 Hz stabilizes diffusion via noise injection, range clipping, and unit-std normalization, decoding without discrete indices [13]. Scheduling remains crucial: CTC blank removal, shallow projectors/MLPs, cross-attention gating, wait–k, and similarity-aware thinning keep time-to-first-token low [62, 84, 85]. Bypassing quantization aids paralinguistics and robustness—Whisper features with a small adaptor suffice for affect understanding and instruction following; alignment-aware connectors that emit vectors directly in the LLM space improve semantic retention over discrete units [7, 18]. Pragmatically, use continuous inputs for perception/reasoning and predict discrete indices or mels for synthesis to bound decoding latency; continuous lexical/syllabic embeddings offer compact semantics without codebooks [19, 13, 62].

3.7 Representative methods and empirical findings

Representative families include RVQ codecs (EnCodec/SoundStream/DAC), semantic units from SSL/supervised encoders (HuBERT/Whisper, syllabic), and hybrid/model-aligned tokenizers (ALMTokenizer, RepCodec, HASRD, XY Tokenizer, LM-SPT, universal tokens) [46, 81, 41]. RVQ codecs with strong decoders (e.g., DAC+Vocos) deliver high PESQ/STOI/MCD, low WER, and high NISQA; AR-first/NAR-refine schedules achieve 3–5

Discrete semantic units remain competitive at sub-kbps budgets: adding ASR objectives during tokenizer training improves invariance/compressibility/linguistic structure, while very large k-means codebooks risk under-utilization and weak compressibility [40]. SyllableLM at 6.25 Hz (8192 units) reaches LibriSpeech WER 7.0/CER 3.2 with 90–300M LMs and strong sWUGGY/sBLIMP/StoryCloze [15]. Interleaved speech–text pretraining corroborates 12.5 Hz as a robust operating point for SLM conditioning and streaming [4]. Large-scale spoken dialogue (dGSLM on J-CHAT) uses HuBERT+k=1000 tokens with HiFi-GAN and XVector prompts at low bitrate [6]. Continuous lexical embeddings (tGSLM) match or surpass discrete GSLM diversity with 5× memory savings and strong zero-shot ABX/sWUGGY/sBLIMP/NED via multi-step heads [76]. Tokenized TTS inside SLMs supports barge-in/duplex via interruption markers [1].

Hybrid/model-aligned tokenizers reduce redundancy and align schedules: AR semantic paths plus NAR EnCodec refinement preserve style with short prompts and yield competitive ASR-BLEU on EnglishSpanish S2ST; FiLM-conditioned hybrids halve tokens/s at equal or better quality by leaning on stronger decoders [5, 13]. Families such as ALMTokenizer, RepCodec, HASRD, XY Tokenizer, and LM-SPT emphasize early-channel semantic supervision, later-channel timbre/prosody, and text-synchronized tokens for editing/parallel generation, aided by PAST-style prompt-aware encoders [8]. Multimodal discrete pretraining (audio+face) improves unimodal reconstruction and downstream emotion without burdening content channels [73]. Efficiency regularities: masked-parallel generation with oracle semantics reveals decoder quality more than codec choice; AR+NAR schedules consistently reduce WER/latency with high-fidelity codecs; 12.5 Hz semantics lighten context and support block-causal streaming; syllabic and 6.25 Hz streams need stronger decoders but preserve intelligibility/similarity with text-aware diffusion/flow. IndicSUPERB and BabySLM provide cross-lingual and developmental probes to contextualize capacity [46, 86, 36]. The practical envelope emerges: kbps codecs (DAC/DM-Codec) lead fidelity; mid/low-rate 12.5 Hz semantics and hybrids deliver competitive WER/MOS with streaming; at ultra-low rate, BSQ tokens plus text-aware diffusion/flow close the reconstruction–generation gap.

3.8 Comparative evaluation axes and open issues

Capacity-controlled comparisons should expose reconstruction quality, semantic fidelity, bitrate/sequence length, downstream SLM performance, robustness, cross-language transfer, codebook health, and generation scheduling. DASB documents a performance gap between discrete tokens and continuous SSL features across tasks, motivating harmonized reporting; STAB provides fast intrinsic diagnostics correlating with downstream behavior [9, 40]. A practical triage combines IndicSUPERB (12 Indic languages; ASR, speaker verification/identification, LID, QbE, KWS), STAB metrics, and fastabx ABX discriminability [40, 53, 36].

Recent advances span LM-SPT (ASR-aligned semantic tokens at 25/12.5/6.25 Hz), PAST (phonetic-aware, streamable), SpeechRAG (direct audio retrieval into a frozen LLM), and SLM bridges that align frozen speech/text foundations with 1

Track codebook health (utilization, entropy/perplexity, dead-codes/collapse, temporal drift), RVQ slice–full consistency, and correlate with IndicSUPERB outcomes; quantify robustness (noise/channel/reverb), redundancy (unit edit distance/context fragmentation), and compressibility (Huffman/BPE/de-dup) with STAB, including TER-based prosody sensitivity [40, 36, 39]. Downstream comparisons should match bit/token rate, decoder backbones, compute, and training data; emphasize cross-language transfer and retrieval diagnostics for homophones and long-range context [21, 43, 36]. Compute/latency reporting—parameters, training FLOPs, inference RTF, auxiliary-branch/scheduling latency—should accompany intrinsic triage (STAB) and DASB-style rankings under constrained budgets [57, 9, 40, 41, 65].

Open issues: - Discrete–continuous gaps at matched bitrate persist on DASB; continuous interfaces remain stronger on several tasks [9]. - No single best segmentation–vocabulary pair across sWUGGY/sBLIMP/ProsAudit/StoryCloze; mixture-of-granularities and adaptive tokenization are promising [63]. - Prosody/non-speech coverage is fragile at low rates; add TER, rhythm/duration, spontaneous/multilingual tests and document de-dup/smoothing effects; very coarse units risk rhythm loss and vocoder dependence [39, 15]. - Channelization vs LM difficulty: more channels aid reconstruction but hurt predictability; standards for channel-aware evaluation/scheduling are needed [12]. - Stability across languages/speakers/conditions is challenging; deeper RVQ layers show lower slice–full consistency, complicating streaming/editing [40, 41, 45, 36]. - Large-vocabulary/multilingual tokenization lacks stable recipes; extend STAB's utilization/entropy normalization to broader, multilingual settings [40]. - Retrieval over speech with homophone robustness and long-window modeling is underexplored [43, 39, 41, 36]. - Extreme-compression validity requires standardized human MOS/CMOS and a reconstruction–generation gap metric (e.g., 16.5

A concise reporting template should state dataset scope (e.g., IndicSUPERB's 1,684-hour Kathbath across 12 languages), task portfolio, tokenizer diagnostics and correlations (STAB), LM-aware choices (e.g., LAST/PAST/LM-SPT/dMel), interface descriptors (rate/bitrate/channels/k and usage), reconstruction/perception metrics (objective + human + SALMon), semantics/LM alignment (resynthesis WER/CER, PNMI/CNMI, LM perplexity), prosody/paralinguistics (TER, rhythm/duration, de-dup/smoothing), robustness/redundancy (STAB invariance/robustness/compressibility), downstream transfer (matched ASR/TTS/S2ST, retrieval diagnostics), and compute/latency (parameters, FLOPs, RTF), using fastabx and suite-level scaffolds for controlled comparisons [40, 22, 21, 54, 49, 53, 36].

4 Speech Encoders and Representations

4.1 Encoder families and layer-wise specialization

Speech encoders expose two LM-facing interfaces: (i) continuous conditioning that projects frame- or segment-level embeddings, and (ii) unitization that discretizes layer-wise features into tokens. SSL families (HuBERT, wav2vec 2.0/XLS-R, WavLM, data2vec) and supervised encoders (Whisper) remain standard backbones; prompt-aware front ends (PAST) and prompt-composition frameworks (SpeechComposer/SpeechGen) specialize features to tasks, while hybrids mix continuous spectra with discrete units to trade context length against LM perplexity/NLL [21, 64, 33, 27]. Depth specialization is consistent: lower layers are acoustic, middle layers phonetic/lexical, and upper layers task-tuned; mid-layer k-means units (100–1k) from HuBERT/wav2vec 2.0 work well for textless modeling and spoken LMs, with SpeechComposer favoring HuBERT-Base L6 [67, 64]. Under coarse, syllabic cadences, data2vec2 with teacher-layer distillation improves reconstruction WER and segment-level invariance over HuBERT; robust preprocessing (LID, diarization, denoising) expands transfer to spontaneous multilingual dialogue [15, 6].

For continuous interfaces, Whisper is a strong frozen front end: thin subsamplers or small projectors suffice for multilingual ASR, affect understanding, and empathetic dialogue; freezing Whisper-large-v3 and stage-wise tuning (encoder→encoder+projector) stabilize alignment [7, 87, 60]. For discrete interfaces, inserting pre-VQ pooling into Whisper yields stable 12.5 Hz tokens suitable for streamable SLMs [4]. Token invariance benefits from LM-aligned and phonetic-aware recipes (LAST/PAST) or training-free dMel, and from enrollment-conditioned selective normalization (SHuBERT), which improves SUPERB and multi-talker ASR with a frozen encoder [22, 21, 54, 88].

Encoder–bitrate co-design is pivotal: mid HuBERT layers (e.g., L6–L9) are strong mid-rate tokenizers; under aggressive compression, data2vec2 plus layer-wise distillation stabilizes syllabic/segmental tokens; Whisper-based discretizers operate robustly at 12.5 Hz [64, 15, 4]. For selection/validation, combine fastabx for phonetic separability, IndicSUPERB for multilingual coverage, contamination-robust ASR testing, and SALMon for acoustic robustness; LM-SPT and PAST offer LM-aligned or phonetic-aware tokens at 25/12.5/6.25 Hz [53, 36, 58, 41, 49, 21].

- Prefer mid-to-upper layers for semantic tokens (HuBERT L6–L9; wav2vec 2.0 mid), selected with LM-aware validation (LAST) and corroborated by SALMon; use SHuBERT/PAST under noise [22, 41, 88, 21, 65].
- Under low rates, favor data2vec2 plus layer-wise distillation to pooled teacher means [15].
- For multilingual/noisy inputs, condition on continuous Whisper features via shallow sub-samplers; freeze then stage-wise tune; hybrids mixing continuous+discrete features control context and NLL [27, 60, 87].
- Discretizing Whisper: average pooling + VQ at 12.5 Hz yields streamable tokens [4].
- For spontaneous multilingual dialogue, combine language-specific SSL encoders with LID/diarization/denoising before clustering [6].

4.2 Fusion and representation selection

Fusion spans layers within and across encoders/modalities; selection decides which features to discretize or project. LM-aware tokenization (LAST) maps speech features toward a text-LM space, improving spoken LM and S2T; moderate segments and larger K boost zero-shot SLU while reducing data/runtime [22, 63]. Middle layers consistently supply semantics: HASRD screens layers (e.g., L8) with a lightweight ASR head, learns convex fusions for reconstruction, and projects out semantics before residual acoustic quantization; transfers from E-Branchformer highlight phonetic emphasis near L8 and linguistic content near deeper layers, with k-means500 balancing granularity and predictability [89, 65]. Probes show a final-layer dip for syntax in wav2vec 2.0/HuBERT, motivating mid-layer conditioning; informed selection via an MLP over time–layer features improves ASR/SID/ER/SE/TTS and interpretability [66, 26]. Text-aligned TASTE queries last-layer keys and shallow values with text tokens to produce subword-synchronous features [78].

For continuous conditioning, aggregating/averaging layers improves coverage of content and paralinguistics; DESTA learns Whisper-layer weights and summarizes with Qformer/CNN adapters to trade fidelity for efficiency, and DM-Codec averages teacher layers (BERT/HuBERT) to supervise RVQ, outperforming single-layer targets [10, 77]. Heterogeneous fusion pairs supervised semantic robustness with SSL acoustic invariance using concatenation+projectors or gated cross-attention; discrete–continuous fusion aligns rates and sums embeddings in a shared LM space for stable parallel generation [90, 91, 92].

- Tokenizers: select mid layers (e.g., HuBERT L8, k500); for mixed tokens, learn convex layer fusions and project out semantics before residual quantization [89, 65].
- Direct conditioning: aggregate/average layers then compress with Qformer queries or CNN subsampling to 60–1,500 states/utterance [10].
- Robustness/targeting: fuse heterogeneous encoders via concatenation+projectors or gated cross-attention to balance semantic vs speaker cues [90, 91].
- Text-aligned SLU/ASR: use attention-based, text-keyed fusion (TASTE) [78].
- Task-adaptive discrete units: adopt informed layer-selection modules that weight per-layer tokens [26].

4.3 Robustness, streaming, and deployment constraints

Real-world SLMs need low latency and stability to noise, reverberation, speaker variation, and emotion across ASR/SID/LID/KWS/QbE. Noise-aware tokenization (NAST), invariance-promoting training, and broad evaluation (SALMon, IndicSUPERB) are central [29, 41, 36]. Streaming front ends trim look-ahead while preserving quality: BESTOW replaces bidirectional encoders with short-right-context or cache-aware FastConformers; CIF-based emission produces boundary-aware, label-synchronous prompts for low-latency ST/ASR [84, 93]. Tokenizers become streamable via block-causal attention and chunked schedules (2 s), with training-free chunk-wise expansion keeping accuracy; duplex agents maintain continuous conditioning with compact encoders and noise injection for robustness [4, 17, 1].

Domain shifts are pronounced: ABX gaps show wav2vec 2.0/XLS-R room/mic sensitivity and speaking-style drift; NAST and speaker-invariant tokenizers reduce unit edit distance under time-stretch/pitch/reverb/DNS noise [46, 29, 17]. SHuBERT's dual-path Barlow Twins with enrollment conditioning improves multi-talker ASR and SI-SDRi; denoising objectives aid syllabic segmentation and duplex operation [88, 94, 1]. Latency reflects encoder context, token/frame rate, projector/LM attention, and scheduling; adjustable contexts, CIF prompts, encoder distillation (GenDistiller), and LM-aligned low-rate tokenizers (LM-SPT/PAST) reduce delay while preserving intelligibility [84, 93, 95, 49, 21].

- Match causalization/context to budget: cache-aware FastConformer [70,13]→[70,0] or bidirectional with 13-frame look-ahead [84].
- Align emission to scheduling: block-causal tokenization (2 s chunks) or CIF; validate with STAB [40, 93].
- Harden invariance with LM-aligned/noise-aware tokenizers, MUSAN/DNS augmentation, time-stretch/pitch-shift, and dual-path Barlow Twins; monitor ABX and Unit Edit Distance [22, 29, 53].
- Favor streamable tokenization/segmentation (LAST or dMel) for unified real-time ASR/TTS on shared LMs [22, 54].
- Optimize for edge: distill encoders and operate at 25/12.5/6.25 Hz with RVQ or compact units to cut memory/latency [77, 49].

4.4 Diagnosing representation quality

Diagnostics should capture phonetic discriminability/invariance, leakage across linguistic/paralinguistic factors, and higher-level competence. As summarized in Figure 3, the suite is organized hierarchically: (1) phonetic discriminability/invariance via fast ABX with MFCC baselines and PanPhon correlations, complemented by layer-wise profiles and snippet/window sweeps to expose leakage across factors; (2) decoder-free token stability and linguistic alignment using TER/MTER under controlled WORLD prosody edits, segment-restricted TER for emphasis, PNMI, and streamable phonetic-supervised tokenizers (PAST) to select rate–latency–fidelity operating points; and (3) beyond-semantic competence and robustness through BoSS affect/dialect probes and SALMon consistency/alignment tasks, extensible to broader multilingual ASR/SID/LID/KWS/QbE coverage. A practical instantiation couples fast ABX, decoder-free token stability, and beyond-semantic tests; layer-wise curves guide layer/rate choices, and PAST defines operating points for rate–latency–fidelity trade-offs [53, 39, 21, 46]. Fast ABX on LibriSpeech (dev-clean/other) reports phoneme vs speaker errors against MFCC baselines and correlates with PanPhon articulatory distances; TER/MTER under controlled WORLD perturbations decouple prosody sensitivity from timbre, PNMI measures phoneme–token alignment, and segment-restricted TER captures word-level emphasis [53, 39]. BoSS adds affect/dialect probes; IndicSUPERB and SALMon broaden multilingual and robustness coverage across ASR/SID/LID/KWS/QbE [47, 36, 41].

- Report within-/across-speaker and within-/any-context ABX with MFCC baselines; include layer-wise profiles and window sweeps [53, 46].
- Use TER/MTER and PNMI at matched bitrates to assess token stability, prosody sensitivity, and linguistic alignment [39].
- Add BoSS affect/dialect probes for empathetic-agent readiness [47].

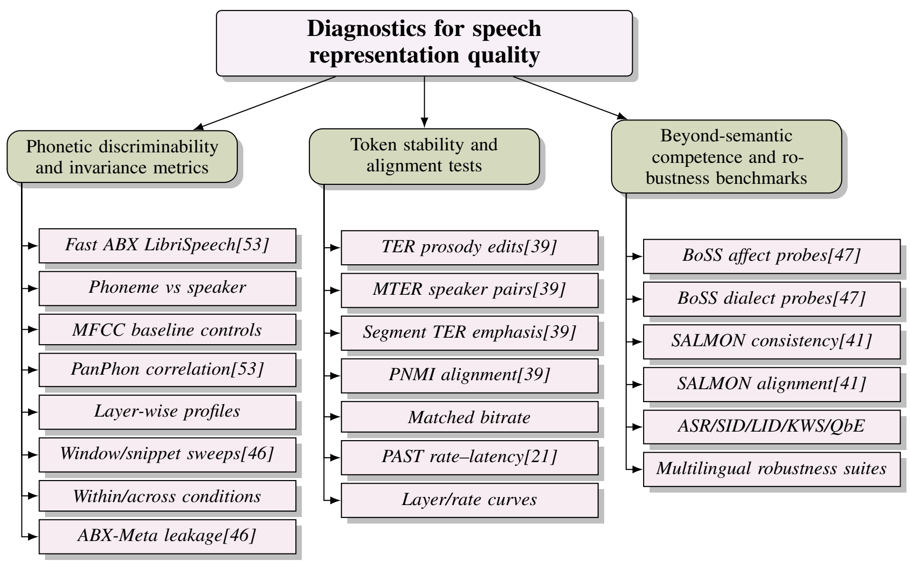

Figure 3: Figure: Hierarchical diagnostics for speech representation quality. The suite organizes (1) phonetic discriminability/invariance via fast ABX with MFCC baselines, PanPhon correlations, layer-wise profiles, and snippet sweeps to expose leakage across factors; (2) decoder-free token stability and linguistic alignment using TER/MTER under controlled WORLD prosody edits, segment-restricted TER for emphasis, PNMI, and PAST operating points to select layer/rate–latency–fidelity trade-offs; and (3) beyond-semantic competence and robustness through BoSS affect/dialect probes and SALMON consistency/alignment tasks, extensible to broader multilingual ASR/SID/LID/KWS/QbE coverage.

4.5 Discretization vs continuous conditioning

Discrete tokens match decoder-only objectives, expose explicit rate/latency control, and enable retrieval/masked modeling; continuous interfaces preserve suprasegmentals, avoid codec loss, and align well to frozen LLMs via thin adapters or query-formers, often improving intelligibility and robustness [96, 97]. Continuous conditioning retains pitch/energy/rate/identity with minimal trainable parameters; bridging frozen Whisper to frozen LLMs via thin projectors and progressive alignment minimizes forgetting, and continuous lexical embeddings (tGSLM) replace hard clusters with a bottleneck; SSL families like data2vec/emotion2vec favor larger k if discretized, tilting toward continuous or high-cardinality discretization [87, 3, 60, 76, 39].

Discrete interfaces provide tunable rates and unified sequence modeling: HuBERT-based 1k vocabularies for S2ST, segment-level k-means to manage context budget, and NAST-style augmentation-invariant units for robustness [5, 63, 29]. Discrete targets can supervise SSL pretraining while serving continuous features at inference (SHuBERT), and discrete indices can outperform regression targets in cross-modal masked modeling (VQ-MAE-AV); ultra-low-rate generation is viable with text-aware BSQ at 6.25 Hz (TaDiCodec) [88, 73, 14]. Codebook size should reflect encoder family: smaller k for HuBERT/ContentVec, larger k for data2vec/emotion2vec [39]. Hybrids are pragmatic for duplex agents: keep lexical progression discrete for predictability/control and inject continuous tokens for style, discourse, and affect to reduce omissions/repetitions and preserve MOS [20, 23, 97].

- Prefer continuous conditioning for high-fidelity comprehension and prosody/affect retention; align with thin adapters and progressive multi-task training without eroding text abilities [20, 3].
- Prefer discrete tokens for explicit rate control, retrieval, and masked modeling; learned codebooks (RepCodec) and training-free dMel improve fidelity/streamability over vanilla k-means [25, 54].
- Use hybrids to combine discrete controllability with continuous fidelity, mitigating discrete inconsistency [8, 23].
- For ultra-low-rate generation, pair BSQ at 6.25 Hz with text-aware diffusion/flow [14].

Evaluation should be modality-aware: for speech UIs report SLU metrics (ASR/SID/LID/KWS/QbE) and alignment-aware diagnostics; for text UIs retain NLU-style tests. Use STAB to rate tokenizers, SALMon for robustness, IndicSUPERB for multilingual coverage, and include retrieval metrics with SpeechRAG-style robustness to ASR errors [18, 40, 41, 36, 43].

4.6 Alignment challenges to text LLMs

Speech–text gaps span information content (tone/prosody/duration), geometry (embedding distributions), timescale (Hz frames vs subwords), and structure (weaker syntax in speech encoders). Probes show speech encoders lag BERT in syntax and can be confounded by lexical artifacts, motivating alignment that reduces non-linguistic variance and clarifies syntactic signals, ideally in the LLM space to handle heterogeneous lengths/pauses [66, 85]. With frozen foundations, uniform subsampling and thin transformers trained by next-token prediction bridge encoders to LLMs; deeper projectors yield diminishing returns, and staged pipelines (modal alignment→conversational tuning) or alignment-aware connectors that pre-distill text-token embeddings mitigate forgetting; optimal-transport regularization improves geometric alignment [3, 52, 18, 85]. Tokenizer-side alignment also helps: LM-SPT avoids frame-synchronous SSL targets at low rates by matching a frozen Whisper's features between original audio and semantic-only reconstructions, and DM-Codec distills text/speech teachers into RVQ latents via cosine alignment without strict time sync [49, 77].

To avoid catastrophic forgetting, favor staged freezing and targeted adaptation: train small projectors with frozen backbones, then freeze connectors and tune the LLM; apply Partial-LoRA to speech-only tokens to add emotion without harming instruction following [3, 18, 7]. Paralinguistic grounding ties acoustic cues to text-native reasoning using time-stamped emotion metadata and by maximizing mutual information between speech-unit and LLM latents, supporting affect/identity retention [37, 47].

- Reconcile rate/geometry by freezing pretrained speech/text models and learning a thin projector; operate at 25/12.5/6.25 Hz [3].
- Enforce tokenizer-side semantic alignment (LM-SPT) for LM-salient low-rate tokens; distill early into RVQ latents (DM-Codec) [49, 77].
- Ground paralinguistics with explicit metadata and implicit QA generation to boost empathetic reasoning [37].
- Use conservative adaptation (frozen backbones, small adapters, staged connector/LLM tuning) to add speech competence while preserving text decoding [3, 18].

Diagnostics should pair syntax-aware probes that control lexical confounds with rate-aware WER, prosody/tone coverage, and alignment metrics computed in the LLM space to verify closure of tone/prosody/duration/syntax gaps without sacrificing text-native behavior.

5 Token to Speech Decoding and Vocoders

5.1 Decoding targets and pathways

Practical decoding targets coalesce into four routes: (i) codec-native inversion that predicts codec indices/latents and invokes a pretrained decoder, (ii) unit-to-waveform vocoding from discrete semantic/mixed tokens, (iii) token-to-mel prediction with a neural vocoder tail, and (iv) hybrid token-to-latent paths that feed a codec decoder. A discrete, LLM-ready token interface—vetted by STAB for downstream suitability—supports controllable streaming and policy-aware scheduling, as in StreamUni [40, 98]. As illustrated in Figure 4, these four routes consolidate into three families: codec-aligned decoding (subsuming hybrid token-conditioned latents) to minimize conversion loss and maintain codec-rate latency; unit vocoder token-to-waveform paths that preserve modularity and enable promptable style control and full-duplex behavior; and token-to-mel pipelines that map tokens to mel spectrograms, leveraging flow/diffusion and mature neural vocoders for chunkable, streaming-friendly synthesis. This organization complements the STAB-vetted, LLM-ready token interface just noted, which underpins controllable streaming and policy-aware scheduling (e.g., StreamUni).

Codec-native inversion removes the spectrogram stage, aligns latency with the codec frame rate, and integrates cleanly with multi-stream RVQ; S2ST systems directly decode generated EnCodec tokens for real-time synthesis [5]. Hybrid token-to-latent variants replace indices with token-conditioned continuous latents—e.g., WaveNet-based diffusion feeding SoundStream—combining discrete predictability with continuous fidelity at 24 kHz [13]. Unit vocoders retain tokenizer modularity and promptable control; a listen-while-speaking SLM uses a GAN-based token-to-waveform vocoder with an acoustic prompt to steer timbre and decouple identity from content for duplex interaction [1]. Token-to-mel pipelines map tokens to time–frequency targets via conditional flow/diffusion and render with HiFi-GAN/BigVGAN/Vocos, enabling chunkable, streaming-friendly S2S and ultra–low-bitrate setups [4, 14].

Consistent with the grouping in Figure 4, route selection hinges on the upstream token interface, latency/RTF, and style control: codec-native inversion minimizes conversion loss and removes a separate vocoder [5]; unit vocoders maximize modularity and promptable identity control [1]; token-to-mel leverages robust conditional flow/diffusion and mature vocoders for streaming [4, 14]; hybrid latent diffusion inherits codec fidelity while easing rate constraints via continuous latents [13].

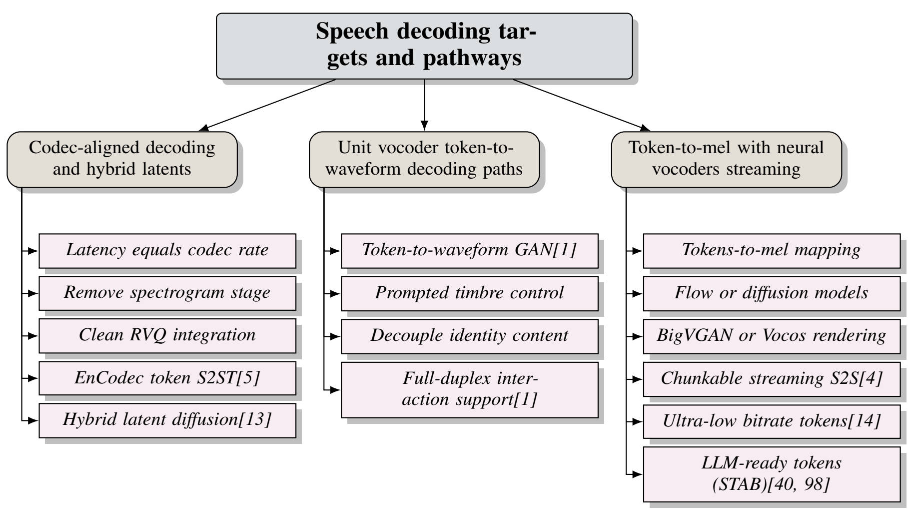

Figure 4: Figure: Decoding targets and pathways. The tree groups practical speech decoding routes into three categories: codec-aligned decoding (including hybrid token-conditioned latents) for minimal conversion loss and codec-rate latency; unit vocoder token-to-waveform paths that preserve modularity and enable promptable style control and full-duplex behavior; and token-to-mel pipelines that map tokens to mel spectrograms, leveraging flow/diffusion and mature neural vocoders for chunkable, streaming-friendly synthesis. The LLM-ready token interface vetted by STAB facilitates controllable streaming and policy-aware scheduling, exemplified by StreamUni.

5.2 Objective families and decoder backbones

Decoder backbones largely fall into: GAN/vocoder pipelines optimized for low-latency rendering, diffusion/flow decoders resilient to aggressive compression and token sparsity, and autoregressive (AR) or semi-AR token decoders that stabilize pacing and alignment; effectiveness depends on token semantics, bitrate, and streaming constraints [28, 68, 83, 13].

GAN/vocoder pipelines. HiFi-GAN remains a strong default for unit-to-waveform and mel inversion due to favorable RTF and perceptual quality; conditioning on x-vectors yields identity-controllable reconstructions from discrete units [6]. Cascaded S2ST stacks commonly predict mels (e.g., FastSpeech2 with PortaSpeech prior/post-net) using mel L1, duration/pitch/energy L2, VAE KL, and flow NLL objectives before HiFi-GAN synthesis [70]. Codec-native variants blend adversarial/feature-matching with reconstruction losses and can adopt learned stochastic decoders at low bitrates [13].

Diffusion/flow decoders. Conditional flow-matching and diffusion map discrete tokens to acoustic targets with stability under low token rates; large-scale interleaved training applies CFM over tokens with HiFi-GAN as terminal vocoder [4]. At ultra-low bitrates, diffusion-only decoding improves intelligibility and perception versus GAN baselines while simplifying optimization [14]. WaveNet-based diffusion with v-parameterization and cosine SNR, integrated into a SoundStream decoder, boosts low-rate quality; moment-matching distillation reduces sampling to 4 steps with minor DNSMOS impact and stable WER [13].

Autoregressive and semi-AR decoders. AR decoders provide tight duration/prosody control, whereas semi-AR/CTC schedules trade some control for parallelism and lower latency. Phonetic-grounded causal modeling (PAST), training-free streamable dMel, unified hierarchical RVQ tokens, and standardized STAB evaluation compose cleanly with GAN or diffusion/flow renderers for stable timing and chunked generation at low rates [21, 54, 83, 40, 13].

Loss design and guidance. Combining adversarial and feature-matching terms for periodicity/timbre with reconstruction/spectral losses for envelopes/prosody works well in codec-native and GAN pipelines [13], while mel-stage decoders leverage mel L1, duration/pitch/energy L2, VAE KL, and flow NLL before HiFi-GAN [70]. Diffusion/flow avoids adversarial instability and is robust at low token rates [14]. At kbps RVQ rates, codec-native or GAN decoders deliver low-latency, identity-preserving synthesis; for semantic/mixed tokens under aggressive compression or streaming, token-conditioned diffusion/flow—with step-distilled samplers and a HiFi-GAN tail when targeting mel—yields higher perceptual quality [13, 14].

5.3 Hybrid continuous–discrete approaches

Conditioning discrete content tokens with continuous vectors for prosody, timbre, and global style consistently improves naturalness and identity retention at low rates. Flow-SLM pairs discrete linguistic tokens with continuous acoustic embeddings [99]; GOAT-TTS keeps content discrete while conditioning on continuous style prompts to reduce buzzy artifacts [100]; DiffSoundStream conditions latent diffusion on semantic and acoustic tokens but decodes via continuous SoundStream latents to recover microstructure under compression [13].

A local–global design pattern is effective: fuse time-synchronous discrete content/prosody with a slowly varying or utterance-level continuous style. VAE-GSLM concatenates discrete semantics, learned paralinguistic latents, and an utterance embedding for DDPM/DDIM decoding, then applies HiFi-GAN, improving MOS over token-only baselines [101]. UniCodec merges content and prosodic token streams and modulates them with a global continuous token via FiLM-style scale/bias, decoupling utterance-level style from segmental content [102]. Instruction-following SLMs expose continuous LLM hidden states to the unit generator for synchronized text-and-unit emission with stable prosody [103], and cascaded S2ST augments discrete streams with a global prosody embedding and scalar trajectories to reduce artifacts in cross-lingual synthesis [70]. Continuous-first modeling in tGSLM learns on continuous lexical/acoustic tokens but currently relies on discrete HuBERT units for synthesis, motivating direct diffusion/vocoder decoders to simplify deployment [76].

Design guidance: adopt local discrete semantics plus learned continuous paralinguistic latents; modulate globally via FiLM/scale–bias; prefer moderate-step diffusion/flow with a robust HiFi-GAN tail for high naturalness under compression [13, 101, 70]. Where explicit control is required, combine global style with discrete content and word-level pause/duration/F0 signals to preserve rhythm and emphasis [70].

5.4 Conditioning, controllability, and style preservation

Controllability separates semantic prediction from acoustic realization so identity, emotion, and prosody are steerable under latency. Affect alignment remains limited (e.g., 52.59

Prompt conditioning and identity control. Short acoustic prompts (3–5 s) seed timbre/global style while leaving content to the LM. In end-to-end S2ST, brief EnCodec-token prompts preserve speaker style across translation, with better WavLM-style similarity after translation-data upsampling [5]. At ultra-low token rates, text-aware diffusion/flow decoders gain from short prompts, improving WER, similarity, and UTMOS without added latency [14]. Unified models combine an <enroll-speech> prompt with x-vectors for zero-shot identity in TTS and conversion [64].

Speaker/style embeddings and role-aware rendering. Fixed-dimensional embeddings injected into decoders or vocoders stabilize identity; multi-speaker dialogue stacks condition a unit vocoder on role-specific embeddings and render stereo with channel swaps for turn-level analysis [6]. Cross-lingual style transfer uses a VAE-derived global prosody embedding with SALN to control style without parallel same-text data [70].

Disentangled channels and localized prosody edits. Mixed-token designs expose semantic and acoustic channels so residual vectors carry prosody/timbre while early channels encode content [47]. Prompt-conditioned acoustic-unit generation preserves identity while the LM focuses on content; residual-channel scheduling enables local prosody manipulation and aligns with multi-stream RVQ [5]. When streams are deduplicated or pushed to lower rates, explicit duration and prosody tokens stabilize rhythm/intonation [70].

Global–local prosody controls and latency. Combining a global prosody embedding with local, text-synchronous edits—pause insertion, duration modification, word-level F0 transfer with speaker normalization—yields better ratings on emphasis, intonation, rhythm, and emotion, directly targeting BoSS affect criteria [70, 47]. Short prompts amortize identity setup and pair well with parallel or semi-AR unit generation; prompt-guided diffusion/flow at low token rates caps latency while maintaining intelligibility and similarity [14]. Reporting should include intelligibility (WER/CER), quality (UTMOS/MOS), and identity/style similarity (e.g., WavLM cosine); role-separated stereo renderings with fixed embeddings aid dialogue auditing [6]. Chunked generation that alternates silent reasoning with speech helps reasoning without extra latency, and synthetic speech–text pairs improve paralinguistic instruction following without degrading core LLM skills [48, 32].

Ecosystem supports conditioning and evaluation at scale: IndicSUPERB offers 1,684 h across 12 languages for six SLU tasks with large gains from language-specific tuning; fastabx provides efficient ABX analysis of learned representations; LM-SPT and PAST learn low-rate, semantically aligned and phonetic-aware units; SpeechRAG aligns speech–text embeddings for ASR-free retrieval and robust QA under noisy transcripts [36, 53, 49, 21, 43].

- Use short, homogeneous prompts to seed identity/style in textless S2ST over neural-codec tokens and in text-aware diffusion/flow or prompt-composition models; continuous speech tokenizers (Cont-SPT) further improve naturalness and continuity [5, 64, 20, 33].
- Inject speaker/style embeddings in decoders/vocoders to preserve identity independently of content; in unit-vocoder stacks, enroll with an x-vector plus an <enroll-speech> prompt for controllable TTS/VC [64].
- Combine global and local controls: apply a VAE-derived global embedding via SALN and add word-level pause, duration, and pitch controls to strengthen emphasis, intonation, and rhythm [70].
- Prefer tokenizations that separate content from acoustics (e.g., LM-SPT semantics with dMel or codec residuals) and localize edits to acoustic/RVQ channels; stabilize low-rate decoding by conditioning codec latents on semantic tokens (DiffSoundStream), mitigating discrete inconsistency, and adding explicit duration/prosody tokens as in ProsodyLM [54, 49, 13, 23, 71].

- Evaluate controllability with identity- and affect-sensitive metrics aligned to task goals; BoSS benchmarks highlight persistent affect gaps that should guide conditioning design [47].
- For multi-party dialogue, condition the vocoder on fixed per-role embeddings and render stereo with channel swaps at turn boundaries for controlled, turn-level analyses [6].

6 Language Modeling over Speech Tokens

Language modeling over speech tokens treats acoustics as sequence elements—semantic units, codec codewords, or short continuous prompts—so a single causal decoder can reason over text and speech while trading off latency, fidelity, and sequencing. To systematize this landscape, we present a hierarchical taxonomy that structures the field into four pillars (see Figure **??**): (1) modeling objectives and architectures—covering AR, NAR, and hybrid decoding with interleaved training and token-conditioned diffusion/flow, accompanied by actionable recipes and diagnostics; (2) modality transfer and LLM adaptation—including continued shared-token pretraining versus cold-start projector bridges, alignment strategies, and PEFT curricula with supporting toolkits; (3) interleaved and duplex modeling exemplars—spanning token-level and linguistic–acoustic interleaving, inference-time budget policies, duplex scheduling with robustness features, and blueprint patterns; and (4) empirical regularities and scaling—highlighting token-rate knees around ≈ 12.5 Hz, capacity/sequence compression effects, task asymmetries, schedule–temperature coupling, token-space data scaling, evaluation suites, and effective configurations. This organization makes explicit the core trade-offs (latency–fidelity–sequencing), guidance on where to place AR vs. NAR computation, how to preserve text priors while adding speech skills, and how interleaving policies and token rates shape compute–quality efficiency, aligning with prior findings on modality transfer, scheduling for mixed streams, and scaling regularities [4, 14].

Figure 5: Hierarchical taxonomy of language modeling over speech tokens

6.1 Modeling objectives and architectures

Decoder-only SLMs realize four regimes under a next-token objective: (i) autoregressive (AR) decoders over discrete codec/semantic tokens or continuous prompts; (ii) non-autoregressive (NAR) decoders that parallelize residual acoustics via masked infilling or multi-codebook heads; (iii) AR+NAR hybrids that place AR on low-entropy channels and NAR on high-rate residuals; and (iv) interleaved dual-token designs that train a single decoder on mixed text–speech streams with a shared vocabulary and schedule-aware loss masking [4]. Token-conditioned diffusion/flow decoders complement these objectives at low token rates to recover prosody and timbre without reintroducing AR latency [14, 13].

AR backbones span textless unit LMs that quantize speech (e.g., HuBERT+k-means) and generate long-form audio via a unit LM plus unit vocoder [6], duplex decoders that predict next audio tokens while ingesting a streaming listening stream with middle-fusion conditioning [1], and decoder-only ASR that feeds CTC-compressed prompts and a sentinel (e.g., <aud>) into a causal decoder trained with joint CTC+CE [62]. NAR and hybrid schedules reduce wall-clock delay by decoding RVQ-1 autoregressively and predicting higher RVQ levels in parallel, preserving intelligibility while cutting exposure bias; end-to-end S2ST and retrieval-augmented speech generation benefit most when codecs prioritize intelligibility over raw reconstruction [43, 28]. Interleaved training with unified text–speech vocabularies enables a single decoder to cover text-only, speech-only, and mixed contexts with modality sentinels and loss masking, while token-to-waveform fidelity is restored via conditional flow/diffusion or vocoder tails, especially for semantic-only or ultra–low-rate tokens [4, 14, 13, 49].

Operational recipes emphasize chunked unspoken reasoning to improve quality without added latency (STITCH), lightweight paralinguistic infusion from synthetic speech–text pairs (DeSTA2), and low-budget training via careful initialization, synthetic interleaving, and preference optimization (Slamming) [48, 32, 104]. Diagnostics and tokenization advances—IndicSUPERB for multilingual SLU, LM-aware tokenization (LM-SPT/LAST), STAB stability probing, and fastabx separability—align interfaces with LM behavior [36, 49, 22, 40, 53]. Actionable patterns: use AR decoder-only LMs for unit-based textless generation [6], inject streaming listening embeddings with middle fusion for duplex agents [1], feed CTC prompts for decoder-only ASR [62], place AR on RVQ-1 and NAR on residuals for low-latency codecs [43, 28], and attach token-conditioned diffusion/flow decoders at low rates [13, 49, 40].

6.2 Modality transfer and LLM adaptation

A practical path to speech-capable models freezes a strong text LLM and adds tiny speech bridges (projectors, adapters, or LoRA), reaching competitive ASR/AST, QA, and mixed-modal dialog with 1

Continued pretraining in a shared token space extends a text LLM to speech without projectors: GLM-4-Voice expands the vocabulary with speech tokens and scales interleaved next-token pretraining, preserving text competence via breadth of mixed-modality data [4], and SpeechComposer adapts a text-only OPT in the same decoder and vocabulary [64]. For speech-only S2ST, warm starts help optimization but final quality depends more on multilingual bidirectionality and parameter sharing than on text-LM initialization [5].

Cold-start bridges map SSL features into a frozen LLM's embedding space with minimal trainable parameters: a USM-to-LLM adaptor enables zero-shot instruction following and SLU [3], production stacks project Whisper-large-v3 into Qwen-class LLMs via linear/MLP projectors with constrained LoRA and Partial-LoRA on speech spans to mitigate forgetting [7], and the same projector approach supports decoder-only ASR and duplex listening embeddings with continued joint training of the speech generator and SSL encoder [62, 1].

Alignment preserves language priors while adding speech skills: KL geometry/policy distillation and KL-regularized RL (e.g., Dr.GRPO) improve speech outputs without collapsing text abilities [7, 61]; MI-based training retains beyond-semantic cues (prosody, identity) while respecting text geometry [47]; synchronized listening embeddings strengthen duplex alignment [1]. Parameter-efficient adaptation and curricula—LoRA on a mostly frozen LLM (typical ranks/: 16–64/16–32), prompt/prefix-tuning for classification-like tasks and few-shot ICL on a fixed 150M GSLM with Random Label Mapping, language-aware prompting to reduce code mixing, progressive unfreezing (encoder $\rightarrow$ encoder+projector $\rightarrow$ projector+LLM with LoRA), and pseudo-prompt LM training for decoder-only ASR—control budgets and limit forgetting [67, 61, 60, 62].

Blueprints and diagnostics—LAST for LM-aware tokenization, VoxEval for spoken QA/knowledge under varied audio, STAB for tokenizer stability, IndicSUPERB for multilingual SLU, and fastabx for ABX probing—accelerate development [22, 45, 40, 36, 53]. Efficient cross-modal transfer aligns speech and text at matched abstraction levels, freezes strong backbones with tiny adapters, scales on large synthetic interleaved token corpora, and uses modality-specific MoE to curb forgetting while preserving native speech latency [105, 106, 3, 4, 11].

6.3 Interleaved and duplex modeling exemplars

Interleaved, duplex-capable SLMs unify perception, reasoning, and speech/text synthesis by consuming and producing mixed token streams in a single causal decoder, typically initialized from text LMs and frozen audio foundations with lightweight adapters or shared dual-token vocabularies for zero-shot instruction following, long-form generation, contextual biasing, and simultaneous listening–speaking at favorable compute [57, 107, 3, 44, 11]. As summarized in Figure 6, we organize this design space into three branches: token-level interleaving (shared vocabularies, adapter-bridged embeddings, synthetic span interleaving, and budget-cycled decoding), linguistic–acoustic planning interleaving (plan-before-speak phases, fixed text:speech ratios, and transcribe$\rightarrow$generate staging), and duplex scheduling controls (simultaneous listen–speak with IRQ tokens, middle-fusion conditioning, and turn-level text guidance). Together, these threads illustrate how recent systems unify perception, planning, and speech synthesis for low-latency, robust spoken interaction. Operationally, these branches instantiate token-level interleaving under a unified next-token objective, linguistic–acoustic interleaving that separates semantic planning from acoustic realization, and duplex scheduling that synchronizes user and assistant channels for overlaps, backchannels, and barge-in.

Token-level interleaving concatenates modality-tagged spans so one decoder covers ASR, TTS, S2ST, and mixed continuation: LLaSM inserts audio patches via boundary tokens [52], SpeechComposer uses a shared vocabulary [64], and large-scale token-space corpora replace span-corrupted text with speech tokens sampled by a text-to-token model (e.g., Poisson spans, 0.3 corruption) to enable

100B–600B-token interleaved pretraining without waveform synthesis [4]. Instruction-following bridges prepend adapted speech embeddings to text before a frozen LLM [3], while inference-time policies reduce latency by cycling text and speech token budgets and by staging transcribe→generate to anchor acoustics to a fresh semantic plan (e.g., 13 text then 26 speech tokens) [16].

Linguistic–acoustic interleaving enforces plan-before-speak schedules using explicit phase tags (chain-of-modality prompting) and streaming schedulers that first output partial transcripts, then complete translations per chunk; fixed text:speech ratios provide interpretable latency–quality knobs and transfer across backbones [4, 98, 16]. Duplex scheduling injects a streaming listening stream while emitting next audio tokens; IRQ tokens pause generation on interruption, and middle-fusion conditioning yields near-ceiling interactivity F1 with strong intelligibility, while turn-level planning that emits brief textual guidance before each speech turn and balances text:speech losses improves semantic quality by over 30

Exemplar blueprints include semantic-first AR planning with low-latency acoustic realization (DualSpeechLM-style), token-LMs augmented by a flow/diffusion decoder (FlowSLM), vector-token predictability with Poisson-span interleaving near 0.3 corruption (VecTokSpeech), and listening-embedding integration with interruption tokens and chunk-wise alternation (Ichigo-style) [4]. When vocabularies cannot be shared, projector-first interleaving delivers mixed contexts without LM expansion [3]; with shared vocabularies, synthetic interleaved corpora efficiently scale mixed-sequence competence [4]. Span markers boost cross-modal continuation, plan-before-speak policies lower initial audio latency, streaming CoT improves S2ST responsiveness, and duplex robustness rises with blockwise listening fusion, IRQ tokens, and turn-level guidance plus conservative decoding temperatures [16, 98, 1, 35].

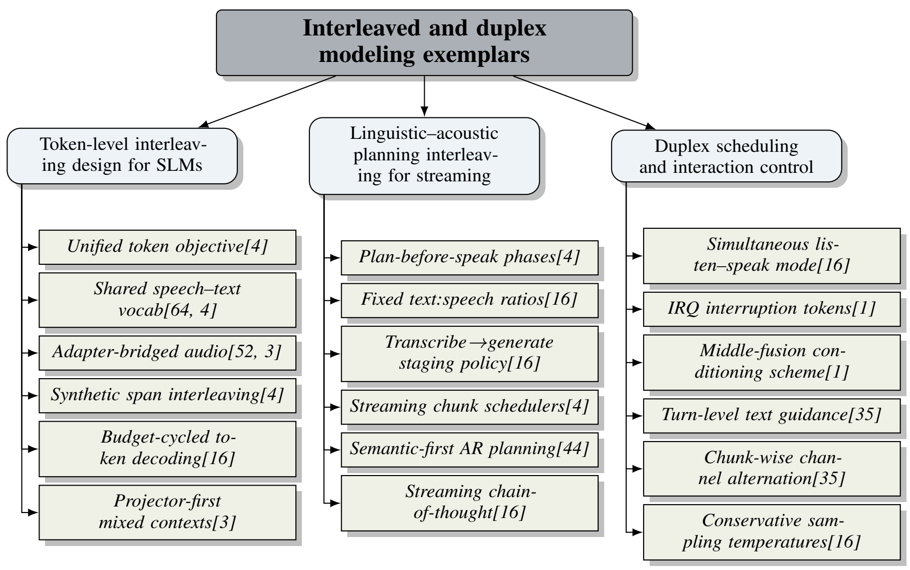

Figure 6: This figure organizes interleaved and duplex SLM design into three branches: token-level interleaving (shared vocabularies, adapter-bridged embeddings, synthetic span interleaving, and budget-cycled decoding), linguistic–acoustic planning interleaving (plan-before-speak phases, fixed text:speech ratios, and transcribe→generate staging), and duplex scheduling controls (simultaneous listen–speak with IRQ tokens, middle-fusion conditioning, and turn-level text guidance). Together, these threads illustrate how recent systems unify perception, planning, and speech synthesis for low-latency, robust spoken interaction.

6.4 Empirical regularities and scaling

Compute–quality depends on token cadence, segmentation granularity, decoder alignment, and capacity splits between the speech front end and language backbone. Interleaved pretraining scales smoothly in token space and exhibits a rate knee near 12.5 Hz, below which further compression hurts performance [4].

Token rate and segmentation: coarser cadences expand headroom at fixed FLOPs—SyllableLM operates at 5.0–8.33 Hz with 30× less LM training compute than AudioLM and 4–4.5× faster inference at equal LM size while maintaining competitive semantics [15]; ultra–low-rate codecs remain generation-friendly when paired with text-aware decoders, with 10–32 diffusion steps forming an efficiency sweet spot and a favorable reconstruction–generation gap for SLMs [14]. Backbone capacity and sequence compression: larger decoders consistently improve recognition in Encoder–Adaptor–LLM pipelines (e.g., Qwen2.5-7B > Qwen2-1.5B; Gemma3-12B > Qwen2.5-7B with identical front ends), while projector pooling/splicing (4:1–5:1, 20→40 ms) halves sequence length with negligible loss [87, 60]. Task asymmetry: scaling disproportionately benefits generative targets—TTS/VC improves faster than ASR as decoder size grows [64], and discrete TTS dWER drops markedly from 389M to 1B parameters—yet cross-domain generalization remains uneven [2, 74].

Scheduling and sampling: planning-aware interleaving favors conservative decoding (e.g., TurnGuide performs best at temperatures 0.8–1.0), whereas speech-only chunking peaks at higher temperatures, revealing a tight schedule–temperature coupling [35]. In-context learning in SLMs often needs meta-training or prompt-tuning; scale alone is insufficient [67]. Data scale and token-space construction: interleaved corpora built directly in token space yield monotonic gains from 0→100B→200B→600B tokens and support the 12.5 Hz operating point for both training and streaming [4].

Capacity planning follows: pick cadence near the knee using prosody-sensitivity and STAB diagnostics (12.5 Hz for semantic/mixed tokens), or, at 5–8.33 Hz, budget stronger text-aware decoders plus explicit silence/syllable modeling [40, 39]. In Encoder–Adaptor–LLM systems, prioritize robust speech backbones with lightweight adapters and reduce sequence length via projector pooling or frame splicing (4–5× to 25/12.5/6.25 Hz); prefer LM-aligned tokenizers (LM-SPT) and adapt by language to avoid semantic dilution at low rates [3, 49, 81]. Allocate compute to model size over extra tokens; interleaved SLMs scale more efficiently than textless SLMs and show slower linguistic scaling than text LMs, making ASR gains hinge on front-end quality and tokenization choices [108, 22, 63, 57, 4]. Tune decoding temperatures to the interleaving policy [35], use diffusion/flow decoders with 10–32 steps for ultra–low-rate synthesis while tracking the reconstruction–generation gap [14], scale interleaved corpora in token space for monotonic returns [4], and include prompt-tuning or meta-training to elicit ICL-like behaviors [67]. Comprehensive evaluation should cover multilingual SLU (IndicSUPERB), robustness/knowledge under varied audio (VoxEval), and tokenizer diagnostics (STAB, fastabx) [36, 45, 40, 53].

Effective configurations pair strong text LMs with rate-aware tokenization near 12.5 Hz, carefully scheduled speech–text interleaving, and compute concentrated in model capacity, trained on large synthetic mixed-token corpora (600B of 1T total tokens), delivering steeper quality–compute gains, competitive spoken QA, and strong semantics at reduced cost [57, 83, 4, 108].

7 Training Paradigms and Modality Alignment

Curricula that endow LMs with speech competence co-optimize five phases—interface pretraining, modality alignment, supervised and instruction consolidation, duplex/streaming specialization, and post-alignment via preferences/RL—under three constraints that govern transfer: interface invariance, compression without semantic erosion, and forgetting control with standardized evaluation. Two pathways dominate: connector-first bridges with frozen foundations and thin adaptors versus continued mixed-modality pretraining with shared vocabularies; both are shaped by AR/NAR decompositions and tokenizer choices that balance latency, bitrate, and drift [22, 3, 4]. The subsections consolidate multi-stage pipelines and alignment blueprints, scalable synthetic supervision, interface consistency with compression, and diagnostics for retention and robustness.

7.1 Multi-stage pipelines

A practical scaffold freezes strong speech encoders and a text LLM, inserts a 1

Key building blocks include semantically aligned or phonetic tokenizers (LM-SPT at 25/12.5/6.25 Hz; PAST with supervised phonetics and a causal streaming variant) [49, 21]; ASR-free SpeechRAG for retrieval under high WER [43]; connector-first SLM with 1

Stage 0 (interface and rate control) fixes tokenizer/codec, cadence, bitrate/codebooks, and inversion targets using STAB-style metrics, while mitigating discrete inconsistency for stable conditioning;

ultra-low-rate diffusion codecs and VQ-MAE-AV represent efficiency and coverage extremes [40, 23, 14, 73]. Stage 1 (alignment) follows connector-first bridges—LM-aware tokens (LAST, LM-SPT) or Whisper→LLM projectors with progressive unfreezing only as needed [22, 49, 60]—or continued pretraining with shared-vocabulary interleaving at 12.5 Hz that scales more efficiently than textless SLMs while preserving knowledge [4, 57, 38, 109]; affect-first alignment can precede paralinguistic grounding (BLSP-Emo) [7]. Stage 2 consolidates with supervised/instruction data—real and synthetic pairs, mined speech, language-specific tuning—and can add auxiliary SER heads for affect; textless SLMs benefit from prompt-tuning warmups with few-shot demonstrations, and decoder-only ASR gains from interleaved LM-only updates plus joint CTC–decoder training [36, 30, 32, 7, 67, 62]. Stage 3 specializes for duplex/streaming via IRQ prediction, barge-in handling, and streaming curricula that interleave partial transcripts/plans before speech emission, and integrates SpeechRAG for listen-and-retrieve [1, 4, 43]. Stage 4 applies preference learning/RL with structural constraints (e.g., KL/DPO proximity, structural tokens, optimal transport) and second-pass rescoring with contextual biasing, few-shot prompts, and checkpoint averaging, achieving sizable WER gains [61, 85, 110].

Forgetting is curbed by progressive unfreezing, freezing connectors during downstream NTP, LoRA restricted to speech spans, retaining LM-only mini-batches for ASR, and amortizing gaps via continued pretraining; reports should include tokenizer geometry/rates, per-stage optimizers/policies, mixed-data composition (1T tokens), streaming hyperparameters, and verifiable-reward setups under standardized benchmarks and STAB diagnostics [60, 67, 4, 40, 36, 73, 14].

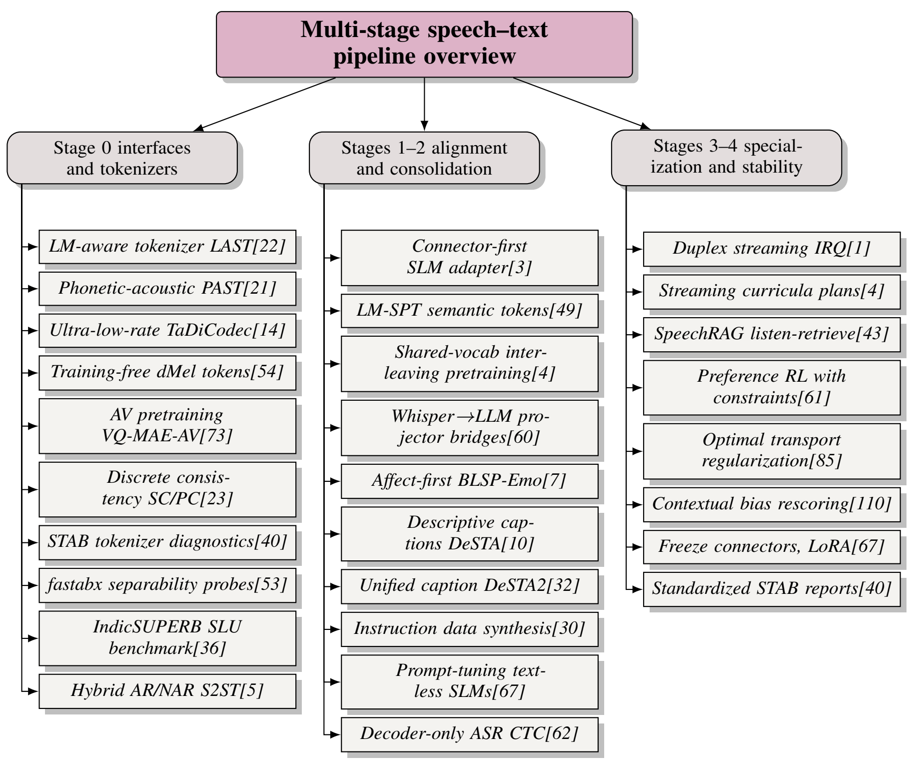

Figure 7: This figure shows a concise two-level map of multi-stage pipelines for speech–text models. Stage 0 aggregates interface and tokenization choices (LM-aware and phonetic tokenizers, ultra-low-rate codecs, audiovisual pretraining, and diagnostics). Stages 1–2 cover modality alignment and consolidation via connector-first adapters, interleaved shared-vocabulary pretraining, caption-based alignment, and paralinguistic grounding. Stages 3–4 highlight specialization and optimization for streaming/duplex use, ASR-free retrieval (SpeechRAG), preference learning with structural constraints, contextual biasing, and stability/reporting practices, with representative works cited at each leaf.

7.2 Alignment mechanisms and blueprints

Alignment spans four layers: LM-aware tokenization that projects acoustics into text-LM space (LAST) [22]; tokenizer-boundary semantic grounding via reconstruction-driven distillation against ASR encoders to yield low-rate units (LM-SPT) [49]; encoder–tokenizer designs that produce decoupled, stable units for joint speech–text modeling [11]; and tokenizer→LM→decoder coupling with multi-token prediction and paralinguistic conditioning [75, 37].

Geometry/timing into the LLM space leverage thin projectors with temporal pooling trained jointly with Whisper and LoRA-adapted LLMs under NTP [60]; KL-based behavior alignment and auxiliary SER heads ground affect (BLSP-Emo) [7]. Projector-free paths extend the LM vocabulary with 12.5 Hz speech tokens and train on interleaved sequences [4]. Decoder-only ASR stabilizes with prompt-length–aware routing to LM-only objectives and joint CTC–decoder training, with shallow fusion at inference [62].

Semantic grounding couples codec and language priors: DM-Codec distills BERT/HuBERT into RVQ token features via cosine and [CLS] guidance alongside reconstruction/adversarial/commitment losses, removing teachers at inference [77]; text-aware diffusion decoders expose and narrow reconstruction–generation gaps at ultra-low rates (TaDiCodec) [14]. Shared-vocabulary interleaving at 12.5 Hz maintains consistent grounding/modeling/inversion targets [4], while AV settings benefit from contrastive correspondence with coupled masking and cross-attention fusion (CAF–CAF) [73].

Stable anchoring favors single-objective alignment of encoder→projector→LLM with temporal pooling for latency control and pseudo-prompt LM objectives to reduce train–test mismatch in decoder-only ASR [60, 62]. Schedule-aware post-alignment down-weights <think> vs <transcribe> losses and uses structural-validity rewards to prevent reward hacking; beyond semantics, BoSS maximizes relevance via neural estimators and MI-preserving constraints so affect/identity/style persist [61, 47].

A compact recipe blends projector-first bridges with KL behavior alignment and affect grounding, shared-vocabulary interleaving, tokenizer-side distillation with text-aware decoding, multimodal fusion choices, and decoder-only ASR routing/CTC settings; diagnostics pair IndicSUPERB, VoxEval, and fastabx with contamination audits [60, 7, 4, 77, 14, 73, 62, 61, 47, 36, 22, 45, 58].

7.3 Synthetic and programmatic supervision at scale

Scaling with modest compute relies on frozen backbones plus 1

Token-space pairing scales efficiently: a text-to-token generator reaches 25k tokens/s on a single H800, enabling 600B mixed tokens at 12.5 Hz for unified NTP and streaming-ready interfaces [4]. Streaming CoT is programmatically built by sampling prefixes from speech/transcript/translation triplets with speech-CoT prompts, and transfers across backbones [98]. Open blends (LLaSO, Omni VoiceTextBlender) synthesize millions of instruction pairs for joint SFT and seed duplex chat skills; compute-aware regimens (Slamming) combine dense TTS pretraining with short, verifiable DPO to surpass uncurated real-audio mixes on single GPUs [111, 103, 104], with bridging-style conversions noting TTS-only distribution shifts [3].

Paralinguistic/contextual supervision conditions targets on transcripts plus emotion labels (BLSP-Emo) [7]; EI-CPM prompts GPT-4o with word-level transcripts (WhisperX) and categorical/dimensional emotions to produce filtered CPQAs, expanded by context-free PQA* [37]. These pipelines broaden coverage over affect, dialect, age, identity, and non-speech vocalizations with TELEVAL/IndicSUPERB assessments [112, 36, 113]. Duplex/IRQ supervision scales with synthetic wake words, interruption timing, and MUSAN noise for precise IRQ labels [1]. Preference optimization in Encoder–Adaptor–LLM ASR uses verifiable two-hypothesis rewards with structural validity/correctness/error-type/detail matching and improvement-over-prior, where RF1+RF2+RF5 combinations perform best; large-scale token interleaving seeds such post-alignment [61, 4].

Standardized SLU (IndicSUPERB), acoustic-awareness evaluation (SALMon), tokenizer diagnostics (STAB), ASR-free RAG, and continuous tokenization (Cont-SPT) accelerate iteration, while known limits include partial linguistic coverage, uneven task gains, ABX tooling friction now reduced by fastabx, TTS-only shifts, safety brittleness without adversarial tests, and budget sensitivity/forgetting mitigated by compute-aware blends and rehearsal/DPO [36, 41, 40, 43, 20, 53, 3, 104]. Low-budget SSL pretraining (e.g., HuBERT on 8 GPUs) lowers barriers but demands contamination controls

[65, 58]. Open, auditable recipes combine 12.5 Hz token-space interleaving, streaming CoT triplets, paralinguistic QA curation with strict filters, duplex IRQ supervision, verifiable post-alignment, and instruction blends that reach sub-3-day training on 4 GPUs with 226 ms streaming latency [4, 98, 37, 1, 61, 111, 103, 104, 32].

7.4 Modality alignment, sequence compression, and interface consistency

A stable speech–LM contract uses invariant interfaces and alignment-aware compression: text-aligned discrete semantic tokens (LM-SPT) or a fixed-geometry continuous token space (continuous tokens/flow matching) predicted by the LM, with learned semantic reduction rather than rigid pooling; reconstruction matches a frozen ASR encoder at 25/12.5/6.25 Hz to preserve lexical anchors and paralinguistics for low-latency streaming and improved S2T/TTS [49, 96]. Interfaces are enforced either by adaptor-first bridges with uniform 4× subsampling into the LLM embedding space [3], shared-vocabulary SLMs with a unified discrete inventory at 12.5 Hz [4], or ultra-low-rate BSQ codecs at 6.25 Hz consumed by AR/masked decoders [14]; RepCodec offers information-rich semantic tokens from HuBERT/data2vec for seamless integration [25].

Compression should not erode semantics: fixing 12.5 Hz during interleaved pretraining standardizes targets and shortens sequences, while 4× subsampling reduces prefilling but motivates alignment-preserving compression; at 6.25 Hz, text-aware decoding closes reconstruction–generation gaps and preserves intelligibility/identity when tokenizer and decoder are co-designed, with OT-based regularization validated by STAB/IndicSUPERB [4, 3, 14, 85, 40, 36]. Lexical anchoring holds under shared 12.5 Hz vocabularies and conservative continuous bridges; ultra-low-rate BSQ with text-aware decoding sustains cross-lingual WER [4, 3, 14]. Continuous tokenizers (Cont-SPT) often capture finer semantics/robustness than discrete tokens; decoupled tokenization with multi-token prediction improves information density and speed, while XY-Tokenizer tightens text alignment and acoustic fidelity with speaker-aware generation [20, 80, 75, 97]. A pragmatic blueprint fixes one interface end-to-end (discrete at a fixed Hz or continuous projection), compresses on the perception path, avoids shifting downstream contracts, audits side-effects (rates, downsampling, alignment-guided vs random), and reports vocabulary/rate consistency across pretraining and fine-tuning [22, 20, 27, 19].

7.5 Diagnostics, ablations, and reporting

A rate- and schedule-aware protocol should standardize interfaces (unit tokenizers; SSR-style segmentation/compression aligned to text tokens), fix compute budgets, and quantify alignment, retention, robustness, and forgetting. Evaluation spans zero-shot cross-modal transfer, spoken QA/knowledge robustness under varied audio (VoxEval), and scaling behavior of interleaved SLMs that favor model size over token count while leveraging synthetic data and low-budget SSL pretraining [45, 57, 18, 65, 11]. Core SLU metrics (ASR, SV/SID, LID, QbE, KWS) should be paired with acoustic slices (noise, emotion, speaker, room), speech–text alignment probes, and tokenizer consistency analyses [21, 22, 54, 36, 41]. Recognition scoring uses IndicSUPERB and STAB with consistent toolchains, fixed decoding, and checkpoint averaging [40, 36]; duplex/TTS fidelity is measured by Whisper-v3 WER on LibriTTS-testsetB without interruptions and turn-taking F1 within [0,1.0 s] windows [1].

Tokenizer robustness should connect STAB axes—speaker/context/language invariance, pitch/speed/noise robustness, compressibility, vocabulary utilization—to downstream trends with multi-seed CIs [40]. For AV emotion, ablations on fusion (self vs cross-attention), loss mixtures, SER heads (attention pooling, mean, Query2Emo), modality contributions, token sizes, and encoder depth consistently favor cross-attention fusion and Query2Emo with both VQ pretraining and encoder fine-tuning [73]. Textless ICL diagnostics verify demonstration use (guessing-rate jumps), track attention progression, and standardize utterance lengths; controlled ablations vary language, task, and representation (FBANK vs SSL encoders), with fastabx isolating phonetic separability on IndicSUPERB [67, 53, 36]. Mixed speech–text generation ablates task-set composition, text-augmented prompting, unseen-speaker conversion, and randomized loss sampling [64]; for S2ST, upsample translation data for style, prefer bidirectional training for ASR-BLEU/style, and warm-start from OPT for speed [5].

Encoder–Adaptor–LLM baselines freeze pretrained encoders/LLMs, train tiny (1

Transparent reporting should include public code/configs/models, full data provenance/statistics, explicit task coverage with standardized benchmarks (IndicSUPERB, STAB, SALMon), compute/training budgets, and exact evaluation setups (tokenizers, contextual-bias prompts/lists, few-shot exemplars, decoding policy) [40, 36, 41, 65, 110]. Real-time duplex UIs should report schedules, frame/token rates, bitrates, vocabulary/codebooks, interleaving policies, beam/sampling settings, WER/CER tools, and baselines; AV/duplex specifics include barge-in window, F1 protocol, streaming encoder and fusion strategy, agent bitrate, SER heads, encoder depth, tokenization (discrete vs continuous), and SSL pretraining/fine-tuning status [114, 3, 41, 36, 20, 34, 65]. Linking STAB/ABX intrinsic metrics to downstream outcomes clarifies how tokenizer choices (e.g., PAST) and chunked reasoning policies (STITCH) shape latency–accuracy profiles [21, 46, 40, 36, 48].

8 System Architectures and Generation Strategies

8.1 Cascaded vs end-to-end vs interleaved

As synthesized in the tree-structured schematic in Figure **??**, three design regimes for spoken dialogue systems are contrasted: cascaded ASR→LLM→TTS pipelines that prioritize controllability yet risk error compounding and paralinguistic loss; unified end-to-end SLMs that reduce hand-offs, support duplex interaction, and rely on thin projector bridges with targeted semantic/temporal/affect alignment to stabilize behavior; and interleaved/parallel decoding that structures when text and speech tokens appear to bound latency while preserving style via AR-on-semantics with NAR residual refinement. The figure also foregrounds evaluation dimensions—multilingual SLU on IndicSUPERB, unsupervised ABX discriminability, and syntax probing—that we use to balance semantic competence against paralinguistic fidelity across regimes.

Cascaded ASR→LLM→TTS systems remain attractive for controllability, debuggability, and component swapability, yet they accumulate errors and lose prosody, speaker, and affect cues when collapsing speech to text; they perform competitively on semantic SLU but lag on paralinguistics and limit coverage for unwritten languages [70, 74]. End-to-end SLMs unify perception, reasoning, and synthesis to cut hand-offs and time-to-first-audio, preserve identity and affect, and enable duplex listening-while-speaking without text-mediated error propagation; thin projector bridges that connect frozen speech encoders to frozen text LLMs deliver strong ASR/AST and zero-shot control with 1

Figure 8: Cascaded, end-to-end, and interleaved speech systems

8.2 Generative regimes and efficient backbones

Generation choices span three axes: RVQ-aware hierarchy, acoustic backbone (token-level AR, hybrid discrete/continuous, flow/ODE/diffusion), and parallelism/factorization (multi-token, grouped decoding, codebook reduction) [12, 27, 28, 19]. Grouped decoding and RQ-style factorization shorten horizons by emitting blocks of residual indices under shared attention.

- Hierarchical RVQ with channel-aware scheduling. Coarse-to-fine schedules devote AR capacity to the first codebook and predict residual channels non-autoregressively, matching or exceeding cascades in S2ST while preserving speaker style; phoneme-synchronous stops stabilize timing under greedy or low-temperature sampling, and intelligibility depends more on quantization/codebook design than the downstream decoder [5, 115, 28]. - Masked infilling and parallel masked generation. Iterative masked decoders avoid long AR chains and hierarchical masking overheads; MaskSR performs full-band masked infilling with nine residual codebooks and parallel multi-token prediction, improving speed and speaker similarity, with quality governed primarily by the decoder backbone [116, 28]. - Few-step flow/ODE and diffusion decoders. Conditional flow-matching or diffusion after a semantic/hybrid token LM reconstructs micro-prosody in 10–32 steps and pairs well with AR or masked LMs, enabling ultra-low-rate token interfaces and explicit quality–latency trade-offs [14, 28]. - Multi-token prediction and long-horizon efficiency. Predicting multiple tokens per step (or per masked iteration) reduces decoding steps with minor quality loss; residual heads can share KV caches with the AR base stream and use factorized classifiers to amortize attention [116]. - Runtime optimizations. FastAdaSP merges adjacent-similar KV states on the fly, compounding gains with grouped decoding and few-step diffusion; constant/linear or single-layer aggressive schedules suit different task sparsities [117].

Design guidance: use AR-on-first/NAR-on-rest RVQ with phoneme-synchronous stopping when balancing intelligibility and latency; prefer masked infilling with codebook summation when step count and speaker similarity dominate; attach few-step flow/diffusion under low token rates; combine multi-token/grouped residual decoding with KV-sequence merging; diagnose bottlenecks via quantization/codebook utilization in AR+NAR and decoder quality in masked-parallel settings [116, 28, 14].

8.3 Sequencing and scheduling taxonomy

Sequencing policies orchestrate semantic planning, recognition, and acoustic realization under latency, duplex, and compute constraints. Three families—token-level interleaving, parallel/phase-decoupled generation with reconciliation, and chunked generation—offer tunable control of time-to-first-audio, stability, and fidelity while countering discrete-codec inconsistencies [23]. Robust evaluation should couple large-scale streaming corpora with IndicSUPERB SLU, fastabx ABX discriminability, and STAB tokenizer diagnostics; LM-aware tokenization (LAST), streamable phonetic-acoustic units (PAST), and LM-SPT consistently reduce frame rates and strengthen SLMs, and SpeechRAG provides ASR-free retrieval robustness under high WER [22, 21, 49, 53, 40, 43, 36]. Lightweight bridges enable zero-shot instruction following, contextual biasing, and dialogue with tight budgets [3].

- Token-level interleaving. Alternate LM-aligned semantic tokens with phonetic/acoustic tokens in a single causal stream; fixed-ratio and plan-first variants bound latency and stabilize semantics. Examples include GLM-4-Voice (e.g., 13 text:26 speech with 0.8 s blocks), SpeechComposer's text-then-speech composition, and block-causal interleaving compatible with 12.5 Hz/2 s streaming tokenizers [16, 64, 4, 21, 49, 40, 35]. - Parallel or phase-decoupled with reconciliation. Generate with an explicit text plan, then reconcile text–acoustic consistency using SALMon-style checks and stable tokenizers; shared-vocabulary interleaving scales efficiently and supports dual-resolution speech (e.g., DrVoice) [57, 41, 92, 40, 64, 16]. - Chunked generation with windowed compute and async I/O. Fixed or overlapping windows control memory and latency for single-GPU budgets; 2 s block-causal decoding matches streaming tokenizers, and simultaneous translation uses 320/640 ms chunks with wait-k and context truncation [4, 98, 104, 65]. - Duplex guardrails and stop policies. Combine input/output-side defenses against audio jailbreaks with activation patching and refusal triggers; IRQ-gated interruption exposes an explicit stop token trained for 0.5 s mean delay and evaluated within [0, 1.0 s] [118, 119, 1]. - Intra-decoder compression schedules. Progressively discretize/downsample hidden states (e.g., dMel intensity bins or SSL k-means) and apply layer-wise merge schedules guided by transfer entropy; validate with ABX and IndicSUPERB [54, 39, 53, 36, 117].

Key knobs include read–write geometry (silent reasoning vs spoken emission), chunk size/cadence, delay control (e.g., wait-k), IRQ/stop, and intra-decoder merging. STITCH leverages long audio chunks to interleave silent reasoning with speech; choose chunk sizes aligned with tokenizer segmentation and latency budgets (0.8 s for rapid dialogue, 2 s for stable inversion, 320/640 ms for simultaneous translation), favor LM-aware tokenization (LAST) with stable units, and apply FastAdaSP merging under any schedule [48, 22, 117, 40, 63, 98]. IndicSUPERB evidences language-specific tuning benefits for ASR/LID and speaker-focused gains from SSL pretraining; full-duplex training supports multi-turn production use [36, 114].

Application templates: low-latency assistants combine SpeechRAG with adapter-based SLMs for zero-shot instruction following and multilingual SLU, using fixed-ratio interleaving with short speech blocks and IRQ-gated stops (optionally 5:5 turn-level guidance), deployable via ESPnet-SpeechLM [43, 3, 113, 36]. For streaming S2ST, pair wait-k with language-pair–tuned chunks; direct speech-token emission avoids text→MT→TTS delays and attains strong BLEU/BLASER 2.0 on CVSS-C, and 2 s text-then-speech blocks add stability; stress-test tokenizers with STAB and adapt per language family [42, 36, 40]. Where strict text fidelity is required, plan-first schedules that predict text before speech anchor acoustic realization; compute-constrained deployments should overlay FastAdaSP-style merging [22, 33, 117].

8.4 Textless S2S and two-branch designs

Two-branch and textless S2S architectures separate a semantic path for linguistic planning from an acoustic path for prosody, timbre, and style, enabling speech-in/speech-out without text intermediates while preserving paralinguistics and supporting unwritten or low-resource languages. The BoSS taxonomy argues for keeping speech-unit embeddings inside the SLM to jointly model affect and prosody with semantics [47]. A unified S2ST LM that predicts HuBERT-based semantic units and EnCodec acoustic streams in a single decoder attains multilingual, bidirectional translation with speaker-style preservation by autoregressing the most informative stream and refining residuals non-autoregressively; syllable-synchronous semantic tokens further compress planning, and unit-vocoder stacks demonstrate textless dialogue continuation [5, 15, 6].

Streaming duplex variants couple a listening branch (SSL encoder + projector) and a speaking branch (decoder-only TTS + vocoder) with an explicit interruption head; middle-layer fusion preserves near-baseline intelligibility and enables barge-in, whereas early fusion degrades WER [1]. When robustness is paramount, a lightweight textual intermediate can be introduced programmatically—e.g., a transcribe→translate Chain-of-Thought that keeps end-to-end training and bounded-lag control [98].

Design choices: pace content with low-rate LM-aligned semantic tokens (LM-SPT or LAST at 25/12.5/6.25 Hz) to reduce context pressure while leaving identity and micro-prosody to the acoustic branch and decoder; favor single-decoder textless S2ST over HuBERT/EnCodec units for tight latency and joint translation–synthesis; interleave listening embeddings at middle layers and regulate emission with IRQ supervision for sub-second synchronization; when strict fidelity or cross-lingual robustness is required, fall back to a plan-first transcribe→translate schedule without abandoning end-to-end learning [49, 22, 11, 1, 98, 71].

9 Expressivity Paralinguistics and Style Control

9.1 Prosody, identity, and emotion modeling

Expressive SLMs such as ProsodyLM, GOAT-SLM, and SeamlessExpressiveLM aim to co-model lexical content with prosody, identity, and affect under bitrate/latency budgets, yet capability probes still surface gaps: BoSS finds weak emotion alignment and limited age-aware adaptation in omni/SLM stacks, and TTS comparisons show an expressivity–identity trade-off—Bark increases F0/rate variance but degrades speaker stability relative to VITS, with only modest gains from scaling [47, 2]. We distill these themes in the hierarchical map shown in Figure 9, which organizes the space along three top-level branches: (1) factorized architectures that preserve global style while controlling local, word-aligned prosody; (2) lightweight, streamable control surfaces and signal-aligned supervision—including adapters, interpretable heads, audio–visual fusion, and phonetic–acoustic tokenizers—that disentangle affect from identity; and (3) practical scaling with frozen foundations, interleaved multitask training, new resources/diagnostics, and rate-aware decoding that ties expressivity to lexical timing. The structure foregrounds the expressivity–identity trade-off, promotes disentangled semantic/acoustic pathways, and situates training/evaluation recipes with respect to BoSS-style capability gaps.

Consistent with branch (1) in Figure 9, a pragmatic factorization separates global style from local prosody: inject an utterance-level embedding via adaptive normalization to preserve timbre/affect and add word-synchronous duration/F0 controls for emphasis, rhythm, and pitch; this improves human judgments, supports cross-lingual transfer without same-text pairs, and aligns with two-branch SLMs that keep a low-entropy semantic stream distinct from an acoustic realization stream [70].

Aligned with branch (2), control surfaces should be lightweight and streamable. Short enrollment prompts provide global levers for identity/emotion/intonation at ultra-low token rates but risk identity drift as expressivity rises [2]. Label-efficient adapters and interpretable heads (e.g., BLSP-Emo, EMOVA, Dual-Adapter SLMs) improve affect control, while audio–visual fusion helps when acoustics alone are ambiguous [7, 72, 120]. Signal-aligned supervision and light bridges between pretrained speech and text foundations are effective: time-stamped categorical/dimensional labels and phonetic-acoustic tokenizers (PAST) strengthen representations, and thin projectors/adapters add 1

Reflecting branch (3), scaling recipes converge on initializing from strong speech/text foundations, freezing them, and training small adapters on interleaved speech–text multitask data (dialog, continuation, QA, ASR/AST, contextual biasing), prioritizing model size over token count and exploiting synthetic data [57, 3]. New resources and diagnostics broaden training/evaluation: IndicSUPERB's Kathbath corpus across 12 languages and six SLU tasks, LM-SPT for LM-aligned low-rate semantic tokens, PAST for streamable phonetic–acoustic units, SpeechRAG for ASR-free QA in high-WER settings, and fastabx for ABX discriminability [36, 49, 21, 43, 53]. Rate-aware decoding helps tether expressivity to words: multi-token prediction accelerates generation (12×), reduces WER, and enables word-aligned prosody controls [71, 75].

Operationally—summarized by the hierarchy in Figure 9 and elaborated below—the above principles yield the following actionable guidelines:

- Factorize style and prosody: inject a global embedding via adaptive normalization and add word-synchronous duration/F0 controls for emphasis/rhythm/pitch; this also supports cross-lingual transfer without parallel text [70].
- Use short enrollment prompts for identity/affect and disentangle semantic vs acoustic streams so lexical tokens remain predictable while acoustic channels carry timbre/prosody; monitor identity drift as expressivity increases [2].
- Add label-efficient emotion adapters and plug-in style heads—BLSP-Emo, EMOVA, Dual-Adapter SLMs—and validate coverage with MAD Speech; leverage audio–visual fusion when available [7, 121, 72, 120].
- Ground affect with time-aligned categorical/dimensional labels and phonetic-acoustic units to teach when prosody should change, improving empathetic QA and speech-in/speech-out reasoning [37, 21, 36].
- Evaluate along expressivity–identity axes with MOS/UTMOS, speaker similarity, SER, diarization/verification, and WER/dWER; BoSS-style testing exposes persistent gaps [47].

9.2 Tokenization for expressivity

Expressivity-aware tokenization should expose controllable prosody, identity, and non-verbals while keeping the semantic path low-rate and predictable. Empirical probes reveal leakage: HuBERT-based units at 50 Hz retain duration/emphasis, and deduplicated content streams still carry tempo cues; coarse syllable tokens flatten prosody and lose identity after k-means, arguing for explicit style streams or conditioning and a capability blueprint with multiple inventories—lexical (V_L), affective (V_AC), contextual-dynamics (V_CD)—plus first-class non-verbal tokens [122, 47]. Text-aligned prosody tokens provide direct, low-rate handles: aggregating shallow ASR features at word spans preserves prosody and speaker traits for high-similarity span-level editing [78]; prosody-sensitive discretization—training k-means on emotional speech with light temporal smoothing—improves responsiveness to emphasis while reducing speaker leakage, though energy cues may need explicit conditioning due to loudness normalization [39].

Multi-stream and RVQ-aware designs disentangle content from style and lower next-token entropy on the semantic path. A unified tokenizer that concentrates linguistic content in RVQ-1 and paralinguistics in later RVQ layers achieves low WER with high speaker similarity; mixed local tokens plus a compact global style vector similarly preserve timbre; audio–visual discrete tokens add prosody–face coupling and protect emotion cues under masking [83, 29, 73]. Codec alignment matters: language-aware codecs improve identity and MOS-Q/S at matched bitrates relative to EnCodec-like baselines, whereas naive mixing of HuBERT with EnCodec increases style leakage, motivating losses that penalize cross-stream bleed or pairing discrete lexical backbones with compact global style vectors [12, 122].

Low-rate constraints favor hybrid, multi-rate tokenization—keep a compact semantic/phonetic stream and allocate prosody/timbre to a slightly higher-rate stream—or adopt continuous tokenizers to preserve low/high-frequency cues, tuning frame rates per language [20, 81, 56, 13]. For editing and S2ST, content/style separation is especially useful: DualSpeechLM uses high-level USToken for understanding with acoustic tokens for generation, StreamUni segments policy/translation stages with speech-CoT, and STAB supplies tokenizer diagnostics predictive of downstream quality [44, 98, 40]. LM-aware training (LAST) aligns discrete speech units with pretrained text LMs, enabling a single

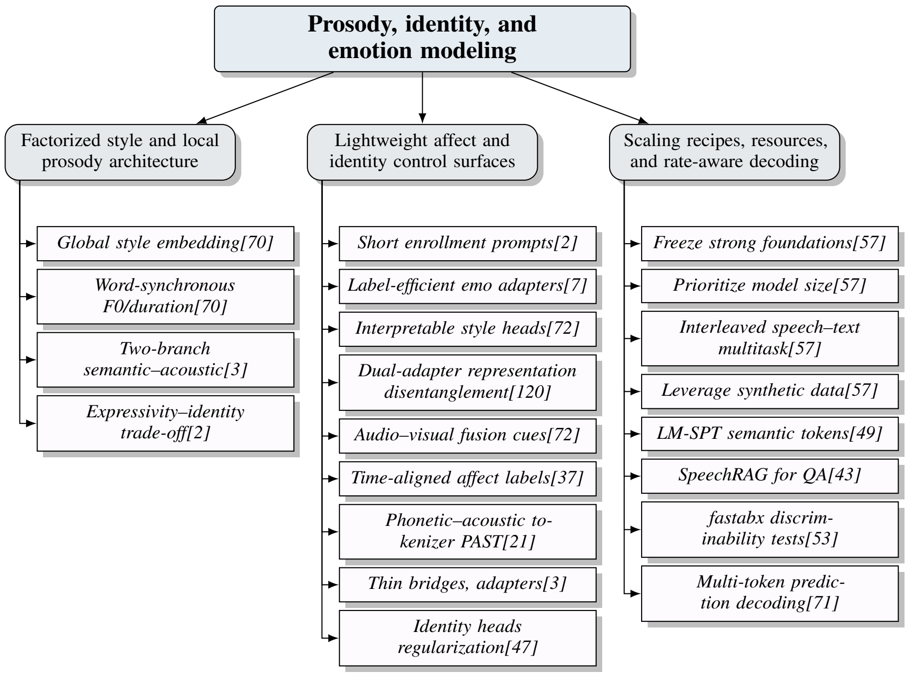

Figure 9: Figure: Hierarchical summary of prosody, identity, and emotion modeling. Top-level branches emphasize: (1) factorized architectures that preserve global style while controlling local, word-aligned prosody; (2) lightweight, streamable control surfaces and supervision that disentangle affect and identity with adapters, interpretable heads, AV fusion, signal-aligned labels, and phonetic–acoustic tokenizers; and (3) practical scaling with frozen foundations, interleaved multitask training, new resources/diagnostics, and rate-aware decoding that ties expressivity to lexical timing. The structure highlights the expressivity–identity trade-off, promotes disentangled semantic/acoustic pathways, and aligns training/evaluation recipes with BoSS-style capability gaps.

LM across speech and text, while PAST, LM-SPT, SpeechRAG, SLM bridging, and IndicSUPERB contribute streamable units, LM-aligned semantics, ASR-free retrieval-generation, and multilingual SLU coverage [22, 21, 49, 43, 3, 36].

- Prefer word-aligned prosody tokens (duration/F0/energy) for contrastive focus, emotion, stress, and cross-utterance consistency; train discretizers on emotional speech with light smoothing to boost prosody sensitivity and limit identity leakage; consider continuous tokenizers when discrete codes are too lossy [71, 39, 20].
- Use multi-stream RVQ-aware interfaces: allocate RVQ-1 or local tokens to content and later layers or a compact global vector to style; encourage disentanglement with masked-channel training and validate via RVQ-1:8 resynthesis (WER, speaker-sim) and SALMon consistency [12, 41, 36].
- When adopting unified codecs, redistribute paralinguistic information to prevent leakage; language-aware codecs are promising defaults [12].
- Do not rely on deduplication to purge prosody; explicitly audit residual tempo/duration coupling when mixing semantic and acoustic units [122].
- Mitigate expressivity loss at low frame rates with hybrid multi-rate or continuous/context-aware tokenizers; if using syllable tokens, add an auxiliary prosody channel and consider diffusion decoding; tune frame rates per language [20, 71, 81, 56, 13].
- Integrate non-verbals and, when possible, audio–visual streams; pair with explicit paralinguistic metadata or QA targets distilled from emotion annotations to strengthen empathetic, context-aware reasoning [21, 47, 37, 26].
- Report style-sensitive metrics alongside intelligibility: Sim-O, MOS-Q/S, MUSHRA; SALMon attribute checks; MAD Speech diversity; ABX-style closeness; prosody-aware

diagnostics (TER under emphasis, span-edit success); and ALLM judges for affect, pace, emphasis, pitch, and non-verbals [41, 121, 46, 123].

9.3 Control interfaces and prompts

Four complementary interfaces govern paralinguistics—text instructions/attribute tags, acoustic prompting with strength scheduling, sequence-level prompt placement, and direct token/control-token manipulation—augmented by decoder-side knobs. Efficacy depends on locality, representation stability, and codec design: discrete codec inconsistencies map similar audio to multiple sequences, causing repetitions/omissions unless mitigated, and compression-first codecs (e.g., EnCodec) can harm semantic fidelity relative to semantic-aware designs like X-Codec [23, 24, 46].

Text instructions and routing provide low-latency, broadly applicable control: instruction-only empathy emerges from alignment, task/language/bitrate routing uses <trans>/<gen>, <src>/<tgt>, and stream-ID tokens <j>, and time-stamped SER metadata can improve affect-aware reasoning when gated against conversational context; a unified surface via <enroll-speech> plus x-vectors steers identity/style across tasks [7, 5, 37, 64]. Acoustic prompts efficiently seed timbre and style, but homogeneity and placement matter: single-speaker, single-emotion prompts or segmented/gated conditioning outperform heterogeneous concatenation; prompt strength scales with the prompt-to-source length ratio, with 0.30 maximizing identity transfer in cross-lingual S2ST without harming translation, and prompt-guided decoding retains steerability at 6.25 Hz [122, 124, 14]. Placing the style prompt in a dedicated segment adjacent to the target with explicit boundaries from the textual plan reduces interference and stabilizes local controls [64].

Direct manipulation unlocks precise trade-offs and edits. Stream-ID tokens select residual codec streams to dial prosodic richness against latency/bandwidth, source-style prompts preserve timbre across S2ST, decoder temperature in flow-matching acts as a global expressivity knob, and role prompts can be composed with attribute tags and acoustic prompts for persona synthesis; when time-stamped SER exists, metadata can bias affect regionally if conflicts are mitigated [5, 96, 7, 37]. Fine-grained word-synchronous controls—pause insertion at punctuation, duration adjustments via alignment ratios, F0 transfer with normalization/interpolation—enable localized edits under strict budgets and complement global prompting [70]. For deployment, IndicSUPERB facilitates multilingual SLU, and STITCH lets models interleave silent reasoning with speech to match latency while improving math reasoning [36, 48]. Evaluations should couple control-compliance with semantic integrity—BLEU/BERTScore for S2ST under fixed prompt ratios/stream-IDs, instruction-only empathy with alignment-matched prompts, word-aligned metrics for pause/duration/F0 edits—and, under ultra-low rates, audits of reconstruction–generation gaps while sweeping prompt conditioning and decoder temperature [7, 70].

- Text instructions and routing: standard empathy prompts; <trans>/<gen>, <src>/<tgt>, and <j> tokens for task/language/bitrate control [7, 5].
- Acoustic prompting: use single-speaker, single-emotion prompts or segmented/gated conditioning; set ratio 0.30 for strong identity transfer; employ prompt-guided decoding at ultra-low rates [122, 124, 14].
- Placement/boundaries: dedicate a prompt segment adjacent to the target speech and separate content/style intents with explicit boundaries [64].
- Local edits: pause, duration, and pitch controls aligned to words under tight budgets [70].
- Decoder-side expressivity: increase flow-matching noise temperature to raise prosodic diversity [96].
- Metadata prompting: inject time-stamped emotion labels when available and gate or down-weight on conflict [37].

9.4 Evaluation and watermarking

Assessment should jointly probe instruction-following for style, paralinguistic fidelity, prosodic diversity, and social cue handling, while enabling provenance checks. Audio-aware LLMs scale open-ended judging: on StyleSet, Gemini-2.5-Pro correlates with humans on voice-style instruction-following and role-play and consistently ranks GPT-4o-audio strongest on style

[123]. Broader suites complement automatic and LLM-based judging: LLaSO-Eval scores emotion/gender/age/accent with closed-set metrics and uses GPT-4o judges with abstention auditing; TELEVAL decouples explicit semantics from paralinguistics and implicit semantics in dialogue to score empathy, dialect-consistent generation, and age-aware style; BoSS fuses GPT-4o style judgments with Emotion2Vec-based audio scoring to reveal dialect and affect controllability gaps [125, 126, 47].

Prosodic diversity can be tracked distributionally by clustering pitch/voicing/energy and comparing synthesized versus training distributions with NDB/JSD; acoustic BPE increases diversity without extra labels [82]. MAD Speech quantifies diversity across identity (voice, gender) and expressivity (emotion, accent, background noise), reporting e.g., SoundStorm > AudioLM on several facets, with Best-of-K reducing and higher temperature increasing diversity; SALMon adds attribute-consistency and text-match checks [121, 41]. Behavioral probes capture spontaneity/domain sensitivity: Bark produces more fillers and longer pauses than VITS and scores higher on read but not conversational speech, indicating domain-specific trade-offs [2]. For provenance and edit detection, SSR-Speech embeds frame-synchronous watermarks (50 Hz) that detect edited regions with 99.9

A reporting checklist improves comparability: include SALMon-style acoustic factors (emotion, background noise, speaker identity, room acoustics); tokenizer/model diagnostics predictive of downstream performance (e.g., STAB) and multi-task benchmarks such as IndicSUPERB; and rigorous provenance (languages, hours, contributors, geographic coverage, conversational setup), following practices from Taiwanese Mandarin spoken LLMs [41, 40, 36, 114]. Aligning intrinsic and extrinsic evaluations—e.g., correlating STAB metrics with IndicSUPERB SLU outcomes across 12 languages—strengthens interpretability and cross-task comparisons, while ALLM judges, TELEVAL, LLaSO-Eval, BoSS, and StyleSet jointly triangulate style controllability versus stochastic diversity [40, 36, 123, 126, 125, 47].

10 Real Time Streaming and Duplex Dialogue

Real-time agents must coordinate perception, planning, and synthesis across concurrent input/output channels under strict latency, bandwidth, and compute limits. A systems-first view treats latency as the primary constraint: decompose end-to-end delay into capture, buffering, perception/prefill, time-to-first-audio, and steady-state generation; then optimize three levers—(i) chunking and read–write schedules that cap look-ahead while preserving coherence, (ii) low-rate, streamable token interfaces that shrink conditioning length and bandwidth, and (iii) budgeted inference that cuts prefill and decoding cost without sacrificing intelligibility or style. These principles structure the subsequent subsections on latency-aware design, streaming policies, duplex turn-taking, and architectures.

10.1 Latency-aware design

Interactive stacks benefit from streamable tokenizers and dual-stream decoders that avoid heavy compression or cascaded MT/TTS, thereby reducing first-packet delay and average latency across ASR, S2ST, and duplex chat [20, 21, 54, 42, 127]. In practice, three levers dominate.

Chunking, buffering, and read–write schedules. With causal tokenizers, short interleaved cycles deliver fast onset: GLM-4-Voice starts speaking after 10 speech tokens with a text-first plan; aligning assistant text insertion to the first speech chunk index k$_{s}tartandsetting_{S}C = 5at12.5Hz(400ms)balancessemanticplanningandresponsivenessinduplexchat$[16, 35]$.ForSimulST, fixedword\text{-}lagkwith320\breve{}640mschunksexposeslatency\breve{}qualityfrontiers, whilesemantictruncatio$

Low-rate tokenization and streamable interfaces. Lower cadence reduces both prefill and per-step cost. A 12.5 Hz discrete interface with block-causal attention in tokenizer/decoder is a robust default for interleaved training and streaming synthesis, sustaining 25k text-to-token throughput on H800 [4]. Ultra-low-rate codecs such as TaDiCodec (6.25 Hz, 0.0875 kbps) with few-step diffusion decoding achieve RTF 0.12–0.29, enabling compute- and bandwidth-constrained operation; training-free, chunk-wise expansion supports online streamability with short windows (T$_{c}hunk1s, T_{s}hift0.4s)andcompactprimarycodebookstoshortenLM\text{-}facingsequences$[14, 63]$.Coarser, syllabicsegmentationprovideslinear\text{-}timepacingcompatiblewith5\breve{}12.5Hzstreams$[94].

Budgeted inference: prefill reduction and fast decoding. Prefill dominates long-context latency; capping it requires shorter conditioning and KV-efficient decoders. Step-size distillation reduces diffusion steps (e.g., 100→4) while maintaining WER/DNSMOS under real-time budgets [13]. LSLM meets duplex constraints with a lightweight discrete interface, a compact decoder (106M), and a streaming encoder (34M), using AR top-p sampling on live embeddings for rapid interruption handling [1]. On recognition paths, CTC sparsity enables online prompting by emitting only salient encoder states [62]. Aligning tokenizer and decoder interfaces (e.g., 12.5 Hz, block-causal attention, chunked emission) simplifies end-to-end control [4].

First-chunk policies and reporting. Deterministic read–write alternation with short text plans and small speech blocks minimizes time-to-first-audio; GLM-4-Voice's 10-token onset and $_{S}C = 5at12.5Hzareconcretetargets[16,\ 35].Reportwall\text{-}clockeffectsandschedulesjointly: fixedword\text{-}lagkandchunksizesinSimulST[98]; token\text{-}to\text{-}speechthresholdsandinterleavingratiosinduplexchat[16]; w$

Tooling and benchmarks. ESPnet-SpeechLM standardizes data/pretraining/inference [113]. Vox-Eval probes spoken QA/reasoning under varied acoustics [45]; SALMon adds acoustic awareness and evaluation robustness [41]; Taiwanese Mandarin spoken LLMs demonstrate real-time, full-duplex multi-turn behavior [114]. IndicSUPERB (1,684 h; 12 languages) highlights SLU gains with language-specific SSL; LM-SPT and PAST yield semantically aligned, streamable tokens; SpeechRAG enables ASR-free retrieval-generation under high WER [36, 49, 21, 43]. For streaming TTS/S2ST, prefer 12.5 Hz discrete/semantic tokens with block-causal attention and interleave textual planning; continuous tokens (Cont-SPT) and diffusion-assisted decoding sustain quality at modest rates [20, 54, 42, 13].

10.2 Streaming policies for ASR/S2ST

Streaming policies decide when to read audio and write tokens, coupling encoder chunking, tokenizer cadence, and decoder scheduling. Four recurring strategies—wait-k, read–write alternation, incremental decoding with request-more-audio control, and duration-synchronous generation—integrate naturally with streamable encoders (Emformer, HuBERT/data2vec), fixed-rate discrete tokenizers (e.g., RepCodec, SpeechTokenizer), and unified SLM decoders [25, 83].

Wait-k and prefix-to-prefix. BESTOW samples KU(3,12) during training and sets task-specific K at inference (e.g., K=6 for ASR), combining look-ahead L=4 with pre-decision steps P=8 and supporting uni/bidirectional streaming to hit latency targets [84]. SimulS2S-LLM derives a test-time wait-k from CIF prompt length, stabilizing prefix decoding via frame-synchronous CTC, incremental beam search, and shallow-fused n-gram LM [93]. In shared tasks, EOS-to-request-more-audio realizes incremental reading with a latency multiplier for soft wait-k; StreamUni further gates generation by transcription growth and prunes served history via semantic truncation [128, 98].

Read–write alternation. In plan-before-speak assistants, emission after every R text tokens yields deterministic cycles and prompt audio onset; first-chunk latency decomposes as LLM(R)+TTS(W)+FlowMatching(W)+Vocoder(2W), exposing tunable trade-offs over R and W [129]. StreamUni complements these cycles with transcription-driven gates and semantic truncation and reports AL/LAAL/StreamLAAL using mWERSegmenter [98].

Incremental decoding with request-more-audio. Textless streaming ST couples an Emformer encoder (e.g., 20/4 or 32/6 segment/right-context) with an RNN-T decoder that streams semantic tokens at 25 Hz with blanks and beam search [42]. EOS-driven request-more-audio loops and latency multipliers stabilize decoding under noise and long-tail inputs; frame-synchronous CTC with shallow fusion maintains bounded lag [128, 93].

Duration-synchronous generation. SyncSpeech aligns emission to predicted durations with look-ahead q=1, enabling true online dual-stream decoding; unlike full-context NAR flow/diffusion, it supports continuous generation under read–write/wait-k [127].

Integration and diagnostics. Emformer chunk geometries provide low-latency features; CIF prompts expose label-synchronous cues for test-time wait-k; RNN-T at 25 Hz yields fixed-rate streams; R and W standardize first-chunk auditing [42, 93, 129]. Report AL/LAAL/StreamLAAL with mWERSegmenter, the LLM/TTS/Flow/Vocoder first-chunk decomposition, and use incremental beam with frame-synchronous CTC and shallow fusion as a stability reference [98, 93, 129].

Design guidelines and complementary advances. Under tight budgets, train with sampled K and small look-ahead (e.g., L=4), pick task-specific K at inference, pair with streaming encoders and incremental CTC beam search, and exploit accessible SSL and codec front-ends [128, 65, 130, 127]. Standardize first-chunk reporting via LLM/TTS/Flow/Vocoder, tune R,W jointly, and favor duration-synchronous emission for continuity [129, 127]. Evaluation should combine IndicSUPERB SLU, VoxEval, tokenizer assessment (STAB), and SALMon stress tests [36, 45, 40, 41]. Robustness improves with dMel tokens for training-free resilience, mitigation of codec DRI to prevent repetition/omission, and alignment-guided sequencing to suppress infinite silences [54, 23, 28, 115].

10.3 Duplex and turn-taking

Full-duplex agents must listen while speaking, support barge-in, govern overlap, schedule backchannels, detect interruptions, and decide turn boundaries via learnable streaming policies realized by duplex S2S architectures that fuse user/agent audio, turn-level text planning, and streaming chain-of-thought [34, 35, 98]. Three paradigms dominate: token-level control in a unified decoder, direct modeling of turn events without external VAD, and distributionally grounded scheduling that matches human turn statistics.

Barge-in and interruption detection. Lightweight interruption heads/tokens enable sub-second reactions. A listening-while-speaking decoder with an IRQ token holds low probability during normal speech and spikes on true interruptions until sampled, achieving interactivity F1 of 95.50

Overlap handling and floor control. Learned overlap caps regulate co-speaking rather than eliminating it. SLIDE inserts short silences to bound predicted overlap (0.3 s when forecasts exceed 0.6 s), producing human-like IPU/pause/gap/overlap distributions and outperforming cascaded baselines [131]. VAD-based diagnostics show generated dialogues can match human distributions even before large-scale augmentation, indicating that streamable scheduling provides a strong prior [132]. For safety-critical or embodied agents, predictable half-duplex remains effective and well-rated [133].

Backchanneling and non-verbal cues. Explicit backchannel policies tied to internal state improve grounding; schedule backchannel tokens under overlap caps and map conversational states to motor behaviors in embodied systems (e.g., listening→nodding, speaking→head motion, idle→breathing) to stabilize floor exchanges [131, 133].

Policy learning vs inference-time control. Next-token–level supervision internalizes turn-taking beyond thresholds. NTPP jointly predicts next-token pairs to model overlaps, backchannels, and interruptions without external VAD; an interruption suite with 400 synthetic scenarios evaluates "speak-up" and interruption responses, and channel swapping validates speaker independence [134]. TurnGuide inserts brief plan-first text just before assistant speech, preserving duplex dynamics and improving coherence, with Candor correlations on IPU/pause/gap/overlap comparable or superior to token-synchronous baselines [35].

Evaluation methodology and training signals. Use pyannote.audio VAD to compute time-normalized per-channel IPU/pause/gap/overlap counts/durations; closeness to human distributions diagnoses scheduling quality [131]. For interruptions, report barge-in success, end-to-end interruption latency, and interactivity F1 under clean/noisy regimes, apply standardized false-alarm rules, stress-test speaker independence via channel swapping, and avoid contaminated test sets [41, 58, 134]. Style–role suites (e.g., StyleSet) cover sequential turn-taking but not overlap/barge-in; duplex-native evaluations and turn-level correlations on conversational corpora (e.g., Candor) fill the gap [123, 35]. Targeted data—barge-in augmentation (truncate speech, insert 0.64 s silence), synthetic interruption suites, inference-only bootstraps (end-of-turn thresholds, silence counters)—combines with integrated policies (overlap caps, IRQ heads, token-pair objectives) and direct duplex S2S fusion to yield natural alternation, timely backchannels, and reliable barge-in under noise with modest data [34, 3, 44]. Operational reporting should include dataset provenance/scale (e.g., IndicSUPERB 1,684 h/12 languages), SLU coverage, tokenizer quality (STAB), and duplex performance under SALMon-style perturbations with contamination-checked holds [36, 40, 41, 114].

10.4 Architectures for streaming and duplex

Deployable stacks share three strata: (i) streamable perception/tokenization that emits low-latency, label- or chunk-synchronous cues; (ii) duplex-aware decoders that interleave listening/speaking with shared caches and channel-aware control; and (iii) chunk-aware synthesis that renders partial acoustic tokens with bounded look-ahead. Representative systems include SALM-Duplex, NTPP, SyncSpeech, Ichigo-style unified decoders, and Stitch, which interleaves unspoken reasoning with spoken chunks to "think while talking"; other paths adapt text LLMs (e.g., TWIST) or unify understanding/generation via dual speech tokens and Chain-of-Condition training (e.g., DualSpeechLM) [48, 38, 44]. A schematic overview is provided in Figure 10, which summarizes deployable streaming-duplex SLM stacks as three coordinated strata and makes explicit the dataflow across them: arrows indicate the progression from low-rate, streamable tokens into interleaved, duplex-aware decoding (e.g., middle fusion, shared KV, and turn planning) and onward to chunked synthesis (e.g., dual-stream TTS and fixed-size playback). By enforcing duration/span-synchronous schedules that keep first-audio low, the depicted pipeline supports "think-while-talking" and responsive barge-in under unified, rate-aware scheduling, mirroring the design space surveyed here.

Streamable encoders and tokenizers. Text-initialized decoders with block-causal streaming attention in the speech tokenizer and block-wise 2 s speech decoding support online S2S under a single vocabulary; training-free chunk-wise expansion further eases duplex deployment [4, 17]. For SimulST, streaming CoT over an LSLM conditions chunk-wise transcription, maintains a truncation queue, and generates translations under delay-controlled rules, achieving state-of-the-art SimulST/StreamST and transferring across base LSLMs [98].

Interleaved, duplex-aware decoders. A listening-while-speaking LM fuses a streaming vq-wav2vec encoder with a decoder-only TTS stack via middle fusion at every block, balancing speech quality and duplex responsiveness (WER 4–4.5

Chunk-aware, low-look-ahead synthesis. SyncSpeech emits duration-synchronized updates in a dual-stream decoder, avoiding full-context NAR passes [127]. Stitch exploits playback slack by chunking language and acoustic decoders, sustaining simultaneous thinking-and-talking under fixed chunk sizes [48]. The same chunking idiom composes with block-causal tokenizers and 2 s decode windows to keep first-audio low and flow continuous [4].

Turn-aware I/O interleaving. TurnGuide interleaves user/assistant channels at the chunk level ($_{S}C = 5at12.5Hz)withUID/AIDmarkers, preservingper\text{-}channeltimingandenablingbarge\text{-}in/overlap; assistantturnsalig$

Blueprint. Perception: select streamable encoders/tokenizers (block-causal tokenizer + 2 s decode windows or training-free chunk expansion). Core: use a single decoder with shared KV, middle fusion for listening, channel embeddings and pair-wise masks for overlap control, and brief text "plan" bursts interleaved with speech. Synthesis: adopt unit/flow/diffusion decoders with duration- or span-synchronous schedules to minimize look-ahead [127]. For SimulST/StreamST, layer streaming CoT and delay-controlled rules on the same backbone [98]. These ingredients underpin SALM-Duplex, NTPP, SyncSpeech, Ichigo-style stacks, and Stitch-like thinking-and-talking, and have been exercised in real-time assistants with low WER, high interactivity F1, and responsive barge-in under unified, rate-aware scheduling [1, 34, 134, 48, 4].

11 Datasets Benchmarks and Tasks

This section consolidates datasets, benchmarks, and task formulations spanning ASR/SLU, S2TT/S2ST, TTS/resynthesis, paralinguistics, audio–visual speech, and duplex dialogue. We systematize corpora by modality, domain, and language coverage; outline common construction/split conventions and tokenization settings; map tasks to datasets/metrics; and compile benchmark suites with multilingual, robustness, streaming, and reproducibility considerations [46, 121, 41, 36, 40].

11.1 Core corpora and data taxonomy

Modality and domain. Resources cover time-aligned ASR/SLU speech–text; S2TT and S2ST (textless and text-conditioned); TTS/vocoder and unit/codec resynthesis; interleaved speech–text instructions for mixed-token training; paralinguistics/speaker; and synchronized audio–visual speech, mapping information types (phonetic, semantic, paralinguistic/prosody), temporal scales, and acoustic conditions to evaluation pathways such as ASR, SV/SID, LID, QbE, and KWS [46, 19, 59]. Read/studio speech supports phonetic probing and high-fidelity resynthesis; conversational/in-the-wild captures overlaps/noise; expressive/emotional material targets prosody/affect; audio–visual corpora probe alignment and fusion robustness [121, 41]. ABX analyses quantify phonetic/speaker/room invariances with efficient tooling [46, 53].

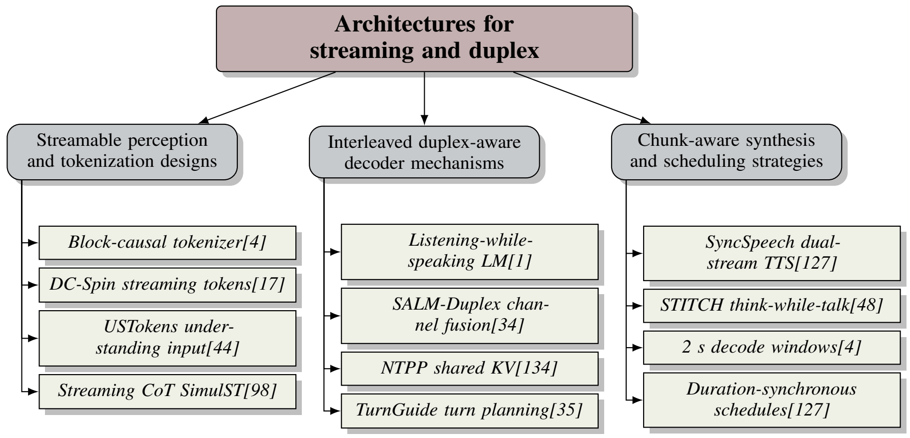

Figure 10: This figure summarizes deployable streaming-duplex SLM stacks as three coordinated strata: (1) streamable perception/tokenization that aligns speech with text cues; (2) duplex-aware decoders that interleave listening and speaking with shared caches and channel-aware control; and (3) chunk-aware synthesis that keeps first-audio low via duration/span-synchronous schedules. Arrows indicate data flow from low-rate, streaming tokens into interleaved decoding (e.g., middle fusion, shared KV, turn planning) and onward to chunked synthesis (e.g., dual-stream TTS, fixed-size playback), enabling think-while-talking and responsive barge-in under unified scheduling.

Language coverage. Corpora span English-only, multilingual, regional families (e.g., Indic), and low-resource settings. IndicSUPERB contributes 1,684 h across 12 languages with six SLU tasks and standardized splits, where language-specific SSL markedly outperforms FBANK, especially for LID and SV/SID [36]. fastabx accelerates cross-lingual ABX probes [53].

Tokenization stacks. LM-aware tokens (LAST), LM-aligned semantic distillation (LM-SPT), phonetic–acoustic tokens (PAST), dMel, and continuous tokenizers (Cont-SPT) provide compact, robust, and streamable interfaces for joint speech–text modeling with improved fidelity/latency [22, 49, 21, 20, 54].

Representative corpora. - English read/conversational: LibriSpeech/LibriLight/LibriHeavy, GigaSpeech; Hub5'00 for conversational evaluation; decoder-only ASR often pairs 5k/2k BPE with LibriSpeech/Fisher text for text-only LM updates [62]. - Multilingual ASR: Common Voice, VoxPopuli, AISHELL, MLS; FLEURS supports tokenizer assessment (STAB) [40]. - Production-style multilingual ASR: MLC-SLM 2025 (1.5k h, 11 languages) with balanced dev; paired speaker sets (CN-Celeb1/2, VoxBlink/2) for embeddings [61]. - S2TT/S2ST: CoVoST2, Europarl-ST, Fisher Es–En; CVSS-C/T for waveform-to-waveform S2ST and style preservation; Heroes/Mined Audiobook for dubbing/audiobook expressivity [5, 42, 70, 98]. - TTS/resynthesis: LibriTTS, VCTK, LJSpeech, Hi-Fi TTS, CMU Arctic for unit/codec decoders, intelligibility (WER/dWER), speaker similarity, and naturalness [22, 3, 11]. - Interleaved speech–text: LibriLight, VoxPopuli, TED-LIUM, People's Speech, Spoken Wikipedia, MLS, plus large text and synthetic speech tokens for unified next-token training [4, 57, 20, 54]. - Speaker-centric: VoxCeleb1/2, CN-Celeb1/2, VoxBlink/2; BoSS adds dialect and non-verbals for beyond-semantic evaluation [61, 47]. - Emotion/prosody: IEMOCAP, RAVDESS, CREMA-D, DFEW, Aff-Wild2; ESD-zh and BoSS non-verbals; ProsodyLM targets word-level prosody [71, 47, 40]. - Audio–visual: VoxCeleb2, GRID, LRS3; AV emotion (RAVDESS/CREMA-D/DFEW/Aff-Wild2); VQ-MAE-AV and BLSP-Emo exemplify multimodal pretraining and empathetic generation [73, 7].

ASR-adjacent stacks. Transcript-free SpeechRAG aligns speech–text embeddings for direct retrieval and audio-conditioned generation, mitigating ASR error propagation; TWIST and tokenizer progress (PAST, LM-SPT, LAST) improve recognition/generation and enable low-rate streaming [43, 38, 21, 49, 22].

Construction and splits. Standard ASR uses LibriSpeech dev/test-clean; contamination in LibriSpeech/Common Voice motivates audited holds; 16 kHz is common for ASR/tokenizers, 24 kHz for TTS/codecs; token frame rate governs information retention and varies by language [58, 81, 28, 24, 82]. Rate–distortion and unit discriminability should be profiled across codebook sizes/frame rates; LM-aware, token-aligned designs preserve semantics at low rates and scale with synthetic data [22, 57, 4, 38, 135].

11.2 Task–dataset–metric mapping

Scope and defaults. We map task families to representative datasets, metrics, and protocol knobs, emphasizing instruction-style setups, rate-/latency-aware controls, and duplex diagnostics [40, 36].

ASR and multilingual ASR. Datasets: LibriSpeech, Common Voice, VoxPopuli, AISHELL, Fisher, MLC-SLM 2025; Dynamic-SUPERB Phase-2 frames ASR as QA-style sequence generation with multi-audio inputs [74, 61]. Metrics: WER/CER/WCER; diarization-aware t-cpWER/t-cpCER and DER when segmentation is joint [63]. Knobs: contextual biasing via entity lists/few-shot and dynamic prompting (9–20

S2TT: offline, SimulST, StreamST. Datasets: CoVoST2, Europarl-ST, MuST-C, SimulST/StreamST shared tasks [98, 84]. Metrics: SacreBLEU/COMET and AL/LAAL/StreamLAAL via SimulEval [98]. Knobs: publish framing/chunking, read–write policy (e.g., wait-k), prefix alignment/gating, tokenization path/rate, codec family/semantic fidelity, and truncation/segment rules to avoid skips [24, 42].

Textless S2ST with style. Datasets: CVSS-C/T; SALMon probes acoustic/text consistency; curated dubbing/audiobook sets support expressivity [41, 70]. Metrics: ASR-BLEU, SALMon ranking, ABX invariance, speaker similarity/EER, prosody measures, MAD Speech diversity, WER/dWER for intelligibility [46, 121, 71, 59]. Knobs: disclose unit/codec, token rate/bitrate, and enrollment prompts [5].

TTS, zero-shot TTS, VC, resynthesis. Datasets: LibriTTS, VCTK, CMU Arctic, VCC, LJSpeech/HiFi-TTS; use uncontaminated LibriTTS-testsetB as relevant [22, 58]. Metrics: WER/dWER, MOS/UTMOS/DNSMOS, FAD, speaker similarity, SALMon for attribute control, MAD diversity, unsupervised ABX [41, 121, 46, 27]. Knobs: 3 s enrollment for zero-shot; distinguish continuation vs cross-speaker; report decoder family/steps for latency [2].

Spoken classification and paralinguistics. Datasets: Speech Commands (KWS), IEMOCAP/RAVDESS/CREMA-D (ER), Voxforge/FLEURS (LID), MUStARD (dialogue acts) [67]. Metrics: accuracy or macro-F1 on clean, contamination-free splits [58]. Knobs: specify few-/zero-shot splits and prompt formats; MUSAN augmentation for interruption/command detection [1].

Enhancement and separation. Datasets: VoiceBank-DEMAND, LibriMix/Libri2Mix, LibriSpeech mixtures with DNS/MUSAN noise and reverberation [54, 29]. Metrics: PESQ, STOI, SI-SDRi, and downstream PER/WER transfer to SUPERB/IndicSUPERB heads [36]. Knobs: label/computation sweeps; supervised vs SSL (HuBERT) front-ends; discrete-token pipelines with universal vocoders and adversarial robustness [65, 26, 119].

Spoken language modeling and interleaved speech–text. Datasets: Spoken StoryCloze/Topic StoryCloze, spoken QA (WebQuestions, LLaMA Questions, TriviaQA), and synthetic interleavings [4]. Metrics: length-normalized likelihood, QA accuracy, WER for intermediate transcripts, LLM-graded outputs. Knobs: modality paths, token rates (e.g., 12.5 Hz), interleaving templates.

Spoken dialogue generation and duplex interaction. Datasets: J-CHAT (68,892 h, diarized, no transcripts) for textless dialog [6]. Metrics: MOS for naturalness/meaningfulness, dWER for unit resynthesis, barge-in success/latency, turn-taking stats (VAD/IPU/overlap), and chunked-reasoning latency [48]. Knobs: diarization-based prompts, unit vocoders + speaker embeddings, long-context prompts, IRQ tokens/end-of-turn thresholds, chunk schedules, token rate/bitrate, and MUSAN noise on listening paths [20, 40].

Social/affective capabilities. BoSS evaluates audio QA, dialect following, multi-turn memory, age adaptation, and emotion comprehension/generation; report zero-shot vs prompted styles [47].

Unified multi-task benchmarking. Combine IndicSUPERB (12 languages; ASR, SV/SID, LID, QbE, KWS), STAB for tokenizer assessment, and VoxEval-style SpeechQA; disclose tokenizer I/O budgets, end-to-end latency, and robustness to acoustic variation [40, 45, 41].

11.3 Benchmark suites and leaderboards

Benchmark	Size	Domain	Task Format	Metric
VoxEval [45]	13,938	Massive Multitask Language Understanding	Multiple-Choice QA	Accuracy, WER
- [2]	4,800	Text-to-Speech	Speech Synthesis	WER, MOS
LLaSO-Eval [125]	15,044	Speech-Language Understanding	Instruction Following	WER; GPT-4o
- [81]	80,000	Automatic Speech Recognition	Speech Recognition	['WER', 'Codebook usage']
- [136]	183,872	Spoken Language Modeling	Speech Resynthesis	UER, WER
IndicSUPERB [36]	1,684	Speech Language Understanding	Automatic Speech Recognition	WER, Accuracy
- [137]	34,000	Audio-Visual Speech Perception	Phoneme Classification	Acc, Peak latency
- [28]	54,000	Text-to-Speech	Text-to-Speech	WER, NISQA

Table 1: Consolidated overview of representative benchmarks for spoken language modeling and speech-centric evaluation. The table enumerates each benchmark's size, domain, task format, and primary metrics, spanning instruction-following suites, automatic speech recognition, text-to-speech, spoken language modeling, and audio–visual speech perception. Reported metrics include accuracy- and error-based measures (e.g., Accuracy, WER, UER, codebook usage) and perceptual/quality scores (e.g., MOS, NISQA), enabling cross-benchmark comparison.

Benchmark	Size	Domain	Task Format	Metric
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IndicSUPERB [36]	1,684	Speech Language Understanding	Automatic Speech Recognition	WER, Accuracy
- [137]	34,000	Audio-Visual Speech Perception	Phoneme Classification	Acc, Peak latency
- [28]	54,000	Text-to-Speech	Text-to-Speech	WER, NISQA

Table 2: Consolidated overview of representative benchmarks for spoken language modeling and speech-centric evaluation. The table enumerates each benchmark's size, domain, task format, and primary metrics, spanning instruction-following suites, automatic speech recognition, text-to-speech, spoken language modeling, and audio–visual speech perception. Reported metrics include accuracy- and error-based measures (e.g., Accuracy, WER, UER, codebook usage) and perceptual/quality scores (e.g., MOS, NISQA), enabling cross-benchmark comparison.

We synthesize recent benchmark suites and leaderboards for spoken language modeling into three complementary branches, as visualized in Figure 11. This organization proceeds from broad, instruction-style capability suites, through dialogue-, style-, and diversity-oriented evaluations, to specialized fronts that stress robustness and system design. The layout in Figure 11 highlights both the complementary coverage across these branches and the logical progression from general competence to interaction quality and, finally, to engineering concerns. Table 3 summarizes the principal benchmarks referenced in this subsection, situating their scale, domains, task formats, and evaluation metrics to support comparison across instruction-style suites, speech recognition and synthesis, spoken language modeling, and robustness-oriented evaluations. To anchor the discussion of benchmark suites and leaderboards, Table 3 consolidates the principal benchmarks referenced in this section, summarizing their domains, task formats, sizes, and evaluation metrics for direct comparison. Table 3 consolidates the principal benchmarks surveyed in this subsection, contextualizing their scale, domains, task formats, and evaluation metrics to support comparisons across instruction-style suites, speech recognition and synthesis tasks, and robustness studies.

Instruction-style, multi-task suites. Dynamic-SUPERB Phase-2 aggregates 180 instruction-following tasks across speech/music/environmental audio with open pipelines, domain-level aggregation, and a

Benchmark	Size	Domain	Task Format	Metric
VoxEval [45]	13,938	Massive Multitask Language Understanding	Multiple-Choice QA	Accuracy, WER
- [2]	4,800	Text-to-Speech	Speech Synthesis	WER, MOS
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Table 3: Consolidated overview of representative benchmarks for spoken language modeling and speech-centric evaluation. The table enumerates each benchmark's size, domain, task format, and primary metrics, spanning instruction-following suites, automatic speech recognition, text-to-speech, spoken language modeling, and audio–visual speech perception. Reported metrics include accuracy- and error-based measures (e.g., Accuracy, WER, UER, codebook usage) and perceptual/quality scores (e.g., MOS, NISQA), enabling cross-benchmark comparison.

compact core; streamable SLMs (e.g., BESTOW) show strong content/speaker skills but headroom in paralinguistics/robustness, and instruction-tuned systems (e.g., DeSTA2) adopt its scoring [74, 84, 32]. In parallel with these global suites, IndicSUPERB provides regional multi-task standardization, extending the first branch to diverse linguistic settings [36].

Dialogue-, style-, and diversity-focused evaluations. TELEVAL targets Chinese interactive scenarios across knowledge QA, dialect following, memory, paralinguistics, and spoken response quality, surfacing failure modes under dialect shifts, low SNR, and packet loss [126]. SALMon offers fast, model-based ranking for acoustic awareness and text–audio alignment, while MAD Speech audits acoustic diversity and mode collapse [41, 121]. Audio-aware LLM judges for style control are also emerging within this branch [123].

Specialized robustness and design fronts. VoxEval delivers speech-in/speech-out knowledge QA across 56 MMLU subjects with 153k audio variants over 26 conditions and spoken math; cascades remain competitive while end-to-end SLMs exhibit large spread, underscoring robustness and spoken reasoning gaps [45]. Discrete-token benchmarks further probe representation choices: DASB standardizes recognition/generation under bitrate control, finding semantic tokens outperform compression tokens yet trail continuous features; STAB measures invariance, robustness, compressibility, and vocabulary dynamics without transcripts, correlating with downstream ASR/AST/ER/SID/LID, and runs quickly on CPU [9, 40]. These studies complement codec/vocoder leaderboards and rate-aware SLM evaluations [8]. LongSpeech-Eval standardizes minute-scale QA and perceptual drift assessments [107], and an SLM-TTS benchmark codifies six axes—speaking style, intelligibility, speaker consistency, prosody, spontaneous behaviors, and human judgments—with public artifacts [2].

Reporting conventions. To ensure comparability across the above branches, use official splits; provide per-task/per-language breakdowns; report multiple seeds with variance; fix scripts and clarify licensing; and prohibit test-time tuning. Disclose instruction modality/provenance, abstention rates, token rate/bitrate, segmentation strategy, and vocabulary size for token interfaces [40, 36, 30].

11.4 Multilinguality, robustness, and streaming

Multilingual breadth. IndicSUPERB (Kathbath: 1,684 h, 12 languages, 1,218 speakers across 203 districts) provides clean/noisy and known/unknown-speaker splits with gender balance, exposing ASR robustness gaps to noise/unknown speakers, heterogeneous SV/SID robustness across encoders, and strong LID robustness [36]. J-CHAT offers 68,892 h of spontaneous Japanese dialogue via Whisper LID, pyannote diarization, and Demucs enhancement, with a reusable in-the-wild recipe [6].

Cross-lingual generalization and alignment. OTReg improves S2TT robustness to domain/language mismatch on CoVoST2/FLEURS while preserving LibriSpeech/MLS performance [85]. Encoder–Adaptor–LLM stacks trained on 1,500 h across 11 languages yield larger gains for

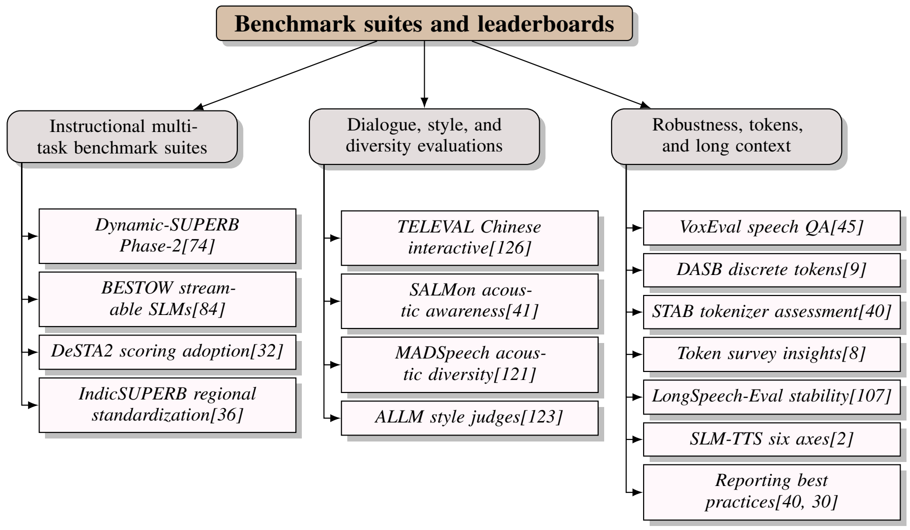

Figure 11: This figure organizes recent benchmark suites and leaderboards for spoken language modeling into three branches: (1) instruction-style, multi-task suites spanning global and regional settings; (2) dialogue-, style-, and diversity-focused evaluations including audio-aware LLM judges; and (3) specialized robustness and design fronts covering end-to-end spoken knowledge QA, discrete-token benchmarks and surveys, long-context stability, SLM-TTS axes, and reporting conventions. The layout highlights complementary coverage and logical progression from broad capability suites to interaction quality and robustness/engineering concerns.

East/Southeast Asian languages under progressive alignment, highlighting language-aware bridging and segmentation effects [60].

Dialect and beyond-semantic robustness. BoSS introduces five Chinese dialects and non-verbals to evaluate recognition and audio responses; augmenting ASR with dialect TTS dramatically reduces WER (e.g., Cantonese 89.54

Paralinguistics. Adaptor-based speech→LLM bridges preserve multilingual affect without discretization; English-only training achieves Chinese CER 33.3 vs 26.4 ground truth, revealing partial transfer and remaining gaps [7, 138].

Streaming and long-form. StreamUni operationalizes streaming chain-of-thought with truncation-only inference on long talks, matching human segmentation COMET across directions/backbones and demonstrating schedule portability [98]. Textless streaming S2ST with an acoustic LM + codec attains state-of-the-art BLEU/latency on CVSS-C [42]. FastLongSpeech improves long-form processing via iterative fusion and dynamic compression [135].

Recommended evaluation matrix. Combine IndicSUPERB for multi-task SLU, SALMon for acoustic awareness and text–audio alignment, STAB for tokenizer profiling, and VoxEval for end-to-end SpeechQA/spoken reasoning; report per-language/dialect deltas (clean/noisy, known/unknown speakers, augmentation), long-form segmentation policies, token rates/bitrates and streaming schedules (chunk size, stride, look-ahead), and pair quality with latency (AL/LAAL/StreamLAAL, time-to-first-audio) [36, 41, 40, 45, 81].

11.5 Data quality and reproducibility

Design principles. Standardized, auditable splits and scoring, contamination control, and open evaluation code enable apples-to-apples comparisons across linguistic and acoustic factors [36, 41].

Splits, stratification, leakage prevention. LLaSO-Eval constructs held-out, stratified splits from LLaSO-Instruct with explicit scope limits; BabySLM enforces voice-disjoint dev/test and balanced phone statistics; IndicSUPERB publishes clean/noisy and known/unknown-speaker splits at 16 kHz with rich metadata; TurnGuide guards against dialog leakage with explicit Fisher/Candor partitions;

Dynamic-SUPERB Phase-2 is test-only and vettable; instruction corpora should filter sources unsuited for vocalization and low-value outputs [125, 86, 35, 74, 52].

Contamination checks and statistics. Pretraining leakage in LibriSpeech/Common Voice can depress perplexity or inflate scores; adopt strict decontamination, non-overlapping holds, bootstrap CIs, and multi-seed aggregation [58]. VoxEval standardizes hyperparameters and applies Friedman/Wilcoxon with Bonferroni [45]. Listening tests require controlled confounds, sufficient raters, and CIs; diversity metrics benefit from multi-seed averaging given label noise [2, 121].

Provenance, licensing, privacy. LLaSO releases alignment/instruction pairs and a reference model; open S2S/VoiceTextBlender efforts release code/data/weights for augmentation and alignment [125, 111]. When full release is infeasible, document restrictions and evaluator calibration (e.g., SpeechGuard), watermark synthetic audio, and enforce ASR-based quality gates post-TTS with reported objective scores [118, 139]. Record maker–checker verification, denoising, and transcript correction; curated dubbing/audiobook sets denoise with DEMUCS and include human filtering [70]. Tokenizer pipelines should specify encoders, codebooks, languages, and retention vs k-means baselines (e.g., RepCodec) [25].

Reproducible scoring and protocol detail. VoxEval automates speech-in/speech-out scoring via Whisper-large-v3 with deterministic extraction and documented augmentation/math-to-speech procedures [45]. Contextual-biasing ASR should publish prompts, few-shot templates, loss balances, and scoring surfaces [110]. Alignment/RL for speech–LMs should disclose preference-pair counts, segmentation policies, steps, and compute [140]. Distribute full evaluation stacks with licensing; Dynamic-SUPERB accepts vetted community contributions [74].

Recommended checklist. Use IndicSUPERB, SALMon, and VoxEval as standard suites; report dataset scale and language coverage; include FBANK and SSL finetunes; evaluate ASR accuracy, acoustic consistency, speaker/language ID, KWS, robustness, and reasoning. Publish stratified, voice-/speaker-disjoint splits and construction scripts; standardize sample rates/metadata; disclose real/synthetic ratios and scope limits; quantify/mitigate test overlap with decontaminated results and standardized tokenizer assessments (STAB) [40]. For streaming/rate-aware models, report token frame rate (25/12.5/6.25 Hz), bitrate, synthesis backend and steps, latency metrics alongside quality, and read–write schedules [81]. When assets cannot be released, provide risk rationales, watermarking, and evaluator calibration. These practices support fair, reproducible comparisons as SLMs adopt LM-aware/aligned tokenizers (LAST, LM-SPT), phonetic–acoustic designs (PAST), and higher-fidelity codecs (RepCodec) [22, 49, 21, 25].

12 Evaluation Metrics and Methodology

12.1 Metric taxonomy and selection by goal

Metric choice should reflect task goals, interfaces, and deployment constraints: recognition/translation/retrieval prioritize semantic fidelity and intelligibility; generation/dialogue emphasize naturalness, identity/style, prosody, controllability, and latency; efficiency is cross-cutting. Interface-aware reports must disclose token/frame rate, bitrate, chunk cadence, and decoding policy, because decoder quality largely shapes perceived naturalness while quantization design and token rate govern intelligibility and LM fit [22, 81, 28].

Recognition and semantic fidelity

- Benchmarks and representation effects. Use SUPERB/IndicSUPERB (ASR, SV/SID, LID, QbE, KWS) with standardized toolchains; language-specific SSL often beats generic baselines, large-scale SSL excels on speaker ID, and continuous features typically surpass discrete tokens for fine-grained meaning. LM-aligned tokenizers (LM-SPT) preserve semantics at 25/12.5/6.25 Hz, shorten sequences, and improve S2T/TTS efficiency; compute-efficient HuBERT lowers the barrier to SSL training [36, 49, 97, 65].
- Intelligibility proxies and diagnostics. Report WER/CER (with Sub/Del/Ins) using a fixed recognizer/decoding policy; Whisper-large-v3 WER on generated audio is common in duplex settings, but proxy bias rises at high WER where ASR-free retrieval (SpeechRAG) is preferable [1, 43, 41]. Pair with substitution-sensitive probes—minimal-pair accuracy, phone confusion, and STAB metrics—since discrete-token TTS can sound natural yet be less

intelligible; phonetic–acoustic tokenizers (PAST) improve reconstruction/SLM outcomes while shifting phonetic fidelity [40, 21, 58, 2].

Translation fidelity

- Text and speech metrics. For S2TT, use SacreBLEU (with signature) and BERTScore; for S2ST, compute ASR-BLEU with a fixed, published ASR recipe, and co-report intelligibility (WER) plus style/identity to expose semantic–paralinguistic trade-offs [5, 49, 65, 36]. Probe representations with ABX and cross-lingual SLU on IndicSUPERB to test phonetic/segmental encoding and robustness [46, 36].

Modeling fit for token LMs

- LM-centric selection and utility. Favor LM-aware tokens (LAST, LM-SPT) that are compact, semantically faithful, and prosody-aware; robust training-free discretizations (dMel) support unified ASR/TTS; word-level prosody tokens capture content–prosody dependencies [22, 49, 54, 71]. Report LM perplexity/NLL/next-token accuracy on held-out speech-token corpora and contextualize with STAB and downstream outcomes (WER/CER, MOS) to avoid over-optimizing surrogates; analyze tokenizer vocabulary and LM size trade-offs as in LAST [40, 22].

Perceptual quality and distributional realism

- Human and proxy measures. Use ITU-T P.800 MOS (mean±CI) with native raters; prefer MUSHRA for controlled comparisons across token rates/decoders; UTMOS/MOSNet/NISQA scale reliably [6, 13].
- Signal- and corpus-level metrics. Report ViSQOL, PESQ/STOI, SDR, DTW-MCD, F0/energy errors, and DNSMOS; use FAD for distribution shift and diversity metrics (MAD Speech) plus embedding ABX to diagnose shifts/confounds across discrete vs continuous tokenization and codec DRI [46, 121, 20, 23]. SALMon ranks acoustic-factor consistency and text–audio alignment; augmentation-invariant discrete representations aid robustness [41, 141].

Controllability: identity, style, prosody, affect

- Identity/style and prosody. Measure speaker/style cosine, verification EER, and co-report ASR-BLEU/WER to reveal content–style trade-offs [59, 11, 49, 36]. Assess F0 RMSE/Voicing/FFE, duration and rate statistics, filler/pause patterns; word-level prosody tokens enable emergent processing, continuous tokenizers (Cont-SPT) better preserve frequency cues, and PAST strengthens prosody under real-time constraints [71, 20, 21, 39].
- Affect and social cues. Report SER Accuracy/F1, dialect/age-style accuracy, and audio-aware LLM rubrics for empathetic dialogue; combine explicit emotion metadata and heterogeneous adapters to disentangle linguistic/paralinguistic factors [7, 120, 37].

Tokenizer/codec diagnostics and efficiency

- Interface budgets. Disclose tokens/s, bitrate, frame/chunk cadence, codebooks/channels, dedup/compression ratios, and decoder steps; interpret metrics at matched operating points—token-rate-aware MUSHRA shows 50 tokens/s can rival or exceed 100 tokens/s under identical settings [13].
- Codebook health and information retention. Track usage entropy, token perplexity, dead/rare codes, and collapse; correlate with downstream metrics (STAB) and stress-test with SALMon; provide rate-aware panels spanning resynthesis WER/dWER, ViSQOL/PESQ/STOI, MOS/MUSHRA/UTMOS, and speaker similarity to quantify reconstruction–generation gaps [40, 41, 21, 33, 3].

Ecosystem anchors include IndicSUPERB (Kathbath; 1,684 h, 12 languages, 6 SLU tasks), fastabx for ABX discriminability, and tokenizers—PAST (supervised phonetic–acoustic, streamable), LM-SPT (LM-aligned semantics at low rates), and dMel (training-free, robust)—that collectively improve reconstruction, semantic alignment, and SLM performance; hybrid discrete–continuous or variationally enriched semantic tokens can further raise naturalness and preference, while SpeechRAG enables ASR-free QA when WER is high [36, 53, 21, 49, 54, 27, 101, 43].

12.2 End-to-end protocols

Protocols prioritize matched rates/bitrates, fixed recognizers/judges, and explicit decoding schedules for reproducible comparisons across speech generation, streaming, and duplex dialogue.

Core hygiene and datasets

- Use open suites with fixed configs: IndicSUPERB for multilingual SLU and compute-accessible SSL (HuBERT on 8 GPUs) [36, 65].

Decoding policies, transcription, and judges

- Fix defaults and sweep a compact grid (e.g., top-p 1.0, 0.99, 0.95, 0.9); with hierarchical RVQ, decode the first residual autoregressively and others non-autoregressively; report WER, NISQA, SIM-O, and human CMOS/SMOS under a fixed budget [28]. For unified speech–text decoders, keep identical sampling across modalities and disclose beam/temperature/top-p [74].
- Transcribe generated speech with a fixed ASR (e.g., Whisper-large-v3), versioned and recipe-published; use dual LLM judges with released prompts/aggregation rules (e.g., GPT-4o settings) for content/paralinguistics [37, 7].

Dataset handling and scoring

- Segment long audio (e.g., 30 s windows, 15 s overlap); use WER for space-delimited languages and CER for logographic/syllabic scripts with consistent toolchains [60].

Instruction-driven ASR/AST and contextual biasing

- Apply LLM second-pass rescoring (bias lists, few-shot prompts, entity-class prediction, dynamic prompting) for 9.6–20

TTS/VC and textless S2ST

- Zero-shot TTS: 3 s prompt+transcript; evaluate WER (fixed ASR), NISQA/CMOS/SMOS, and speaker/style similarity; normalize timbre (e.g., FreeVC) and synthesize multiple samples per condition [2]. Unified SLMs with LM-aligned tokens (LM-SPT/LAST) via small adapters over frozen speech/text backbones support interleaved text–speech with a fixed vocoder and disclosed decoding policy [22, 3, 49]. For textless S2ST, report ASR-BLEU, speaker/style similarity, dWER/WER, and NISQA/CMOS; include bilingual human evaluations with calibration and per-item medians [5, 28, 70].

Prosody stress tests and perturbations

- Use WORLD-based emphasis/range edits (e.g., 1.15, 2.2) and compute Token Edit Rate with SSL+k-means tokenizers (k100,500,2000) under segment/full-utterance constraints; co-report NISQA/WER/SIM-O at fixed decoding [28].

Streaming, chunking, and duplex dialogue

- Long-form streaming ST: SimulEval with chunk sizes (320/640 ms), mWERSegmenter, LAAL/StreamLAAL; sweep delay k and report latency–quality frontiers/schedules [98, 42]. Duplex agents: report interruption P/R/F1, WER on generated speech, first-response latency under declared interleave/chunk schedules [1, 35, 34]. Speech continuation: prompt 5 s, generate 25–120 s with fixed beam and unit/neural vocoder; score with GPT-based semantics and WER [6, 35].

Diagnostics and disclosure

- Include fastabx ABX and STAB profiling; disclose prompts, recognizer/judge versions, decoding settings, interleave/chunk sizes, and interface descriptors; run sampling ablations (WER/NISQA/SIM-O/CMOS vs top-p) and justify operating points [53, 40, 60, 28].

12.3 Bitrate- and token-rate–aware evaluation

Treat frame/token cadence r (Hz) and bitrate b (kbps/bps) as independent levers; hold decoders, sampling, judges, and hardware fixed while sweeping r and b.

Axes and controls

- Factors: model/tokenizer family and size (FBANK vs wav2vec2/HuBERT/SLM), tokenizer choice (STAB), input granularity (snippet/utterance), language (e.g., 12 Indic), task (ASR, SV/SID, LID, QbE, KWS), and acoustic condition with curated ABX metadata [53, 40, 46, 36, 113].
- Token/frame rate r: sweep 75/50/25/17.7/12.5/8.33/6.25/5.0 Hz; syllabic pacing at 5.0–8.33 Hz (60–91 bps) probes ultra-low-rate semantics; large interleaved pretraining suggests 12.5 Hz as a semantic knee [15, 4].
- Bitrate b: at fixed r, vary RVQ channels/bitwidths or K; at 50 Hz, tiers (3/1.5/0.75 kbps) expose WER/WIL vs ViSQOL/STOI trade-offs [28, 77]. Sequence compression via dMel/PAST, dedup/audio-BPE, or attention-based thinning should report reduction ratios and layer placements [54, 21, 43].

Dependent variables and panels

- Semantics: WER/CER, WIL, unit-resynthesis dWER, zero-resource probes (sWUGGY/sBLIMP/StoryCloze) under a fixed ASR [28].
- Acoustics: ViSQOL, PESQ/STOI, DNSMOS/NISQA, MOS/MUSHRA, DTW-MCD, speaker/style cosine; ABX and SALMon for phonetic/extra-linguistic and acoustic-factor consistency [46, 41, 53].
- Prosody/diversity: duration/F0 preservation, NDB/JS over pitch/voicing/energy, response to controlled prosody perturbations, and word-level prosody tokens [71, 39].
- Efficiency/latency: LM perplexity/NLL at matched sequence lengths, prefill time, decoder RTF, time-to-first-audio, token-reduction vs accuracy; interleaved speech–text pretraining scales compute more efficiently than textless SLMs [57, 142, 40].

Fair ablations and stability

- Align datasets/splits/preprocessing/languages so STAB and fastabx correlate with IndicSUPERB downstream performance; fix decoder/vocoder, sampling, and ASR across r/b sweeps [40, 53, 36]. Bitrate ablations: fix r (e.g., 50 Hz), vary b (3/1.5/0.75 kbps). Token-rate ablations: fix b where possible, vary r only; avoid mismatches (50 vs 75 Hz) at fixed b [77, 28].
- Plot semantic and perceptual metrics vs r/b at matched operating points; semantically informed codecs can preserve intelligibility better at the same rate [24]. At fixed r, increase RVQ channels/bits to raise b and reconstruction quality; check for WER/CER plateaus [28]. Mitigate DRI to stabilize sequence lengths; dMel provides streamable, OOD-resilient tokens; quantify behavior via STAB [54, 23, 40].
- Enforce fixed-length budgets (e.g., 30–50 tokens) for textless LMs to avoid confounds [67]. Compare learned codecs (RepCodec/Language-Codec) to frame-rate or mel-channel quantization (dMel); include SALMon tests and report dedup/aBPE or merging ratios [25, 12, 81, 41]. Mixed semantic–acoustic streams at base 12.5 Hz with depth 12.5–112.5 tokens/s show diffusion decoding can keep DNSMOS stable as depth decreases and that 50 tokens/s can surpass 100 tokens/s in MUSHRA [13, 4]. At ultra-low rates (e.g., 6.25 Hz, 0.0875 kbps), report intelligibility/identity and reconstruction–generation gaps (e.g., TaDiCodec) [14].

Minimum reporting

- Specify tasks/benchmarks (IndicSUPERB, VoxEval), languages, audio conditions, modalities, metrics, pretraining/fine-tuning, and interface descriptors (r, b, codebooks/channels, dedup/aBPE, encoder layers) [45, 36, 12, 81, 25]. Provide ablations: frame-rate grid

(50/25/12.5/8.33/6.25/5.0 Hz), bitrate tiers at 50 Hz (3/1.5/0.75 kbps), and 12.5 Hz acoustic-depth sweeps with stability diagnostics (ultra-low-rate WER gap), length-control effects, and caveats about rate mismatches [53, 40, 41, 14, 67, 28].

12.4 Latency metrics and methodology

We adopt a structured view of latency evaluation that is summarized by the hierarchical map in Figure 12. At the top level, the map organizes (i) core latency definitions and protocols—RTF, TTFT, streaming lag, and duplex IRQ—anchored to shared long-audio workloads; (ii) compute accounting and path-wise latency decomposition spanning encoder buffering, tokenization/adapters, LLM prefill/decoding, and token-to-waveform synthesis, including policy and token-rate effects; and (iii) reporting fields and normalization guidance, covering required fields and duplex sheets, acoustic standardization, dataset/split and tokenization-alignment controls, quality–latency curves, and full compute/time disclosure. Citations throughout indicate exemplar methods and benchmarks supporting each element. Against this backdrop, latency must reflect interface (discrete vs continuous; r/b), streaming policy (wait-k, read–write, chunking), and duplex behavior under fixed hardware and decoding; shared definitions and long-audio workloads enable apples-to-apples comparisons [126].

Definitions and measures (tier i in Figure 12)

- Real-time factor (RTF): wall-clock/output duration; report global and length-stratified values with FLOPs/memory and batch size. A standard recipe reports throughput (tokens/s), prefill/decoding latency, RTF, and max batch on 120/240 s audio under fixed beams/hardware [117].
- Time-to-first-token (TTFT): separate text and speech. Decoder-only ASR with prompt compression improves CPU RTF (0.24–0.29) vs encoder–decoder (0.53–0.58) [62]. Publish first audio-chunk time and steady-state throughput under interleaving [117].
- Streaming lag: use AL, LAAL, StreamLAAL for SimulST/StreamST; report BLEU/COMET vs lag at fixed chunk sizes/delay k [98]. Duplex interruption/overlap: pre-register a success window, compute success/latency/false-alarms, and train/report IRQ timing with a common delay criterion [1].

Compute accounting and path-wise latency (tier ii in Figure 12)

- Break down encoder/right-context buffering, tokenizer/adaptor, LLM prefill/decoding, and token-to-waveform decoding (AR/NAR, diffusion/flow); publish audio-only and LLM-inclusive first-packet variants with fixed device/precision and no batching [36, 40]. For unified LSLMs (e.g., StreamUni), report policy/segmentation handling and LAAL/StreamLAAL vs quality across wait-k/prefix and chunked read–write schedules [98]. Include token-rate effects (e.g., 12.5 vs 6.25 Hz) with identical decoders and segmentation overhead when using syllabic pacing [15].

Reporting fields and duplex sheets (tier iii in Figure 12)

- Minimum fields: TTFT (text/speech), first audio-chunk time, length-stratified RTF, prefill vs decoding latency, tokens/s per modality, memory-bound batch, peak VRAM, and AL/LAAL/StreamLAAL with chunk size and k; compare CPU/GPU RTFs for decoder-only vs encoder–decoder ASR under matched params [62]. Duplex sheet: interruption window, actuation delay, success/false-alarms, latency distributions with standardized VAD/onset detection under matched SALMon-like acoustic conditions; include tokenizer-aware controls (NAST, STAB) [29, 41, 40].

Interpretation and normalization (normalization guidance within tier iii in Figure 12)

- Normalize dataset/splits (IndicSUPERB), streaming window/lookahead, frame hop/sample rate, decoding settings, hardware/batch size, and tokenization (discrete vs continuous, vocab), as r and speech–text alignment (LAST, Cont-SPT, OTReg) directly affect latency/throughput [22, 20, 85]. Co-report quality–latency curves for streaming ST (BLEU/COMET/BLASER 2.0 vs AL/LAAL/StreamLAAL) and interleaved assistants (time-to-first-audio-token and sustained audio-token throughput vs task quality) [57, 98, 84, 42]. Provide compute/timing (hardware, training/finetune time, latency/throughput, FLOPs, memory-bound batch) and contrast distilled vs teacher front-ends when relevant [104].

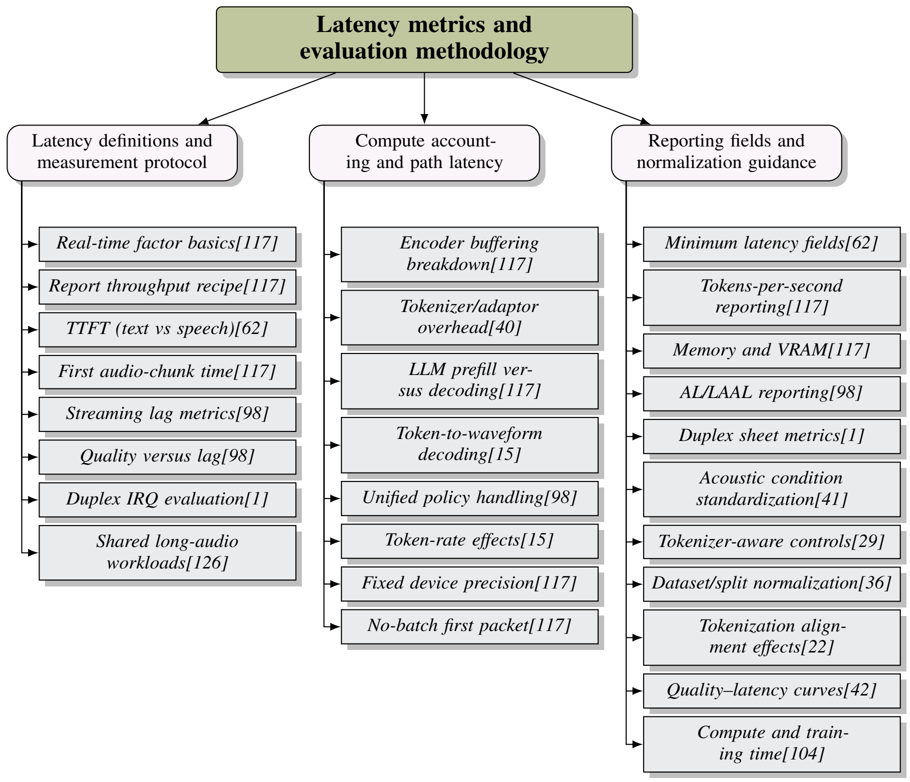

Figure 12: Figure: Hierarchical map of latency evaluation for speech/SLM systems. Top-level categories organize (1) core latency definitions and protocols (RTF, TTFT, streaming lag, duplex IRQ) anchored to shared long-audio workloads; (2) compute accounting and path-wise latency decomposition across encoder buffering, tokenization/adapters, LLM prefill/decoding, and token-to-waveform synthesis with policy and token-rate effects; and (3) reporting fields and normalization guidance, covering required fields, duplex sheets, acoustic standardization, dataset/split and tokenization alignment controls, quality–latency curves, and compute/time disclosure. Citations indicate exemplar methods and benchmarks supporting each element.

12.5 Human-in-the-loop and statistical rigor

Human panels and audio-aware LLM (ALLM) judges must be calibrated, auditable, and paired with prespecified statistics and agreement analyses, stratifying language/speaker/task coverage following IndicSUPERB and validating intrinsic metrics (STAB) against downstream tasks.

Judge selection, calibration, and blinding

- Fix judge configuration (model/version, prompts, temperature, trials, aggregation). BoSS uses GPT-4o with three trials and power-scaled averaging and recommends auditor diversity; calibrate ALLM to human judgments, report human–ALLM and human–human correlations, sensitivity to judge settings/data shifts, and use dual-judge setups with an open-source counter-judge [47]. Randomize and blind evaluators; use speaker-stratified multi-seed splits and balanced AB/BA orders [40, 65].

Panel design, instruments, and reliability

- Target 60 native raters with balanced loads (8 items/rater), 1–5 MOS scales across dimensions (naturalness, meaningfulness), and 95

Statistics, confounds, and replication

- Use two-sided Wilcoxon with Holm–Bonferroni and effect sizes (rank-biserial r/Cliff's delta) for nonparametric panels; when MOS is approximately normal (checked via Shapiro–Wilk/QQ-plots), use two-sided t-tests with 95

Reporting checklist and defaults

- Specify datasets (e.g., Kathbath 1,684 h, 12 languages), tasks (ASR, SV/SID, LID, QbE, KWS), baselines (FBANK vs SSL), tokenizer diagnostics (STAB) and correlations to downstream, and include SALMon and VoxEval where applicable [36, 40, 41, 45]. Provide judge metadata (model/version, prompts, temperature, trials/aggregation), human–ALLM calibration, per-sample ratings/transcripts/temperatures, code/seeds, and bias disclosures [47]. Practical defaults: evaluate with IndicSUPERB, select tokenizers via STAB, and prefer adapter-based SLMs that freeze speech/text backbones (1

13 Efficiency Compression and Deployment

13.1 Parameter and training efficiency

Freezing strong speech/text backbones and training thin bridges or PEFT modules delivers the best cost–quality frontier. A 156M, 2-layer adapter aligning a 2B USM encoder to a 13B mT0-MT maintains text skills and stable next-token modeling while avoiding full-finetuning drift [3]; prompt-tuning adapts textless GSLMs for classification with <0.1

LM-aware and phonetic–acoustic tokenizers (LAST, LM-SPT, PAST) plus training-free dMel compress the interface into LM-aligned or phoneme-faithful units at low frame rates, enabling a single text LM to process speech and text for zero-shot instruction following, ASR-free SpeechRAG, and multilingual transfer validated on IndicSUPERB; fastabx offers efficient ABX diagnostics guiding unit design [22, 49, 21, 54, 43, 53, 36, 40]. Compact cores can replace cascades: MSLM-S2ST uses 306M AR + 306M NAR LMs to jointly translate and generate units [5]; a 24M dialogue LM sustains long-form textless generation [6]. Efficient front ends and inference-time token-merging reduce memory by 7× and raise throughput 1.8× without hurting ER/SQA; long-speech compression and optimal-transport regularization strengthen long-form efficiency and cross-dataset generalization [117, 135, 85].

Compute-efficient recipes widen access: Slam trains a competitive SLM in 24 hours on a single academic GPU with careful initialization, synthetic data, and preference optimization [104]; optimized HuBERT reproduces Large-model results with 8 GPUs, sometimes skipping the first SSL iteration via ASR bootstrapping [65]. In multi-stage curricula (ASR→TTS→SQA), experience replay, LoRA-scale discounting, and model merging mitigate forgetting [31]. Interleaved speech–text pretraining scales more efficiently than textless SLMs and favors spending on model size over tokens [57]. IndicSUPERB and SALMon standardize multilingual SLU and acoustic-robustness evaluation [36, 41].

Actionable configurations: - Frozen USM-2B + mT0-MT-13B with 156M adapter [3]. - Partial-LoRA for affect (rank 16, =16 on K/Q/V/O; 4×A100 for 2.6 days) [7]. - LoRA + 4-bit quantization for 12B decoders (=32) [60]. - Staged unfreezing: projector-only, then rank-16, =32 LoRA on Whisper encoder + 9B LLM [61]. - Compact S2ST: 306M AR + 306M NAR [5]. - Small dialogue SLM: 23.64M parameters [6]. - Parameter-light SSL front ends (95–100M) with token-merging inference [65, 117, 57].

13.2 Model and sequence efficiency

Three levers compose: rate-aware token interfaces, compressed conditioning paths, and schedule-aware decoding/rendering. Moderate cadences shrink sequences while preserving semantics and control—12.5 Hz supports long-context mixed speech–text pretraining without waveform synthesis [4]; 6.25 Hz with few-step text-aware flow/diffusion maintains intelligibility/identity at much lower sequence cost [14]. Hybrid semantic–acoustic tokenizers keep cadence fixed but vary per-frame acoustic depth; 50 tps (N$_s = 1, N_a = 3) matchesorexceeds100tpsonWER, DNSMOS, andMUSHRA[13].Duplexstacksremaincompactbyfusingasmallst$

Conditioning compression trims prefill and attention: 4× frame subsampling before a thin adaptor and projector pooling (4:1–5:1) reduce injected states with minimal loss and, with FlashAttention-2/ZeRO-2, fit modest GPUs [3, 60]. Decoder-only ASR benefits from label-synchronous compression via CTC blank removal, shrinking inputs to 14.6

On discrete streams, run-length de-dup and acoustic BPE shorten strings without changing interfaces; compressibility diagnostics (Huffman/BPE/de-dup) often favor moderate vocabularies at similar accuracy, informing rate-aware K selection [40]. Knowledge-distilled tokens shift the rate–quality frontier: supervising early RVQ levels with text/SSL teachers increases informativeness without extra runtime, and AR on RVQ-1 with NAR on residual levels improves intelligibility and shortens acoustic horizons for TTS/S2ST [77, 5].

Parallelization spends AR capacity on semantics and amortizes high-rate acoustics through NAR refinement or masked infilling; projector pooling and CTC blank removal shrink prefill, while delay-controlled emission and semantic truncation bound latency [98]. Rendering favors codec-native inversion or lightweight vocoders; few-step text-aware diffusion at ultra-low rates achieves favorable real-time factors [14]. Tooling such as fastabx and FastAdaSP enables large-scale multi-task experiments with up to 7× memory reduction and 1.8× throughput gains at negligible accuracy cost [53, 117]. Across 12.5/6.25 Hz interleaving [4], conditioning compression [3, 60, 62], distillation with AR/NAR splits [77, 5], acoustic-BPE shortening [40], and lightweight rendering [14], prefill and per-step latency drop while intelligibility and style control remain strong.

Practical defaults: - Token rate: 12.5 Hz for mixed training; hybrid tokens with $\mathrm{N}_s = 1, N_a = 3(50tps).Dropto6.25Hzonlywithtext\text{-}awaredecoders$[13, 14]$. - Conditioningcompression : 4subsamplingor4\breve{}5projectorpoolingwithZeRO\text{-}2/FlashAttention\text{-}2; CTCblankremovalfordecoder\text{-}onlyASR$[3, 60, 62]$. - Parallelizelongsequenceswithtokenreduction(FastAdaSP)andefficientABX(fastabx); allocateARtosem$ 53]$. - Distilltokensforinformativenessandkeeprendererslight$[77, 14].

13.3 Fast decoding and streaming

Latency falls by emitting more tokens per step, cutting per-token compute, and shortening horizons. Grouped multi-token prediction (MTP) amortizes attention and pairs with RVQ residual-level parallelism (AR base, NAR residuals); masked infilling adds parallelism in few iterations [116], and k-frame decoding (k4) boosts throughput without quality loss [54]. Mixed stacks profit from parallel speech–text emission and NAR vocoders [92]; read–write interleaving with short planning bursts yields low first-audio delay [129]. Dual-stream TTS initiates speech by the second input token and emits all speech tokens per text token in one step [127].

Speculative bursts (10 tokens) reduce time-to-first-chunk [143]; block-wise 2 s decoding matches streamable tokenizers and duplex schedules [4]. Cache-aware inference thins sequences and stabilizes incremental decoding—FastAdaSP merges adjacent input slices before KV construction for 1.44–1.84× throughput and 50

Lower token rates shrink decoder steps and KV growth: syllable-synchronous tokens at 5–8.33 Hz accelerate boundary extraction [15]; 6.25 Hz tokens with few-step text-aware diffusion reach end-to-end RTF 0.12–0.29 [14]. Acoustic BPE accelerates AR continuation by 2.8–5.0× on 20 s segments [68]. Streamable tokenizers emit semantic tokens online and expand chunk-wise into acoustic tokens and waveform segments via an acoustic LM and codec/diffusion decoder [22, 54, 13, 40, 42]. Renderer choice is decisive: single-pass GAN/vocoders minimize last-mile latency when feasible [2], while step-distilled diffusion compresses sampling steps with minor quality trade-offs [13].

A practical low-latency stack composes grouped MTP or k-frame decoding, residual-level NAR prediction, speculative bursts, FastAdaSP sequence thinning, per-chunk KV updates with fixed beams, a streamable tokenizer with chunk-wise expansion, low-rate tokens (6.25–12.5 Hz) with few-step text-aware diffusion or a NAR vocoder, and block-wise 2 s decoding—yielding 1.44–1.84× throughput gains and 50

13.4 Bandwidth and edge–server trade-offs

Deployments must balance link bitrate, VRAM/KV growth, and server FLOPs while supporting duplex, low-latency interaction. Low-rate token streams are viable: semantic-conditioned DiffSound-

Stream matches SoundStream quality at 50 tps with four diffusion steps [13]; textless streaming S2ST achieves strong BLEU/latency/BLASER-2.0 by emitting discrete tokens end-to-end [42]. XY-Tokenizer improves the semantic–acoustic trade-off with high speaker similarity at comparable rates [80], and SALM-Duplex caps downlink near 0.6 kbps with high UTMOS and speaker similarity, establishing a sub-kbps working point; 0.3 kbps modes remain intelligible for poor connectivity [34, 144]. A practical ladder is 0.3→0.6→1.24 kbps, with language-dependent adjustments for prosodic density [39, 81].

Separating perception from generation decouples encoder compute from codec bitrate and eases heterogeneous client integration; semantic-conditioned tokenization shifts information into the semantic path to lighten the acoustic stream when both are transmitted [34, 13]. On constrained VRAM, Slam proves single-GPU training in 24 h [104]; FastAdaSP enables up to 7× memory reduction and 1.8× throughput gains for higher concurrency [117]. Mixed speech–text decoders fit within 19 GB VRAM on 8B backbones for on-prem single-GPU serving; quantization (4–8 bit) and expert routing by language/task further reduce VRAM/FLOPs [51, 117]. Interleaved speech–text pretraining improves scaling efficiency and shifts optimal spend toward larger models; speech-native RAG bypasses ASR cascades, reducing latency and compounding errors [57, 43, 3, 104].

Operational policies: - Mid-rate default 50 tps (0.55–0.6 kbps) with capacity for textless streaming S2ST; burst to 1.24 kbps for prosody-critical spans; fall back to 0.3 kbps under congestion [42, 13, 144]. - Memory/concurrency: track per-session KV/VRAM and max batch; quantify 10–50- Reporting: justify edge vs server execution using IndicSUPERB, STAB diagnostics, SALMon robustness, and interleaved scaling dynamics [36, 40, 41, 57].

Blueprints: - Ultra-low-bandwidth field agent: on-device ESPnet-SpeechLM with dMel tokens for SLU; server-side mixed SLM with FastAdaSP concurrency; 0.6 kbps downlink [54, 113, 36, 117]. - Mobile default: 50 tps semantic-conditioned acoustic tokens with short expressive bursts; low memory via token-merging; compatible with discrete streaming [9, 13, 117, 42]. - On-prem single-GPU: 24-hour Slam training, IndicSUPERB evaluation, int8/4-bit quantization, duplex 0.6 kbps [104, 36, 51].

13.5 Implementation recipes and benchmarks

As a visual synopsis, Figure 13 presents a hierarchical map of the implementation recipes and benchmarks surveyed in this subsection. The top-level branches group best practices into tokenization/alignment, practical modeling with compute-efficient recipes, and evaluation with open benchmarks; the leaf nodes condense these into concise, cited action items—such as efficient k-means, Optimal Transport Regularization (OTReg), single-GPU Slam, SSR-Connector, and the STAB and VoxEval suites—guiding reproducible, rate- and latency-aware development and fair reporting.

Rate- and latency-aware practice centers on open toolchains, standardized token interfaces, fixed decoding policies, and matched compute budgets. Resources: IndicSUPERB/Kathbath (1,684 h; 12 languages; ASR, SV/SI, LID, QbE, KWS), SALMon (noise, emotion, identity, room), STAB tokenizer diagnostics, VoxEval SpeechQA [36, 41, 40, 45]. Compute-efficient baselines include 24 h single-GPU Slam, 8-GPU HuBERT SSL pipelines, and SpeechGen-style adapters on GTX-2080-class GPUs [104, 65, 33]; LLM-based contextual biasing improves second-pass ASR by 9.6–20

We elaborate each branch of Figure 13 below.

Tokenization and alignment: - Efficient unitization via k-means (moderately coarse segmentation; larger codebooks) cuts training data 50- In-memory labeling and cross-node k-means avoid TB-scale storage, keeping clustering memory sub-GB; ESPnet configs document budgets [65]. - Optimal Transport Regularization narrows the speech–text gap and improves multilingual generalization with a lightweight, label-free term; a two-stage schedule (2+3 epochs; AdamW 1e4→1e6; $_{s}pr = 0.1)isreproducible$[85].

Unified modeling: - SUTLM mixes speech-unit LM, paired speech–text (cross/auto-regressive), and text-only LM over a shared vocabulary; at inference, zero non-target modality logits to prevent leakage [11, 105, 19]. - Language-Codec trains a language-aware tokenizer/codec with LM-conditioned and masked-codec objectives for robust, streamable tokens; open runs use 2M steps on 8×A100-40GB (AdamW 2e4) [12, 145, 25]. - Omni-modal alignment (EMOVA) and SSR-Connector add controllable style/emotion and compress speech to text granularity to avoid forgetting [72, 18].

Representative blueprints: - SpeechComposer unifies ASR, TTS, speech/text LM, and composite tasks in a decoder-only model initialized from OPT-125M/350M with HuBERT-Base k-means tokens and HiFi-GAN; released configs/checkpoints aid reproduction [64]. - Low-rate semantic pacing (5 Hz) via SyllableLM matches or outperforms SotA with 30× less training compute and 4× faster inference; report RTF on fixed 32×25 s batches with 16 CPU threads [15]. - Single-stage joint SFT with LoRA mixing ASR/AST/SQA and conversational data, with continual text-only checks to audit forgetting, is tractable on 4×A6000 over 5 days [21, 31].

Evaluation and reporting: - Prefer intrinsic first, downstream second: run STAB's Apache Beam suite in 15 CPU-minutes per metric per tokenizer before costly downstream sweeps (e.g., DASB) [40, 9]. - Standardize decoding (beam, temperature/top-p, averaging windows, padding/splicing) and fix token frame/bit rates and AR/NAR schedules for fair timing/quality across tokenizers and languages [40, 39, 36]. - Report RTF, TTFT, tokens/s, WER/BLEU/MOS proxies, VRAM class, concurrency, training wall-clock/GPU-hours, optimizer/schedule, precision/parallelism, and seeds [104, 65, 57]. - Publish tokenizer pipelines and memory footprints (on-the-fly labeling, Kaldiio sampling, k-means node counts) and attach code/models/eval harnesses for RepCodec, SALMon, VoxEval, and DASB [65, 25, 41, 45, 9].

These practices—paired with interleaved speech–text scaling that prioritizes model size, LM-aware tokenization (LAST/LM-SPT/PAST/dMel), and bandwidth-aware streaming policies—enable reproducible, compute-light development of duplex, barge-in-capable agents under tight latency and memory budgets [22, 49, 21, 54, 57]. Read together with the high-level structure in Figure 13, the section provides both a compact roadmap and the concrete, cited steps needed for faithful reproduction and fair comparison.

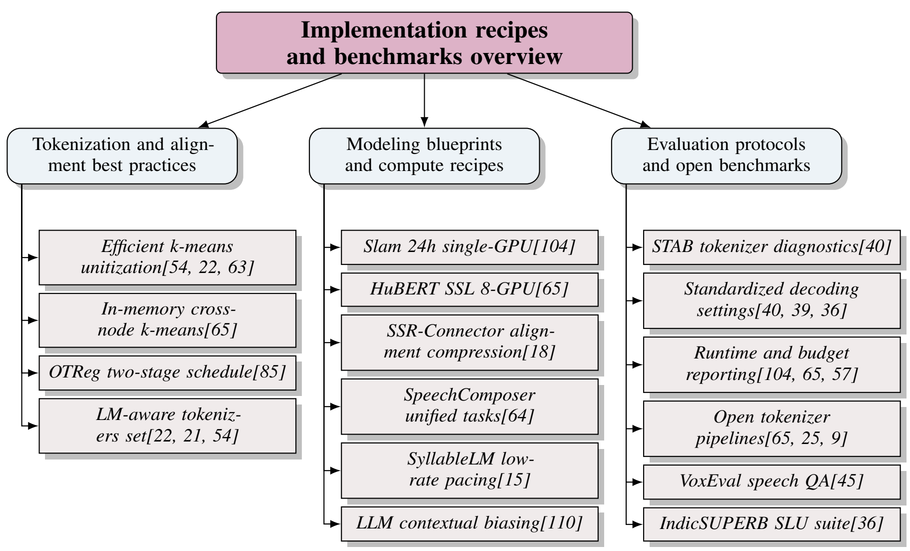

Figure 13: Figure: Hierarchical map of implementation recipes and benchmarks for spoken language modeling. Top-level categories summarize best practices in tokenization/alignment, practical modeling and compute-efficient recipes, and evaluation with open benchmarks. The leaves highlight concise, cited action items (e.g., efficient k-means, OTReg, Slam, SSR-Connector, STAB, VoxEval) to guide reproducible, rate/latency-aware development and fair reporting.

14 Safety Robustness and Ethics

14.1 Threat models and vulnerabilities

SLMs introduce attack surfaces at the audio encoder, modality connector, and unified decoder absent in text-only LLMs. White-box, gradient-based waveform perturbations reliably bypass safety with 79–97

14.2 Defense-in-depth

To situate the following safeguards within a unified lifecycle view, Figure 14 depicts a conceptual defense-in-depth tree that (i) groups pre-deployment measures—data curation and training-time alignment—to improve coverage, robustness, and safety; (ii) consolidates runtime defenses and provenance tools that harden inputs/outputs and enable post-hoc mitigation and verification; and (iii) summarizes a reference stack with recommended components and reporting standards. Arrows in the figure trace the progression from preparation to inference-time protection and auditing, emphasizing layered, complementary controls across the model lifecycle. We instantiate these branches as follows.

A layered stack spans data, training, decoding, and post-hoc defenses with provenance throughout:

- Data-time safeguards. Curate multilingual, accent-diverse corpora; audit tokenizers with STAB; use IndicSUPERB's stratified coverage to identify language-specific gaps [40, 36]. Prefer LM-aligned, robust units (LM-SPT, PAST, NAST, LAST) that support low rates and streamability [49, 21, 29, 22]; use ASR-free SpeechRAG in high-WER domains [43]. Watermark synthetic audio (AudioSeal) and apply misuse-aware filtering (KE-Omni); add negative exemplars that map non-speech/inaudible inputs to safe refusals [139, 51].
- Training-time safeguards. Combine safety filtering and refusal shaping with compute-efficient SSL (e.g., HuBERT on 8 GPUs) and standardized tokenizer assessment (STAB); mitigate forgetting via replay, model merging, and LoRA scaling; enable watermark-by-default to ease provenance [65, 40, 31, 146, 138].
- Decoding-time safeguards. Harden inputs and constrain outputs: time-domain noise flooding at 48–60 dB SNR drops jailbreak success from 90% to 0–8% [118]; apply deny-lists and dialect-aware refusals [51]; insert watermarks at generation (SSR-Speech) for frame-level audits [146].
- Post-hoc defenses and provenance. Activation patching (SPIRIT) detects noise-sensitive neurons and blocks audio jailbreaks with 98.5–99.9% success and negligible utility loss, outperforming denoising [119]. Use SSR-Speech for edited-region localization and AudioSeal for corpus-scale verification; disclose residual risks and restrict weight release when misuse remains material [146, 139, 107].

Completing the lifecycle emphasized in Figure 14, the reference stack (rightmost branch in the figure) combines robust discrete units (NAST/PAST/LM-SPT; DASB indicates discrete tokens often beat codecs but can trail continuous features at some rates) [9, 26, 39, 21, 49, 29], dual watermarking (AudioSeal, SSR-Speech) [139, 146], input noise flooding plus activation patching [118, 119], and standardized reporting of jailbreak/defense success with GPT-Score and WER trade-offs [118, 119, 40]. For noisy or ASR-free pipelines, hybrid representations and retrieval conditioning (e.g., Whisper-GPT, SpeechRAG) stabilize behavior [27, 43]. In concert with the pre-deployment and runtime controls outlined above and visualized in Figure 14, this stack operationalizes defense-in-depth from preparation through inference-time auditing.

14.3 Provenance, impersonation, and privacy

Rising fidelity in synthesis/conversion heightens deepfake risks as models learn prosody and paralinguistics and remain robust under acoustic shifts [101, 27, 41, 141]. Make provenance first-class: frame-synchronous SSR-Speech watermarks enable edited-region localization at tens-of-Hz token rates, while AudioSeal watermarks preserve corpus-scale traceability; pair watermark-on-by-default synthesis with cryptographic request signing and immutable logs linking prompts, enrollment assets, model/version, and watermark keys [146, 139, 124].

Voice conversion/cloning requires explicit consent and anti-spoofing. Tokenizers such as UniCodec permit conversion by swapping a global token, creating a direct impersonation channel [102]. Enforce liveness-verified enrollment, per-request consent tokens bound to target speakers, identity allow/deny-lists, and mandatory watermarks; integrate anti-spoof detectors at enrollment and inference and quarantine suspicious traffic [54]. When identity is unnecessary, use privacy-preserving virtual voices and restrict model release for high-risk systems; surface impersonation risks, consent status, and takedown hooks in UI/docs [139, 107, 124].

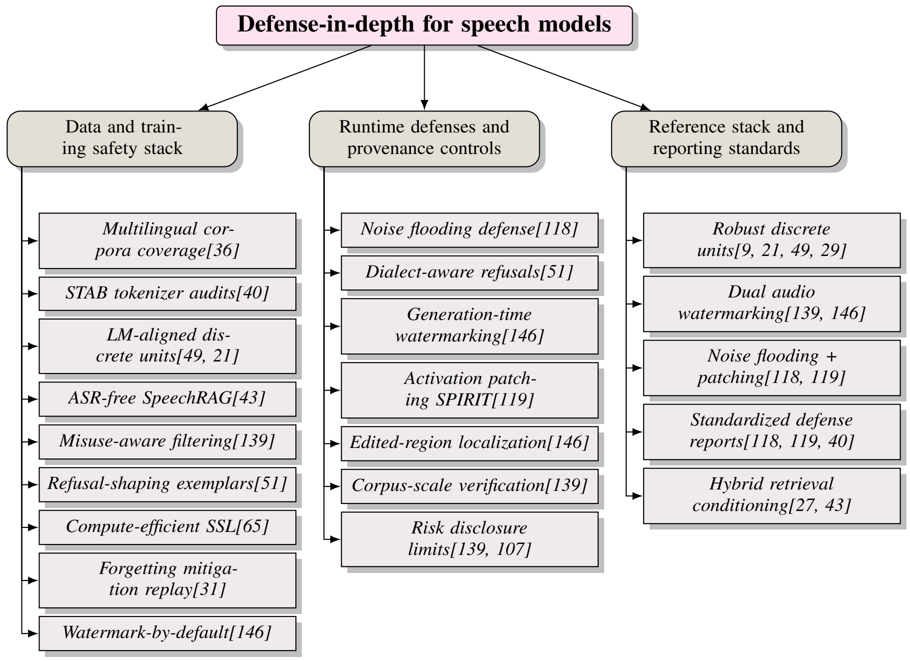

Figure 14: Figure: Defense-in-depth for speech-language systems. The tree organizes safeguards into (1) data and training measures that improve coverage, robustness, and safety alignment; (2) runtime defenses and provenance tools that harden inputs/outputs and enable post-hoc mitigation and verification; and (3) a reference stack with recommended components and reporting standards. Arrows in the conceptual figure flow from pre-deployment preparation to inference-time protection and auditing, emphasizing layered, complementary controls across the lifecycle.

Evaluate watermark detection at frame/segment granularity, anti-spoof robustness (replay/synthesis/cross-channel), consent ablations, and style-transfer errors across demographics [146, 54, 124]. A deployment blueprint couples LM-aware tokenization (LAST), multilingual SLU validation (IndicSUPERB), and acoustic robustness (SALMon) [22, 36, 41], while acknowledging adversarial audio jailbreak rates near 90

14.4 Robust and invariant representations

Representations should encode linguistic content while remaining invariant to speaker, channel/room, device, moderate prosody changes, non-speech acoustics, and adversarial perturbations. SALMon documents consistent degradations from noise, identity, emotion, and room effects; discrete codecs exhibit instability that induces omissions/repetitions, mitigated by augmentation-invariant tokens and continuous targets [41, 23, 96, 46].

Diagnostics. Decoder-free Token Edit Rate (TER/MTER) on discrete sequences, with feature smoothing and k-means corpus selection, quantify speaker/perturbation invariance; data2vec-audio-base units perform strongly [39]. STAB adds chrF robustness under pitch/speed perturbations and additive noise (10 dB SNR) and shows ASR auxiliary losses (USM-v2) improve speaker invariance; metadata-driven ABX across microphones/rooms exposes extra-linguistic leakage [40].

Mixture robustness and selective invariance. SHuBERT pre-trains HuBERT with target-speaker conditioning and dual-path decorrelation, aligning clean/mixture views, suppressing interference while preserving clean accuracy, transferring to low-resource settings, and strengthening downstream discrete unitizers and continuous prompts [88, 26, 27, 141].

Tokenization and codec targets. NAST produces local units plus a global residual stable under time-stretch, pitch-shift, reverberation, and noise, disentangling style [29]. USM-based tokenizers validated with STAB—and boosted by ASR losses—show content-oriented supervision increases invariance [40]. dMel tokens align to perceptual time–frequency structure, improving robustness to music/background speech at comparable bitrates over HuBERT-k-means and SpeechTokenizer; self-segmented syllable-synchronous embeddings provide rate-controlled, cross-lingual stability [54].

Training levers and evaluation. Combine content-grounded supervision at the tokenizer (ASR-style losses), enrollment conditioning and decorrelation (SHuBERT-like), and prosody-aware smoothing/data selection, with augmentation for room/channel diversity [40, 39]. Report TER/MTER under spectral-envelope warping and cross-speaker same-text pairs, STAB's chrF robustness and invariance probes, mixture robustness via cross-correlation, codec/token robustness under music/background speech, and matched-rate intelligibility/prosody slices [40, 39, 54].

14.5 Fairness and evaluator bias

Fairness requires disentangling true demographic/language disparities from evaluation artifacts such as contamination, tokenizer mismatch, and judge bias, and adopting multilingual, stratified, end-to-end evaluations. Large gaps persist across accents, L1/L2 status, and conditions; zero-shot WER can differ by an order of magnitude, and ICL narrows but does not close gaps and can harm some varieties [147]. IndicSUPERB's gender- and region-aware design supports auditable coverage; Dynamic-SUPERB Phase-2 and simultaneous ST pipelines show propagation of upstream demographic/domain biases and teacher errors [36, 74, 128].

Evaluator bias. Audio-aware LLM judges (ALLMs) scale evaluation but exhibit self-bias when grading same-family systems and show limited resolution among weaker models; cross-family judging and calibration reduce errors [123]. TELEVAL mitigates bias via string matching for factoid QA, separate text/audio scoring, and objective audio metrics, with triad GPT-4o judging for open-ended tasks [126]. Align-SLM and BoSS warn that single-ASR reliance and surrogate judges skew outcomes; dataset-centric coverage (e.g., Chinese focus) and binary gender assumptions demand explicit caveats [140, 47, 121].

A fairness-aware playbook combines standardized multilingual benchmarks (IndicSUPERB), tokenizer audits (STAB), end-to-end speech QA under varied acoustics (VoxEval), and acoustic-robustness probes (SALMon), within reproducible and compute-accessible SSL recipes [36, 40, 45, 41, 65]. Recommended practices:

- Stratified reporting with confidence intervals and effect sizes across accent/region, language/dialect, L1/L2, inclusive gender labels, and acoustic conditions; add STAB/ABX diagnostics to attribute disparities [40, 46, 148].
- Multi-judge, multi-tool scoring: cross-family ALLM ensembles, human calibration with disclosed correlations, objective metrics where possible, and sensitivity to the chosen ASR [126, 140].
- Provenance for synthetic data and domain matching: enumerate sources/teachers, deduplicate public corpora, release configs, and quantify domain shift via ablations [125, 3, 58].
- Group-aware adaptation with guardrails: report per-group deltas and flag negative or negligible gains from ICL/adaptation [147].

Contamination audits, tokenizer benchmarking with STAB, and LM-aware tokenization controlling vocabulary/rate (LAST, LM-SPT) reduce evaluator bias and yield more defensible fairness conclusions [22, 49, 40, 58].

14.6 Reporting standards and red teaming

Safety claims should be auditable and comparable. An evaluation card specifies robustness assumptions (acoustic confounds, adversarial budgets), metrics (Safety/Relevance/Helpfulness, JSR, SPR/SNR), benchmarks (IndicSUPERB, STAB, SALMon), judges (human and calibrated ALLMs), and governance (data provenance, contributor coverage, code/data release) [40, 36, 41]. Threat models must state access and budgets (white-/black-box, targeted/untargeted, SPR/SNR constraints, device/channel assumptions) and report per-category breakdowns with confidence intervals and pinned aggregation policies [118]. Release spoken red-team sets with speaker/accent/language/device metadata and licensing; pin judge models/prompts, calibrate to human labels (F180

A minimal red-teaming protocol uses ESPnet-SpeechLM for reproducible preprocessing, pretraining, inference, and evaluation; exercises tokenizer choices (Cont-SPT, LAST) and multilingual

SLU coverage (ASR, LID, KWS, QbE, SID/SV) via IndicSUPERB; and probes reasoning under latency with chunked think-while-talking [113, 20, 22, 36, 48]. Construct bilingual/multilingual seed sets (harmful/benign, varied delivery), generate multi-budget adversarial variants with per-sample SPR/SNR, and evaluate Safety/Relevance/Helpfulness/JSR using calibrated ALLMs correlated with standardized speech metrics (IndicSUPERB, STAB, SALMon), with pre-registered thresholds and human adjudication of low-confidence cases; include cross-accent/cross-lingual stress and expand coverage iteratively with IRB approval [118, 40, 41, 133].

Transparency and stewardship require open-sourcing datasets, models, configs, and code; reporting compute budgets and seeds; releasing harmful/benign sets, judge configs, scoring scripts, and per-sample annotations; and standardizing metric names/calculations with per-category/accent/language breakdowns [118]. Maintain multilingual, diverse benchmarks (IndicSUPERB), tokenizer assessments (STAB), spoken QA (VoxEval), and prosody-sensitive unit probes, and run iterative red-team rounds that recalibrate judges, expand datasets, stress-test named-entity handling, and publish full compute/language coverage disclosures [36, 40, 45, 39, 41, 110, 65].

15 Open Toolkits Frameworks and Reproducibility

15.1 5C overview: Code, Config, Compute, Corpora, Comparison

The 5C lens—Code, Config, Compute, Corpora, Comparison—specifies the open artifacts, rate-/latency-aware settings, and disclosure norms required for reproducible and cross-lab comparable speech–AI. As visualized in Figure 15, the overview is organized into three complementary pillars: (i) open stack exemplars (tokenizers/codecs and corpora), (ii) standardized evaluation anchors (low-cost, rate-/robustness-aware diagnostics), and (iii) benchmark-card disclosures (coverage, conditions, comparability, and compute/cost reporting). Together, these pillars link open, inspectable pipelines to comparable cross-lab evaluation via explicit, auditable reporting. It emphasizes end-to-end, inspectable stacks spanning tokenizers/codecs and LM bridges, interleaved SLMs, decoders/vocoders, and a standardized evaluation harness with dataset hygiene and statistical testing, exemplified by streamable dMel discretization with parallel encode/decode, RepCodec's high-fidelity semantic tokens, and LLaSO's public Align/Instruct/Eval sets with a 3.8B reference model [54, 25, 125].

For low-cost, comparable diagnostics (pillar ii), the framework anchors to STAB for tokenizer assessment, SALMon for acoustic-factor sensitivity, IndicSUPERB for multilingual SLU, Dynamic-SUPERB Phase-2 for community-expanded audio tasks, and VoxEval for speech-in/speech-out QA and spoken reasoning [40, 41, 36, 74, 45]. Finally, in keeping with pillar (iii), a 5C benchmark card accompanies results with Coverage, Conditions, Capabilities, Comparability, and Compute/Cost fields, mandating explicit token/frame rates, bitrates, schedules, hardware/precision, and auditable scorer scripts [9, 45, 40, 36, 104].

15.2 Code: Open ecosystems and stacks

Open ecosystems now release code, configs, and checkpoints with pinned evaluators and interfaces. LM-aligned tokenizers/codecs (LAST, RepCodec) retain semantics better than k-means; interleaved SLMs reuse a single LM for speech+text; IndicSUPERB supplies reproducible SLU baselines and leaderboards across 12 languages; toolchains integrate ESPnet/fairseq/NeMo/SpeechBrain/s3prl and log token/frame rates, bitrates, and streaming schedules [22, 25, 36]. Representative releases include: ESPnet recipes for encoders, tokenizers, decoders/vocoders, and decoder-only ASR with optional shallow-fusion LMs [62]; SpeechLM toolkits that standardize data prep, pre-training, inference, and evaluation, with ProsodyLM for word-level prosody tokens and ESPnet-SpeechLM covering 15+ tasks [71, 113]; LLaSO's public model, datasets, and evaluation suite [125]; OpenS2S for streaming, empathetic interleaved decoding [149]; SimulST/duplex SLM repos with fixed interleaving and read–write policies and first-audio latency under SimulEval [21, 98, 42, 84]; AV tokenizers (VQ-MAE-AV) and streamable phonetic tokenizers (PAST) for controlled fusion under fixed token budgets [73, 21]; evaluation harnesses integrating IndicSUPERB, STAB, and fastabx for scalable probes [36, 40, 53]; Encoder–Adaptor–LLM bridges, diarization toolkits, and unified scoring pipelines for replicable ASR/SQA ablations [61]; and community suites expanding discrete-token leaderboards (DASB), speech QA (VoxEval), and audio tasks (Dynamic-SUPERB Phase-2) [9, 45, 74]. Ready-to-run tokenizer training (memory-aware k-means), interleaved speech–text pretraining in token space, and mixed-vocabulary duplex stacks should ship with fixed configs/checkpoints, and STITCH-

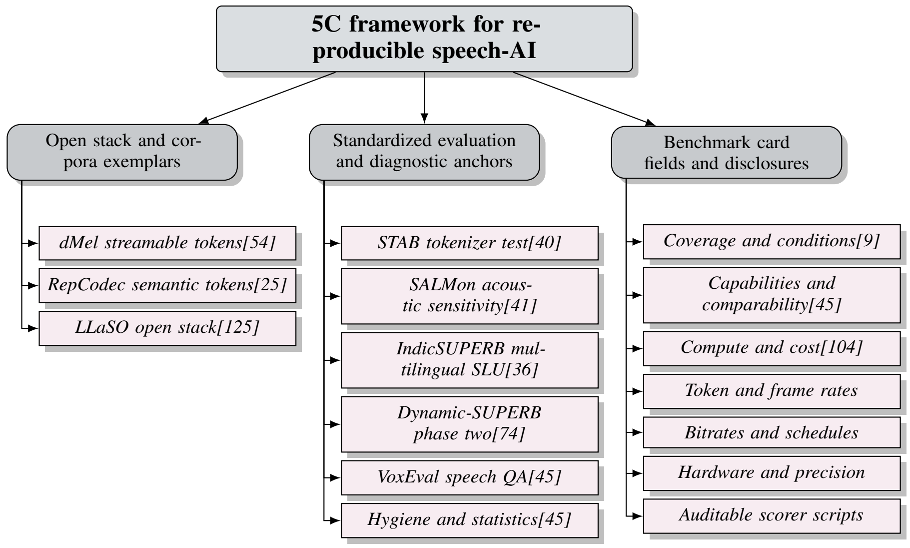

Figure 15: This figure shows a two-level 5C overview for reproducible speech–AI, grouping content into: (1) open stack exemplars (tokenizers/codecs and corpora), (2) standardized evaluation anchors (low-cost, rate-/robustness-aware diagnostics), and (3) benchmark card disclosures (coverage, conditions, comparability, and compute/cost reporting). Together, these pillars link open, inspectable pipelines to comparable cross-lab evaluation via explicit, auditable reporting.

style think-while-talking experiments must publish prompts, seeds, and chunking/scheduling scripts [65, 4, 51, 48].

15.3 Config: Reusable building blocks

Reusable, declarative configs separate datasets/task templates from model hyperparameters; fix tokenization, prompting, compute, decoding, streaming, and scoring defaults; and encode determinism (seeds, kernels, sharding, exemplar selection) to make few-shot and streaming runs exactly reproducible [54, 48]. Recommended structure: - Project manifest and task templates: a data.json of corpus paths, sampling rates, language tags, splits; reusable YAMLs for ASR/AST/S2ST/TTS/VC/unit LM/interleaved SLM/KWS/SER/LID and streaming variants across 15+ tasks [113]. - Tokenization block (LM-facing contract): LM-aware or ASR-guided semantic tokenizers (LAST, LM-SPT, RepCodec, dMel) with explicit frame$_rate_hz, bitrate_kbps, codebookgeometry, segmentation, dedup/aBPE, andupstreamfeatures; trackqualitywithS$ $rate/languageeffects$[22, 49, 25, 81, 40]. $-PromptingandICL: enumeratedprompttemplates, deterministicexemplarselection, contextualbiasinglists; multilingualsupportaligned$ $VoxPopuli$[147, 67, 110, 36]. $-Trainingblock: resource-awareoptimizers/schedules(bf16, gradaccumulation), HuBERT-styleSSLon8GPUswithoptionalASRbootstrapandpinnedLR/numelpackers/seeds$[65, 147]. $-Inference/streaming: unifiedsegmentationandread\breve{}writepolicies(wait-k/ratio), chunksizes, decodingknobs(beam/temperature/top-p)withstandardizedlatencyreporting(TTFT, first-audio, RTF)forcascaded, textlessS2ST, andSpeechLLMs$[98, 84, 42].$-Evaluation/scoring: pinnedASRforWER/CER, BLEU/COMETfortext, NISQA/MOSandSIM-Oforspeechquality/semantics, audio-awareLLMjudgeswithfixedseeds/prompts, fastabxABXsettings, andSTA$ 40, 53, 123, 36].$Configinheritanceshouldlockvettedbasestrategies(e.g., STAB-validateddMel/semantictokens; AdamWwith32kwarmupto5e-4, bf16; 45Mnumelsampler)whileoverlaysswapdatasets/prompts/decodingforASR/AST/dialog/QAorhierarchic$ $conditionmetadata$[54, 40, 33, 41].

15.4 Corpora: Curation and licensing

Ethical provenance, explicit licenses, reproducible splits, and auditable pipelines are prerequisites for fair comparison. Fieldwork collections like Pangloss provide CC licenses, informed consent, DOIs, and rich metadata enabling factor-isolated ABX diagnostics [46]. Kathbath, gathered via Karya with maker–checker verification, underpins IndicSUPERB (1,684 h; 12 languages; 1,218 contributors; 203 districts) with standardized WAVs and task-ready splits for ASR, SV/SID, LID, QbE, and KWS [36]. Curation checklist: - Licenses and scope: pin dataset/model/tool licenses, adhere to original splits (e.g., DASB), and publish dataset/model cards detailing uses and composition [9]. - Provenance and metadata: document text sources (IndicCorp, LDC-IL, Wikipedia), languages/scripts, sampling/normalization, seeds; release per-language keyword/QbE inventories, OOV/coverage stats; log speaker/microphone/room/genre to support robust splits [46, 36]. - Standardization: resample to 16 kHz mono; pin tokenizers/codecs and SSL layers (PAST, RepCodec, Cont-SPT); release preprocessing, and verify acoustic/content fidelity with SALMon [21, 25, 20, 41]. - Splits and audits: provide language-balanced train/valid/test-known/test-unknown/noisy and streaming subsets; publish de-dup logs and overlap checks to prevent contamination; disclose synthetic supervision recipes and hours by condition [40, 58, 98]. - Benchmarks and eval: release task cards with metrics and judge calibration sets; align with IndicSUPERB, STAB, SALMon, and VoxEval; report stratified CIs and effect sizes [40, 41, 45].

15.5 Comparison: Evaluation and testing

Comparability depends on open suites with fixed toolchains, pinned configs, and statistical reporting. Core pillars: IndicSUPERB for multilingual SLU, STAB for tokenizer diagnostics, SALMon for acoustic robustness, VoxEval for speech-in/speech-out QA, and DASB for discrete-token benchmarking under bitrate control; LLaSO-Eval unifies schemas and scorers; TELEVAL fixes decoding/backends for spoken dialogue; fastabx standardizes ABX pipelines [36, 40, 41, 45, 9, 125, 126, 53]. Required practices: - Pin interfaces and schedules: record token/frame rate, bitrate, codec depth/vocab, dedup/aBPE, decoder family/steps, beam/temperature/top-p, chunk sizes, read–write/wait-k; fix audio conditions, I/O modalities, and scoring dimensions [81, 45, 126]. - Rate-aware ablations: sweep rate/bitrate with fixed decoders; report WER/CER/BLEU, MOS/MUSHRA/NISQA/FAD, and identity/style metrics while holding prompting/policies constant [26, 81]. - Statistical rigor: paired, distribution-free tests with FDR control; bootstrapped CIs/effect sizes; human–ALLM agreement and judge calibration; forbid post-selection [113, 123]. - Minimum report: benchmark/task names and splits, corruption/conditions, decoding/schedules, judges/metrics, interface descriptors; supply code links, commit hashes, runnable recipes [28, 53]. - Leaderboards: aggregate across discriminative/generative tasks and bitrate tiers; require seeds, tokenizer/decoding configs, judge specs, I/O schemas, compute metadata; adopt SALMon's likelihood protocol as APIs converge [9, 41].

15.6 Compute: Budgets and resource table

Transparent compute disclosure normalizes comparisons across tokenizers/codecs, interleaved SLMs, streaming/duplex stacks, and judged evaluations. Report per-stage device type/count, precision, optimizer/schedule, steps/epochs, effective batch size, device-hours, throughput/utilization, and FLOPs or ISO-FLOP equivalents, with wall-clock on named hardware; pin software stacks (PyTorch/Transformers, FSDP/ZeRO, FlashAttention/TensorRT-LLM) and memory tactics; token-reduction (e.g., FastAdaSP) should publish reproducible memory/throughput gains and configs [104, 65, 117]. Resource-card blueprint (one line per stage): - System/stage and roles: tokenizer/codec (LAST, LM-SPT, dMel) at 6.25–25 Hz; SLM pretraining that reuses a text LM; SFT/alignment; decoder/vocoder; streaming/duplex policy; links to IndicSUPERB, STAB, VoxEval, SALMon, and fastabx [22, 49, 54, 40, 45, 41, 53]. - Data footprint: hours/tokens, sequence lengths, per-language/task breakdowns on Kathbath/IndicSUPERB; curriculum phases/data mixes [36]. - Hardware/precision and kernels: accelerator model×count, bf16/fp16/int8/QLoRA, checkpointing/packing, parallelism, kernel accelerators; report MFU/TFLOPs/s and speedups [57, 117]. - Optimization: AdamW with warmup/decay, weight decay, gradient clipping; optional OTReg to align speech–text embeddings for cross-dataset generalization [85]. - Batching and lengths: micro-batch, accumulation, effective batch; audio durations and token/frame counts; note segmentation width and vocab effects on efficiency [22]. - Time/utilization/FLOPs and latency: wall-clock, device-hours, throughput, MFU, total FLOPs; RTF and streaming latency for interactive use [104, 36]. - Artifacts: repo URLs, tags/commits, pinned configs, checkpoints, tokenizer files, evaluation scripts, dataset IDs and checksums [125, 40, 41]. Tracks should scale by budget: small uses modest labeled subsets and FBANK baselines on IndicSUPERB; medium adds SSL (e.g., wav2vec2) and tokenizers (dMel, Cont-SPT);

large unlocks full multilingual coverage with SALMon acoustic tests and optional contextual biasing. Include single-/few-GPU recipes (e.g., Slam: 24 h on one GPU; HuBERT on 8 GPUs) with pinned schedulers/decoders, fixed evaluation envelopes, and safety-eval compute/attack budgets (PGD steps, SNRs, judge runs) [54, 20, 41, 104, 65, 118]. Worked exemplars should state hardware/epochs per stage for empathetic alignment (BLSP-Emo), AV tokenization (VQ-MAE-AV), multilingual bridges (Encoder–Adaptor–LLM), and textless dialogue LMs on J-CHAT with dataset links and global statistics [7, 73, 61, 6].

16 Open Challenges and Future Directions

This section distills open problems and concrete paths for streamable, duplex speech–language systems that retain text competence, scale multilingually, and operate under realistic rate–compute budgets across five axes: interfaces that disentangle lexical content from residual acoustics; long-context memory for multi-turn interaction; low-resource and multilingual generalization; alignment to text knowledge without catastrophic forgetting; and a rate–capacity perspective linking token cadence, per-token entropy, and decoder scheduling with evaluation guardrails.

16.1 Unified vs multi-stream tokens and disentanglement

Interface choice sets the entropy presented to the semantic planner and the controllability of prosody/speaker identity at a given bitrate, thereby constraining streaming and duplex schedules. Factorized designs split a low-entropy semantic stream from acoustic/codec tokens, enabling conditional decoding that reduces redundancy, preserves controllability at low rates, and supports direct streaming S2ST without ASR→MT→TTS cascades or unified LSLMs that interleave segmentation, policy, and translation [98, 42, 13]. Unified designs remain competitive when decoders are text/prompt-aware and aligned: semantics and speaker/style are pushed into conditioning, permitting ultra-low-rate operation with intelligibility and style retention across ASR, TTS, VC, enhancement, and mixed speech–text generation, albeit with potential residual entanglement inside a single stream [14, 64].

A capability-centric decomposition suggests explicit channels for lexical content $(\mathrm{V}_L), affect(V_AC), contextualdynamics(V_CD), implicitsemantics(V_IS), andnon\text{-}verbals(e.g., laughter, backchanne$

Bitrate-controlled evaluation should probe: lexical predictability and resynthesis WER/CER (semantic path); speaker/style similarity, zero-shot prompt stability, prosody editability (duration/F0) and non-verbal timing (social cues); and latency/schedule compliance for streaming/duplex operation. Unified systems should audit leakage of tempo/affect into content tokens and quantify the conditioning strength needed to match explicit style channels, while mixed-token systems should report cross-stream redundancy and token-budget reductions from semantic→acoustic conditioning [13]. Design guidance: when bandwidth and simplicity dominate, use a unified low-bitrate stream decoded with text/prompt-aware models—achieving intelligible, style-consistent speech at tens of tokens/s with a single promptable interface [83, 44, 13, 42, 102]; when expressivity and social competence are primary, separate $\mathrm{V}_L from V_AC/V_CD/V_IS and condition acoustics on semantics for rate efficiency [13, 47]. A forward path combines both: keep semantics unified and predictable, attach an explicit thin control stream for aff$

16.2 Long context and conversational memory

Duplex agents must reason over minutes of audio and preserve state across turns while remaining streamable. Current stacks highlight windowing and segmentation limits: decoder-only speech LMs stabilize in-context learning at 4 demonstrations and prefer fixed-length formatting [67]; Encoder–Adaptor–LLM pipelines commonly bound context to 30 s segments with overlap, constraining cross-turn dependencies [60]; lightweight history already helps (e.g., two-utterance augmentation improves mid-training WER/CER) [61]. Streaming translation depends on punctuation cues and heuristic boundaries, motivating trainable semantic boundary detectors, adaptive delay policies, and joint speech-encoder–LSLM training [98]. Community evaluation is beginning to reflect longer-audio and multi-audio protocols [74]. These constraints, and the modular remedies they invite, are synthesized in Figure 16, which organizes (i) current streaming limitations (few-shot ICL saturation, segmentation and boundary heuristics), (ii) a duplex-aware memory blueprint (small local windows plus sparse, turn-synchronous summaries with prosody-aware compression and joint encoder–LSLM policy alignment), and (iii) a phased curriculum with evaluation and reporting (fixed-window warmup, boundary/delay supervision, distant supervision, progressive unfreezing, and memory-centric probes/metrics). Taken together, the chart maps constraints to design choices and training–evaluation practices for robust, low-latency duplex dialogue.

Aligned with the memory blueprint highlighted in Figure 16, a streaming-/duplex-aware memory maintains (i) a small, fast local context for token-level decisions and (ii) sparse, turn-synchronous summaries in a persistent, channel-tagged store so overlaps/interruptions do not erase state. Summaries should be rate-aware, combining variable-rate compression with anchors that preserve prosody/non-verbals; write/evict/retrieve policies are trained with latency objectives and informed by learned boundaries and adaptive delay [98]. Mechanistically: derive turn-synchronous anchors from streaming representations; keep local windows fixed-length for stable in-context behavior [67]; smooth boundaries via overlap–add or masked streaming schedules; retrieve conditioned on speaker and turn under duplex constraints; and align memory control with streaming policy via joint encoder–LSLM training [98]. Practical recipes: prepend one–two prior utterances for conversational ASR [61]; for assistants, store channel-tagged turn summaries and retrieve each turn while writing on longer segments but inferring on short windows [60]; for simultaneous translation, replace punctuation triggers with learned boundaries and adaptive delay [98].

Building on the curriculum and evaluation axis depicted in Figure 16, a phased curriculum introduces memory without destabilizing streaming: start with fixed-length local windows [67]; add supervision for boundary detection and delay control [98]; incorporate distant supervision for memory writes from transcript segmentation or multi-audio alignment [74]; and progressively unfreeze the encoder to manage compute and latency [60]. Evaluation tracks should include multi-turn instruction tasks with minute-scale memory probes and multi-source reasoning [74], recall–lag curves and learned boundary/delay policies under noise [98], with/without-context ablations reflecting two-utterance gains [61], and speech in-context scaling beyond four demonstrations with variable-length prompts [67]. Reporting minimums: segment size/overlap; boundary detector and delay policy; memory write/eviction rates, lookup latency, cache hits; duplex channel tagging/conflict resolution; quality–latency deltas with/without memory; SLM initialization/interleaving; compute budget/allocation; synthetic data and TextLM family; continual-learning stages and retention metrics; forgetting mitigation; and reproducibility metadata [57, 31, 65].

16.3 Low-resource and multilingual generalization

Multilingual robustness hinges on joint advances in interfaces, training, data, and evaluation. Interleaved pretraining is still En/Zh-heavy, limiting cross-lingual coverage and prosodic diversity [4]; expressive S2ST studies urge typologically diverse pairs (including tonal languages) and direct S2ST for unwritten languages [70]. Low-rate, streamable, linguistically aligned semantic streams stabilize semantics and reduce supervision pressure across languages [78]; speech-only routes scale without orthography [5]; discrete unit streams transfer cross-lingually but show zero-shot gaps on unseen languages, calling for stronger invariances and adaptation [67]. Multimodal distillation into residual-quantized latents and multi-teacher alignment reduce ASR dependence and strengthen language-robust content latents under noise and style variability [77]. Adapter-based bridges can deliver competitive conversational WER/CER across 11 languages with thin projectors over frozen encoders/LLMs, and modest multilingual additions yield further gains [87].

A practical curriculum couples: broader multilingual interleaved pretraining beyond En/Zh [4]; speech-only S2ST scaling for unwritten languages [5]; and codec/token-side semantic distillation with multi-teacher ensembles [77]. Promptable, in-context adaptation with frozen backbones and thin multilingual adapters, plus LM-aware tokenization, supports label-lean transfer [67, 22, 3, 85]. Data should prioritize human-recorded multilingual dialogue and expressive S2ST with code-switching, disfluencies, and diverse acoustics, complemented by synthetic instruction data to mitigate distribution shifts [52, 4, 70, 77].

An actionable blueprint integrates IndicSUPERB for multi-task SLU across 12 Indic languages [36]; LM-aware tokenization (LAST, LM-SPT) aligning speech tokens to a single LM at 25/12.5/6.25 Hz [22, 49]; VoxEval for spoken QA and robustness; SALMon for acoustic awareness; and fastabx for ABX discriminability [45, 41, 53]. Interfaces should emit low-rate semantic tokens directly in streaming S2ST and synthesize with an acoustic LM and codec to shorten sequences and improve BLEU/BLASER 2.0 on CVSS-C [42, 49]. Training can warm-start from text LMs, scale to

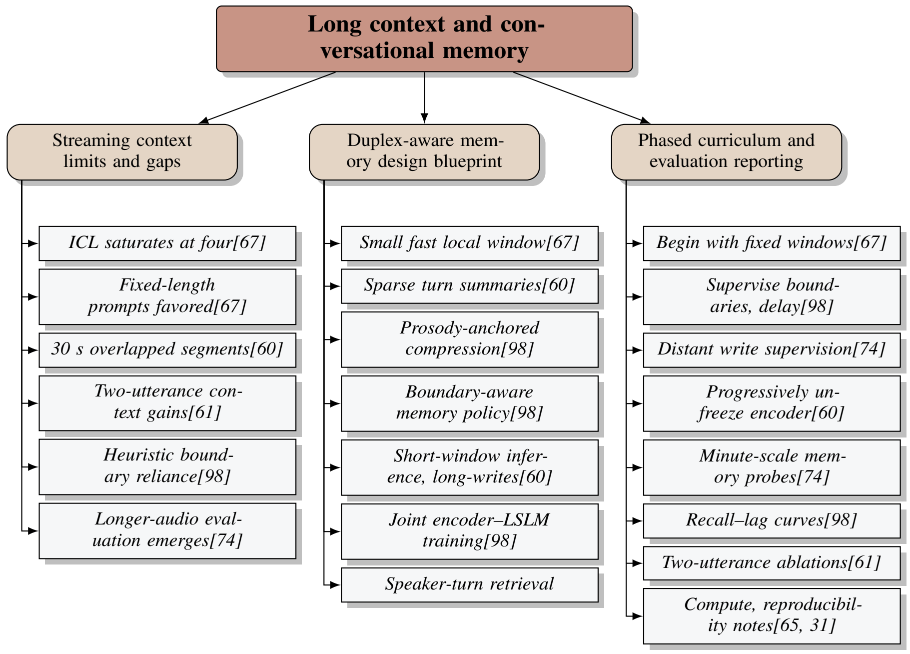

Figure 16: Figure: Duplex-aware long-context memory for streaming speech agents. The chart organizes (1) current streaming limitations (few-shot ICL saturation, segmentation and boundary heuristics), (2) a practical memory blueprint (small local windows plus sparse, turn-synchronous summaries with prosody-aware compression and joint encoder–LSLM policy alignment), and (3) a phased curriculum with evaluation/reporting (fixed-window warmup, boundary/delay supervision, distant supervision, progressive unfreezing, and memory-centric probes/metrics). Together, these components map constraints to design choices and training–evaluation practices for robust, low-latency duplex dialogue.

O(10^{12})$interleavedtokensviatext\text{-}to\text{-}tokensynthesis, andextendspeech\text{-}onlyS2STwithmulti\text{-}teacheralignment$[38, 22, 4]$.Streaming/duplexevaluationshouldstressadaptiveinterleavingandbarge\text{-}inundermultilingualnoise/overlap$
42, 3]$, withtargetsofimprovedunseen\text{-}languagezero\text{-}shotaccuracyoverSVCbaselines$[67]$, lowermultilingualstream$

16.4 Alignment with text knowledge without forgetting

Warm-starting from a pretrained text LLM and localizing updates to small speech-side adapters or discounted-LoRA while freezing the language core preserves instruction-following and world knowledge during speech alignment; forgetting is better mitigated via experience replay, occasional model merging, and regularizers that maintain the base model's geometry/behavior [31, 38, 10, 3]. Naive cross-modal finetuning degrades text abilities, and warm starts are insufficient without geometric/behavioral alignment [5].

Practical levers include LM-aware tokenization (LAST) and tokenizer diagnostics (STAB) that predict downstream quality [22, 40], chunked think-and-talk decoding (STITCH) for silent reasoning at constant latency [48], and adapter bridges that enable ASR/AST and zero-shot instruction following with frozen backbones [3]. SpeechRAG shows that aligning speech–text embeddings for ASR-free retrieval/generation can outperform cascades when WER is high [43]. A retention-aware curriculum: Stage A aligns geometry with a speech→text projector into the frozen LLM embedding space plus KL distillation to the base decoder [22, 3, 38]; Stage B jointly finetunes speech–text tasks (ASR, ST/S2ST, SQA) interleaved with text-only SFT for rehearsal [5, 111, 3]; Stage C densifies textual conditioning before speech (plan-before-speak) [35]; Stage D deepens on multilingual SLU (IndicSUPERB) and refines tokens with LAST or dMel [36, 22, 54]; Stage E preserves MI between LLM latents and speech units to retain prosody/affect [47]. Additional tactics include prompt bottlenecks for decoder-only ASR with pseudo-prompt curricula [62], behavior KL to the base policy [7], and RL safeguards with KL/hypothesis-similarity constraints when process-based rewards are sparse/ineffective [61].

Evaluation should combine IndicSUPERB, STAB, augmentation-invariant tokenizer checks, and retention audits on text-only tasks [40, 141, 36].

16.5 Toward a theory of rate vs capacity

Treat the interface as an information pipe with load R = r·H bits/s, where r is token cadence and H is per-token entropy after abstraction; decoder capacity B(,C,S) depends on model size/architecture, compute, and schedule (autoregressive vs masked/grouped vs diffusion). Mismatch yields saturation when R>B, while underuse and unstable codes arise when RB or codebooks misalign; higher-information tokens mitigate waste [25, 65]. Mixed speech–text pretraining reveals a semantic knee near 12.5 Hz for moderate decoders, with smaller models showing modality competition [4]. Diffusion-backed decoders shift detail into the schedule and maintain DNSMOS/WER as tokens/s shrink via variable acoustic depth [13]; text-aware diffusion reaches low WER and high SIM at 6.25 Hz with both large AR and compact MGM decoders, exposing quality–efficiency contours via 10–32 diffusion steps [14].

Design pillars: cadence shaping and abstraction alignment to fit R within B at target scale, reducing H through structure-preserving abstraction rather than blunt downsampling; dual-channel apportionment of B across lexical and paralinguistic streams with cross-layer alignment to each stream's entropy budget; and scheduler-level control of bits-per-step (diffusion steps, acoustic depth, multi-token prediction) to keep online R compatible with B under streaming [4, 13, 14]. A measurement program should run iso-rate sweeps (vary decoder expressivity/schedule at fixed r), iso-capacity sweeps (vary r at fixed /S to locate knees near 12.5 Hz and validate 6.25 Hz feasibility), and cross-modal competition tests that vary interleaving and size to map interference thresholds; reporting must include R = r·H estimates, schedule descriptors, and downstream quality/latency to place systems on a shared rate–capacity plot [4, 13, 14]. Actionable defaults: 12.5 Hz for mixed streams with entropy reduction via segmental abstraction/quantization; 6.25 Hz for large/hybrid stacks when tokens are generation-aligned and decoders operate at 10–32 diffusion steps; capacity apportioned across lexical/acoustic channels with abstractions aligned accordingly, using schedule-level knobs to maintain streaming compliance.

16.6 Trade-offs and limitations

Choices in interfaces, tokenization, decoders, alignment curricula, and evaluation co-determine quality, latency, robustness, and validity. Thin adapters (1

Tokenization should disentangle content from speaker/paralinguistics while meeting codec–decoder constraints: dMel provides training-free discretization with parallel decoding [54]; LAST aligns tokens to text-LM objectives [22]; PAST jointly models phonetic/acoustic tokens with a streamable variant [21]; and SSL-based semantic tokens plus universal vocoders with layer attention enhance OOD robustness [26]. Language-aware codecs remain speech-only at 24 kHz, lose accuracy beyond four channels, and lack streaming/channel-budget evaluations; online inversion and channel balancing remain open [12]. Text-aligned discretization improves editability but inherits ASR-encoder domain limits and needs co-training with streaming, duration-aware vocoders to close rhythm/prosody gaps [62].

Paralinguistic grounding benefits from explicit metadata (emotion/prosody) and implicit supervision (QA pairs), improving LLM-judged empathetic QA by 38–46

Actionable guardrails: standardize on IndicSUPERB and STAB, preserve benchmark assets, and track text-only capabilities before/after each alignment stage [40, 36]. Stress-test duplex stacks with barge-in and multilingual noise/overlap; compare continuous projectors versus discrete-token conditioning at matched rates and, if discrete lags, increase text rehearsal, apply MI-preserving unit alignment, or route semantics through prompt bottlenecks [1, 47, 62]. For ASR, strengthen CTC prompts (e.g., blank-frame removal), provision decoder capacity, and report compression–WER trade-offs [62]. For S2ST, ablate codec channel budgets and prompt/context lengths; adopt adaptive channel capping; report streaming latency (average lagging, first-packet delay) and online inversion latency; and compare against textless streaming transducers and IWSLT-style simultaneous ST with LM-aligned tokenizers at 25/12.5/6.25 Hz [5, 49, 128, 42, 127]. Standardize cross-codec comparisons at matched rates/domains/bitrates with public AR/NAR/masked-parallel baselines [28]. Expand multilingual coverage beyond English/EsEn and disclose style/prompt regimes; augment human evaluation with ensembles of audio-aware LLM judges and ProsodyLM-style content–prosody metrics, plus SALMon-style acoustic consistency and broader SLU tasks [71, 123, 41, 36]. Pair capability work with open, multilingual datasets and compute-efficient SSL to promote inclusive and equitable speech technology [65, 36, 47, 114, 46].

17 Conclusion

17.1 Key takeaways and recommendations

- Token interfaces and rates
  - Default to mixed speech–text modeling at roughly 12.5 Hz to balance semantic fidelity, sequence length, and streaming practicality; always report tokens/s and effective bitrate alongside quality and latency.
  - Under tight budgets, adopt coarse, text- or syllable-aligned content tokens and route prosody/timbre through a complementary acoustic path; at extreme budgets, 6.25 Hz is feasible only with a text-/prompt-aware decoder and explicit tracking of the reconstruction–generation gap.
  - With RVQ codecs, reduce acoustic depth per frame before lowering cadence, and condition the acoustic stream on semantics to remove redundancy and improve predictability at a fixed bitrate.
- Modeling and decoding
  - Pair autoregressive decoding on semantics (or the first RVQ channel) with non-autoregressive refinement on residual channels to cut wall-clock latency while preserving intelligibility; use grouped/multi-token prediction or masked infilling to further shrink steps.
  - Favor few-step flow/diffusion decoders at low token rates; otherwise rely on codec-native inversion or lightweight vocoders to bound rendering latency.
  - For duplex agents, inject streaming listening features at every block (middle fusion), expose an explicit interruption token/head, and interleave short text "plan" bursts with speech chunks to minimize time-to-first-audio without sacrificing coherence.
- Alignment and training
  - First bridge speech to a frozen text LLM via a thin projector; then apply constrained LoRA on speech spans and rehearse text-only batches to preserve text competence and limit drift.
  - Stabilize decoder-only ASR and mixed decoding with label-/prompt-synchronous compression (e.g., CTC blank removal, short pseudo-prompts) and a plan-before-speak schedule that continually anchors acoustics to fresh semantics.
  - Elevate beyond-semantic channels to primary learning targets: integrate relevance-driven processing objectives and mutual-information constraints so prosody, affect, and contextual dynamics remain decodable and controllable.
- Disentanglement and control
  - Keep lexical tokens low-entropy; route prosody, identity, and non-verbals through a control plane (global style embedding or short acoustic prompt plus an explicitly conditioned acoustic stream).
  - Build encoders/decoders that preserve affective cues, contextual dynamics, and implicit semantics end to end, enabling controllable beyond-semantic attributes and personalization through explicit character and style controls.
- Evaluation methodology
  - Center reports on bitrate and cadence: disclose token/frame rate, bitrate, channel/codebook geometry, and post-compression ratios; pair semantic metrics (WER/CER/BLEU) with perceptual (MOS/MUSHRA/UTMOS, FAD) and identity/style similarity; publish reconstruction–generation gaps at low rates.
  - For streaming and simultaneous settings, standardize latency panels—time-to-first-token/audio, real-time factor, AL/LAAL/StreamLAAL—under fixed chunk sizes and read–write or wait-k schedules.

- Use calibrated human raters and pinned audio-aware LLM judges; report inter-rater reliability, human–judge correlations, and multi-seed statistics; broaden multilingual and dialectal coverage and strengthen human-evaluated protocols.

- Deployment and streaming
  - Target a bandwidth ladder for robust duplex on constrained links (0.3/0.6/1.2 kbps), with a midrate default and brief quality bursts for emphasis; co-report VRAM envelopes, throughput, and first-chunk latency on named hardware.
  - Use block-causal tokenizers and block-wise decoding for steady streaming; add sequence-thinning/KV-cache optimizations and grouped multi-token decoding for throughput; harden duplex with explicit interruption handling and overlap caps.
- Safety, robustness, and provenance
  - Employ defense-in-depth: input hardening for adversarial audio, runtime guardrails, and activation-level patching where appropriate; enable watermark-on-by-default synthesis with frame-level detectors to localize edited segments.
  - Report Safety/Helpfulness/Relevance and jailbreak/defense success rates with calibrated judges and per-category breakdowns, including attack budgets and SNR/SPR settings.
- Open tooling and reproducibility
  - Follow 5C: ship code and evaluators; pin rate-/schedule-aware configs; disclose compute and latency sheets; release licensed corpora with contamination checks; compare under open suites with statistical rigor.
  - For beyond-semantic capabilities, prioritize multilingual benchmarks, richer human-evaluated protocols, controllable paralinguistic generation, and explicit personalization hooks in model cards and APIs.

17.2 Community resources and standardization

A durable speech–AI ecosystem depends on open toolchains, auditable corpora, and rate-/latency–aware benchmarks that make tokenizers, SLMs, and duplex systems comparable across labs. Community baselines already span discrete-token evaluation, intrinsic tokenizer screening, instruction-style multitasking, spoken knowledge/reasoning, and dialogue capability; elevating these to first-class standards enables faster, cumulative progress.

Actionable community commitments

- Open stacks and ready-to-run recipes
  - Publish end-to-end code, pinned configs, and checkpoints for tokenizers/codecs, Encoder–Adaptor–LLM bridges, interleaved decoders, and renderers with rate-aware defaults. Provide runnable duplex-ready stacks on a single GPU and reference streaming scripts with fixed chunk/lag policies to reproduce latency–quality curves.
- Curated corpora with auditable provenance
  - Release large, multilingual, spontaneous corpora with licensing, maker–checker pipelines, stratified splits, and turn-level metadata. Standardize streaming chain-of-thought and long-form truncation datasets for simultaneous translation, and document domain-specific resources for expressive speech-to-speech with denoising and transcript-correction protocols.
- Standardized, rate-aware evaluations and leaderboards
  - Adopt a shared suite spanning bitrate-controlled downstream tasks, intrinsic tokenizer probes at CPU cost, instruction-style multitasking, spoken knowledge/QA, dialogue under noise/dialects, and ABX diagnostics. Require per-condition reporting (token/frame rate, bitrate, chunk/lag policy) with multi-seed statistics, published aggregation rules, and judge prompts.
- A shared benchmark card and 5C reporting

- Mandate a rate-/latency–aware benchmark card: interface descriptors (Hz, kbps, codebooks/channels), decoding/scheduling (beam/temperature/top-p, block sizes, wait-*k*/read–write ratios), fixed recognizers/judges with calibration, and compute/runtime sheets (TTFT, first-audio time, RTF, VRAM). Use the 5C lens—Code, Config, Compute, Corpora, Comparison—for apples-to-apples replication and ranking across models and rates.

- Streaming and duplex standardization
  - Fix reference streaming policies—chunk sizes, wait-*k* and CIF gating, and duplex interleaves (plan-before-speak cycles with explicit interrupt windows)—and report AL/LAAL/StreamLAAL with BLEU/COMET or WER under matched schedules.
- Safety, provenance, and red teaming
  - Ship watermark-on-by-default decoders and detectors for edited-region provenance; publish adversarial audio corpora and defense baselines with Attack/Defense Success Rates and SNR/SPR budgets; standardize judge pipelines and per-category Safety/Relevance/Helpfulness breakdowns on spoken red-team suites.
- Open instruction and evaluation judges
  - Release auditable instruction resources and pinned judge prompts, calibrating audio-aware LLM judges to human raters with reported correlations and confidence intervals; encourage dual-judge setups (proprietary and open-source) to reduce self-bias.

Community registries and governance

- Establish a registry of speech tokenizers/codecs with machine-readable interface manifests (Hz, kbps, codebook geometry, streaming support), intrinsic probe summaries, and links to bitrate-controlled downstream results.
- Maintain a catalog of duplex-ready SLMs with standardized duplex metrics (interactivity F1, barge-in latency) and streaming sheets, plus a multilingual dataset index listing licensing, splits, and coverage.
- Require model and dataset cards to include rate-/latency–aware descriptors, safety posture (watermarking, adversarial defenses), and judge calibration artifacts for reproducible, risk-aware deployment.

By converging on open recipes and corpora, adopting shared rate-aware benchmark suites, and institutionalizing 5C benchmark cards, the community can make heterogeneous speech–AI stacks comparable and reproducible, advancing tokenization, modeling, streaming/duplex, and safety in lockstep.

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